FishBot v0.4

Exact Combinatorial Inference, Opponent Modelling, and the Limits of the Uniform Posterior in Six-Player Literature

Abstract

Literature (also called Fish) is a six-player, two-team, imperfect-information card game in which players interrogate opponents for specific cards and score by declaring the exact distribution of a six-card half-suit among their own team. It is a team game with hidden information and no legal communication channel, so neither the two-player zero-sum machinery that solved poker nor the fully cooperative machinery developed for Hanabi applies directly.

The previous version of this engine ended by naming its own bottleneck: its belief sampler was documented as *not uniform* over the deals consistent with the public record, so every probability it reported inherited an unquantified bias. This paper removes that bias, and reports what happened.

We first show that hidden-state inference in Literature reduces to counting assignments in a degree-constrained bipartite system, which is $\#P$-complete in general but tiny in the dimension that matters here: real positions have 25.1 unlocated cards on average but only 4.1 *distinct candidate sets*, so cards are exchangeable and the problem collapses onto a count matrix with at most 9 rows across the corpus we measured. A dynamic program over that structure returns the exact partition function, exact marginals, exact uniform sampling and exact joint probabilities, validated against exhaustive enumeration. The one part it cannot absorb — the disjunctive constraints certifying that an asker held a card of the half-suit — we handle by importance sampling with closed-form proposal densities, after measuring why the two obvious alternatives fail.

The result is that the inference is now provably correct, and **it changed nothing**: against the previous champion, with the ask objective held fixed, the set differential was -0.22 per duplicate deal-pair, 95% CI $[-0.96, +0.52]$.

To find out why, we built a benchmark the previous version lacked: because the hidden hands exist in simulation, a posterior can be scored directly on how much probability it places on where the cards actually are. Over 517 decisions and 13,640 uncertain-card predictions, the *exactly uniform* posterior was *worse* at predicting reality than the biased sampler it replaced (negative log-likelihood 1.362 against 1.341 at a matched budget). The uniform posterior is a correct theorem about a false hypothesis — that players' choices carry no information beyond legality — and the old sampler's bias happened to encode a crude opponent model.

Replacing the accident with an explicit one — a single parameter modelling the hypothesis that a player asks in a half-suit in proportion to how many cards of it they hold — beats both on posterior accuracy (1.325) and is worth **+1.85 sets per duplicate deal-pair** in play, 95% CI $[+1.23, +2.46]$, essentially flat over a threefold range of the parameter. For scale, the entire objective improvement separating the previous champion from its own baseline was 1.26 sets per pair.

The resulting engine beats the previous champion by **+1.85 sets per deal-pair** $[+1.54, +2.17]$ over 500 duplicate deals, wins every pair against the two weakest policies in the ladder, and improves agreement with provably optimal play in exactly solved positions from 67.6% to 72.5%. Nothing else we tried helped: an eleven-term ask objective, a learned half-suit value, a rollout-regression objective, and a widened exact endgame solver are all reported as nulls, and the one screening cell that did resolve failed to replicate twice. We also build the one search design the variance diagnosis leaves open — a lookahead over the posterior rather than over sampled layouts — and prove that its natural form is exactly

greedy at every depth, so that only the quota coupling between candidates can make such a search non-trivial. Settling whether the coupled form helps took 9,700 duplicate deal-pairs across four rounds; the answer, from a pre-registered test over the final 6000, is $+0.104$ sets per deal-pair, 95% CI $[+0.020, +0.189]$. The route there is the more useful finding, and the estimate over rounds is the shape of it: $+0.570$ at the 200-pair screening cell that started this, $+0.153$ pooled over the unselected cells, $+0.104$ pre-registered. Three runs of the identical configuration returned $+0.570$, -0.044 and $+0.386$ sets per deal-pair, disagreeing by more than their own intervals allow ($Q = 7.03$, $p = 0.030$), which suggested a between-run variance component that no interval in this paper models. We measured that directly — 24 blocks of an A/A, where the true differential is exactly 0 — and it does not exist: coverage of the nominal 95% interval is 23/24, $Q = 18.41$ on 23 df at $p = 0.735$, and the estimated between-run standard deviation is 0.0000. The intervals are honest and the three-cell disagreement was the one-in-thirty event. Establishing that required giving the harness the ability to observe its own null, which it did not have: under seat-based seeding an A/A is bit-identical in both halves of a duplicate pair and its differential is exactly zero for every deal. Two later results bear on where the remaining strength is. Tripling the posterior sampling budget, from 160 draws to 480, is worth **+0.340 sets per deal-pair**, 95% CI $[+0.243, +0.436]$ over a pre-registered 6000-pair run with all six blocks positive and $I^2 = 0\%$ — the largest effect in this study after the opponent model itself, and more than three times the belief-space search. It is affordable because a decision costs 3.57 ms of fixed work plus 6.3 μs per draw, so tripling the sampler costs 1.38× rather than 3×; profiling locates that fixed term in per-card dispatch inside a sequential sampler, where there is nothing to hoist. Since the objective ranks barely better than $\mathbb{P}$(success) alone and nearly every term added beside it is null, we read this as evidence that the leverage in this game is in the belief rather than in what is computed from it.

The second is a retraction. This paper closed its objective-learning line on the claim that the belief tracker *cannot* attach to a determinized mid-game position. It can, in a few lines. On identical positions, worlds and root asks, finishing rollouts with the full engine instead of a public-information heuristic raises the target's slope on $\mathbb{P}$(success) from $+0.040 \pm 0.045$ to $+0.681 \pm 0.142$, a paired difference of $+0.641 \pm 0.145$; the heuristic needs 181 plies to finish a position the engine finishes in 26. The diagnosis was right about the mechanism and wrong that it was a wall. We report it as the fifth appearance in this study of the same error — reasoning about a quantity instead of measuring it — and the first in which it closed a research direction rather than merely inflating a number, because a negative claim generates no experiment.

We also report several methodological corrections, including a bias in the previous harness's handling of timed-out games that we demonstrate can place the true value outside a reported 95% confidence interval; a measurement showing that the previous version's 600-pair experiment cells could not resolve effects below about 0.44 sets per pair, so its smaller null results were uninformative rather than negative; and the finding that the per-pair standard deviation those calculations rest on is not a property of the game at all but of how often the two policies being compared diverge, which systematically over-sizes exactly the near-identical comparisons that most need care.

1 Introduction

Literature — also called Fish or Canadian Fish — is a six-player, two-team, imperfect-information card game. Players interrogate opponents for specific cards and score by *declaring* the exact distribution of a six-card half-suit among the members of their own team. It sits in a corner of the game-AI landscape that is unusually poorly served: it is a team game with hidden information and *no legal communication channel*, so neither the two-player zero-sum machinery that solved poker nor the fully cooperative machinery developed for Hanabi applies directly.

This paper is the fourth iteration of an engine for the game. The previous version, v0.3, established three results that frame everything here. First, exact belief tracking dominates: an agent that merely chains every logical implication of the public record beats a public-information heuristic by 11.86 sets per duplicate deal-pair. Second, and more surprisingly, search does not help, and the reason is measurable: the standard deviation of a position's value across possible hidden layouts is about 2.4× the gap between the best and the worst candidate move, so any search that evaluates different candidates against different sampled layouts is ranking luck. Third, acting on that diagnosis — by improving the *objective* rather than the depth of search — produced the study's largest single gain.

v0.3 closed by naming its own bottlenecks, and this paper attacks them directly. Its belief sampler was documented as *not uniform* over the set of deals consistent with the public record, so every probability the engine reported inherited an unquantified bias. Its ask objective was a hand-weighted sum of three terms. Its claim policy was a fixed confidence threshold. The obvious research programme is therefore: make the inference exact, widen and then learn the objective, and make claiming decision-theoretic.

We carried that programme out, and the headline result is not the one we expected.

Contributions.

1. **Exact combinatorial inference.** We show that hidden-state inference in Literature reduces to counting assignments in a degree-constrained bipartite system, and that although the general problem is $\#P$-complete, the specific instances arising in this game are tiny in the dimension that matters: the number of *distinct candidate sets* is on average 4.1 even when 25.1 cards are unlocated. We give an exact dynamic program over that structure that returns the partition function, exact marginals, and exact uniform sampling, and validate it against exhaustive enumeration (Section 5).

2. **A measured account of the disjunctive constraints.** The constraint set contains clauses ("the asker held at least one card of that half-suit") that the dynamic program does not absorb. We quantify three ways of handling them and show why the obvious two fail, then give an importance sampler whose proposal density is computable in closed form and whose self-normalised weights are consistent for the target (Section 6).

3. **Posterior quality measured against ground truth.** Every belief metric in the previous version was indirect. Because the hidden hands exist in the simulator, we can score a posterior directly on how much probability it places on where the cards actually are. We report negative log-likelihood, Brier score, top-1 accuracy and calibration for every inference configuration (Section 8).

4. **The central negative-then-positive result.** Making the posterior *exactly* correct under the standard assumption — that players' choices carry no information beyond legality — made it measurably *worse* at predicting reality than the biased sampler it replaced, and produced no improvement in play. The reason is identifiable: the old sampler's bias was accidentally informative, encoding a crude opponent model. Replacing the accident with an explicit, single-parameter model of how players choose which half-suit to ask in recovers the loss and then beats both (Sections 7 and 14).

5. **A wider, ablatable ask objective**, including two terms that the literature suggests are unstudied: the *exposure* of one's own already- public cards to the opponent who would receive the turn on a failed ask, and the *signalling* value of an ask, since the choice of half-suit is the only legal communication channel in the game and is simultaneously read by the opposition (Section 10).

6. **Methodological corrections** to the previous evaluation harness, including a bias in how timed-out games were handled that we demonstrate can put the true value outside a reported 95% confidence interval (Section 13); a measurement showing the previous version's experiment cells could not resolve effects below about 0.44 sets per pair, so several

of its reported nulls were uninformative rather than negative; and a worked example of a screening study manufacturing a false positive, caught by replication (Section 14.10).

7. **A widened exact solver**, which is the source of two structural facts about the game: the number of live cards is always exactly six times the number of unresolved half-suits, and the perfect-information value has the closed form sign(mover) $\times (2f - m)$. The second means the previous version's headline absolute result — 100% agreement with optimal play in information-resolved positions — was a shallower certificate than it read, and we reproduce it on a corpus that is 79% two-half-suit (Section 9).

8. **The search the variance diagnosis leaves open, and why its obvious form cannot work.** Every search this project has tried samples layouts, and the measured spread across them is why they all failed. A lookahead over the posterior instead is deterministic given the belief, so that variance is structurally absent. We show (Proposition 3) that the natural version of it — one whose transition edits only the asked card's row — is *exactly* greedy at every depth by an exchange argument, so it is a no-op by construction rather than by measurement, and only the quota coupling between candidates can make such a search non-trivial. Whether the coupled version helps is still open, and the attempt to resolve it produced a methodological result we consider more valuable than the search would have been: a direct measurement, over 24 blocks of an A/A, that this harness's duplicate-deal intervals cover at their nominal rate and carry no between-run variance component — which required first making the harness able to observe its own null at all (Sections 12 and 14.11).

2 Related work

Literature itself. We could find no peer-reviewed AI work on Literature, Canadian Fish or Russian Fish. Several open-source implementations with bots exist; the best documented uses Q-learning on a four-player variant with a first-order knowledge encoding. The game family did recently enter a standardised testbed: Goadrich et al. (2026) include Go Fish, and report that "Go Fish and Crazy Eights show little separation between MCTS and random play, likely because CardStock's determinizations account only for visibility and not for information inferred from action history." That is a published negative result about exactly the failure mode this paper attacks from the other side — they identify that action history carries information their determinizations ignore; we quantify how much, and put it in.

Determinization and its pathologies. Evaluating an imperfect-information position by sampling consistent worlds and solving each one perfectly is the standard technique, and its two failure modes are named: *strategy fusion*, where the searcher implicitly assumes it can act differently in states it cannot distinguish, and *non-locality* (Frank and Basin, 1998). Long et al. (2010) give the conditions under which it nonetheless works, parameterised by leaf correlation, bias, and a disambiguation factor. Two consequences shape this work. First, our disambiguation factor is unusually high — every ask is a hard constraint on the deal — which is favourable. Second, and decisively, the Claim action is a strategy-fusion trap: inside a determinization the searcher knows the true layout, so every claim evaluates as a certain gain and the declared assignment differs between sampled worlds. Claims in v0.4 are therefore never selected by search. Cowling et al. (2012) introduce Information Set MCTS and state plainly that they do not consider belief distributions; v0.3's own experiments found ISMCTS losing badly to its own prior, which is consistent with both that and with the variance diagnosis it measured.

Inference from what opponents chose. The closest published line is Solinas et al. (2019) and Rebstock et al. (2019), who infer hidden cards in trick-taking games from the actions players took, using a learned policy as the likelihood. Their factorised belief does not enforce hand-size constraints; they note the constraint could be added and do not add it. Our setting lets us do both at once: the quota constraints are handled exactly by the dynamic program of Section 5, and the choice likelihood enters as an importance weight on top. Solinas et al. (2023) prove that exhaustive history filtering is polynomial only when the public tree is sparse; Literature's is dense, so their route does not port — but the count abstraction makes the constraint side tractable anyway.

Counting and sampling with fixed margins. The combinatorial problem underneath is counting bipartite degree-constrained subgraphs, equivalently a restricted permanent, $\#P$-complete in general (Valiant, 1979), with a celebrated approximation scheme (Jerrum et al., 2004) and a literature on sequential importance sampling for contingency tables (Chen et al., 2005) — along with cautionary results showing SIS can be exponentially bad on some instances (Bezáková et al., 2012). We need none of that machinery, because the instances here are small in the dimension that matters. The constraint-programming counterpart is Régin's global cardinality constraint (Régin, 1996), which achieves arc consistency by a flow argument; our dynamic program is stronger in that it also *counts*, which is what claim probabilities require.

Team games without communication. Literature is a two-team zero-sum game with pre-play agreement and no in-play communication, which is exactly Celli and Gatti's regime; the solution concept is a team maximin equilibrium with correlation, not a Nash equilibrium (von Stengel and Koller, 1997; Celli and Gatti, 2018). The convergence guarantees that underwrite two-player zero-sum solvers do not transfer. Hanabi (Bard et al., 2020) is the closest studied game in spirit, and one of its methodological contributions transfers cleanly: cross-play between independently trained partners as the instrument for detecting conventions that work only against copies of oneself (Hu et al., 2020). What does not transfer is the remedy, because Hanabi is fully cooperative while Literature's only communication channel — the choice of which half-suit to ask in — is *costly* and read by the opposition as well as by one's partner. We found no published treatment of that combination, and it is the motivation for the `signal` term of Section 10.

3 The game

Six players sit in a circle in alternating team order: seats $\{0, 2, 4\}$ against $\{1, 3, 5\}$. The deck is partitioned into *half-suits* of six cards each. Our primary variant uses 54 cards and nine half-suits: the eight natural half-suits (2–7 and 9–A of each suit) plus a ninth composed of the four 8s and two individually identified jokers, a Red Joker and a Black Joker. Each player is dealt nine cards. The classic 48-card, eight-half-suit variant is also supported.

On turn, a player either **asks** or **claims**.

An **ask** names an opponent and a specific card. It is legal only if the asker holds at least one card of that card's half-suit and does not hold the card itself. If the opponent has it, the card transfers *face up* and the asker keeps the turn; otherwise the turn passes to the opponent who was asked.

A **claim** names a half-suit and declares which teammate holds each of its six cards. If the declaration is exactly right, the claiming team scores the half-suit. If any of the six is with an opponent, the opposing team scores it. If all six are within the claiming team but the declared distribution is wrong, the half-suit is nulled and nobody scores. The requirement to name the exact distribution, not merely to assert possession, is what makes Literature a game about inference rather than collection.

A player with no cards on move passes the turn to a teammate who has cards. If an entire team is out of cards, the other team must claim out the remaining half-suits without asking.

Remark 1 (Termination is a property of the players, not the rules). *Nothing in the rules forces progress: two opponents can pass a card back and forth and return to an identical position, so the state graph contains cycles. Consequently any claim that "every game ends" describes an agent's stalling rule rather than Literature itself. Our agents carry an explicit progress rule and our harness imposes an action cap; Section 13 explains why the cap must be* scored *rather than discarded.*

Remark 2 (Solution concept). *Literature is a two-team zero-sum game with pre-play agreement and no in-play communication. The appropriate solution concept is therefore a team-maximin equilibrium with correlation (TMECor) in the sense of Celli and Gatti (2018), not a Nash equilibrium, and the convergence guarantees available for two-player zero-sum games do not transfer. We compute no equilibrium and make no equilibrium claim; every strength statement in this paper is either relative (against a named opponent) or absolute against an exactly solved subgame.*

4 Information integrity

An imperfect-information study is worthless if the agents can see the hidden state, and the failure mode is usually subtle rather than flagrant. The boundary is enforced four ways.

Structural. The engine's `GameState` never reaches a policy. Policies consume only an `Observation`: their own hand, the public event log, public hand counts, resolved half-suits, whose turn it is, and the rules.

Proved by reconstruction. `Observation.reconstruct` rebuilds a seat's entire view from nothing but (rules, that seat's initial hand, the public event log). Tests assert the engine-provided observation is *identical* to that reconstruction at every step of full games, for all six seats. In an audit conducted for this paper the property was re-verified over 183,750 seat-plies across eight rule configurations with zero mismatches.

Proved by invariance. Every quantity v0.4 computes — posterior marginals, all eleven ask-objective terms, and the end-to-end policy action — is asserted to be bit-identical when the hidden cards of the other five seats are permuted into a different layout that is equally consistent with the public record. Candidate permutations are drawn from the acting seat's own belief sampler and then validated by an *independent* replay of the entire public history, so the test does not merely assume the sampler is correct.

Separation of training from acting. Ground truth may be used to *score* (Section 8 scores posteriors against the true hands) and to train; no acting policy ever sees it. That line is what makes the distinction meaningful rather than decorative.

5 Exact combinatorial inference

5.1 The inference problem, stated exactly

The observation that shapes the whole engine is that *every card movement in Literature is public*. A successful ask names the card that moved; a claim reveals all six locations; nothing changes hands unobserved. It follows that the current location of every card is a deterministic function of the *initial deal* and the public log, so all hidden-state inference reduces to constraints on the initial deal. Those constraints are of exactly three kinds:

1. **Candidate sets.** Per card c, a set $S_c \subseteq \{0, \dots, 5\}$ of players who may have been dealt it. A failed ask excludes the target; a successful ask pins the card; asking for a card excludes the asker (under standard no-bluff rules); a claim pins all six.
2. **Exact counts.** Every player was dealt exactly q cards (9 or 8); after propagation, player p has a residual quota q_p of the cards that remain unlocated.
3. **Disjunctions.** A legal ask certifies that the asker held *at least one* card of that half-suit at that moment: an OR-constraint over the cards of the half-suit not yet publicly located.

Write F for the set of *free* cards (those not publicly located and not pinned by propagation). A *world* is a map $f : F \to \{0, \dots, 5\}$ with

$$f(c) \in S_c \quad \forall c \in F, \qquad |f^{-1}(p)| = q_p \quad \forall p, \tag{1}$$

that additionally satisfies every OR-constraint. Let Ω denote the set of such worlds and $\Omega_0 \supseteq \Omega$ the set satisfying only (1).

Proposition 1 (Uniform posterior under uninformative choice)**.** *Suppose the deal is uniform over all $\binom{54}{9,9,9,9,9,9}$ partitions, and suppose that at every decision point each player's choice of action depends on their private hand only through which actions are legal. Then the posterior over worlds given the public history is uniform on Ω.*

Proof sketch. Let h denote the public history and w a world. The prior is uniform, so $\mathbb{P}(w \mid h) \propto \mathbb{P}(h \mid w)$. Factor $\mathbb{P}(h \mid w)$ over the actions in h. Under the stated assumption each factor depends on w only through the legality of the taken action, and card transfers are deterministic given w and the action. Hence $\mathbb{P}(h \mid w)$ takes the same value for every w that makes all of h legal and reproduces every observed outcome, which is precisely Ω, and is 0 otherwise. $\square$

Proposition 1 is the target that v0.3's sampler was *intended* to hit and, by its own documentation, missed. It is also, as Section 8 shows, not the target one should want — because its hypothesis is false, and interestingly so.

5.2 Why the counting is tractable

Counting the elements of Ω_0 is an instance of counting perfect matchings in a degree-constrained bipartite graph, equivalently a restricted permanent, which is $\#P$-complete in general (Valiant, 1979). The instances arising here, however, are small in the only dimension that matters. Measured over the 360 decision points of the stored corpus (`results/infer_position_stats.json`):

quantity	mean	median	p90	max
free cards $\|F\|$	25.1	25	42	45
distinct candidate sets	**4.1**	**4**	**6**	**9**
active OR-constraints	4.7	5	9	14
cards per OR-constraint	3.17	3	5	5

Cards sharing a candidate set are *exchangeable*. Grouping them collapses the problem from an assignment over $|F| \leq 45$ labelled cards onto a *group-count matrix* $k \in \mathbb{N}^{G \times 6}$, where k_{gp} is the number of cards of group g dealt to player p. G is small: at most 9 across those 360 positions, with a 99th percentile of 8. That is an *observation about this game*, not a bound — nothing here proves G cannot be larger, and the dynamic program does not assume it is not.

5.3 The dynamic program

Let group g have candidate set S_g and size n_g, so $\sum_g n_g = |F|$. The number of worlds realising a given k is a product of multinomials, hence

$$Z \;=\; |\Omega_0| \;=\; \sum_{\substack{k\,:\,\sum_p k_{gp}=n_g\ \forall g\\ \sum_g k_{gp}=q_p\ \forall p,\quad k_{gp}=0 \text{ if } p\notin S_g}} \;\prod_{g=1}^{G} \frac{n_g!}{\prod_p k_{gp}!}. \tag{2}$$

We evaluate (2) by dynamic programming over *players*, carrying as state the vector $r \in \prod_g\{0,\dots,n_g\}$ of cards still unassigned in each group. Define $\Phi_0(r) = \mathbf{1}[r=(n_1,\dots,n_G)]$ and

$$\Phi_{p+1}(r') \;=\; \sum_{\substack{\kappa\,:\,\sum_{g\in A_p}\kappa_g=q_p\\ r'=r-\kappa,\ \kappa_g\le r_g}} \Phi_p(r)\prod_{g\in A_p}\binom{r_g}{\kappa_g}, \qquad A_p=\{g : p\in S_g\}, \tag{3}$$

so that $Z=\Phi_6(0)$. The product of binomials telescopes across the six player steps to exactly the multinomial in (2), so the recursion is exact rather than approximate.

Two implementation notes make (3) fast. First, the constraint $\sum_g \kappa_g = q_p$ is enforced with an auxiliary axis counting cards taken so far, so the compositions κ are never enumerated: player p's step is $O(\sum_{g\in A_p}\min(n_g,q_p))$ array operations. Second, the state space $\prod_g(n_g+1)$ is small in practice, because G is small and not because $|F|$ is. An earlier draft put it at "median 272, mean 577 over real positions"; those figures back nothing in `results/infer_position_stats.json`, which stored the *quota* lattice $\prod_p(q_p+1)$ under a similar name and never the group product at all. The group product is now measured alongside it, and the two are different quantities rather than two estimates of one.

What the dynamic program yields.

- **Partition function** $Z=|\Omega_0|$, i.e. the exact size of the information set, computed in closed form. This makes quantities that are normally estimated (such as the disambiguation factor of Long et al., 2010) directly computable.
- **Exact marginals.** A backward pass B_p mirroring (3) gives $\mathbb{E}[k_{gp}]$ by a forward-backward sweep that carries one accumulator per group, so all $6G$ expectations come out of a single pass. Since cards within a group are exchangeable, $\mathbb{P}(f(c)=p)=\mathbb{E}[k_{g(c)p}]/n_{g(c)}$.
- **Exact uniform sampling.** With B available, each player's whole take is drawn from its exact conditional, so a draw is exactly uniform on Ω_0.
- **Exact joint probabilities.** Pinning named cards reduces n_g and q_p by one each, so $\mathbb{P}$(a declared claim distribution is exactly right) is a ratio of two evaluations of (2). Because cards are labelled throughout, no extra combinatorial factor is needed.

Validation. On 400 randomly generated systems the dynamic program reproduced the brute-force count *and* every per-card marginal to within 10^{-9}, 400/400; the pinned-count routine matched brute force on 200/200. An exact uniform sampler drawn 40,000 times matched the exact marginals to 0.0000 at four decimal places.

Remark 3 (Completeness)**.** *Because a marginal of zero is a proof of impossibility, the dynamic program is a* complete *propagator for the candidate-set-plus-quota subsystem: any placement it assigns probability zero is genuinely impossible, and any placement it assigns probability one is genuinely forced. v0.3's propagator was explicitly documented as sound but not complete for this subsystem. The disjunctive constraints remain outside this guarantee.*

6 The disjunctive constraints

The OR-constraints are the part of the system (1) that the dynamic program does not absorb. Three routes are available, and we measured all three rather than choosing on intuition.

(a) Ignore them. On 40 real positions, exact marginals computed on Ω_0 differed from the truth on Ω by a mean L_1 error of 0.127 per card — *worse* than v0.3's biased sampler at 0.074. So the clauses carry a great deal of information and cannot be dropped.

(b) Rejection. Draws that are exactly uniform on Ω_0 can be filtered to be exactly uniform on Ω. A typical clause covers 3.35 live cards, each landing with the relevant player about one time in five, so a clause holds with probability roughly 0.53 and five independent clauses hold about 4% of the time. Measured acceptance on real positions: mean 0.28, median 0.17, and below 0.05 on 16% of positions. Rejection is therefore exact but unaffordable.

(c) Fold them into the dynamic program. Refining the groups by OR-membership keeps the cards within a block exchangeable and makes each clause a local condition on one player's step, which can be enforced by inclusion– exclusion over that player's clauses. The refined state space, however, has a bad tail: median 5,760 but p90 230,000 and p99 2.7×10^7. Exact in the median case, hopeless in the tail.

6.1 Sequential importance sampling with closed-form densities

We therefore sample. Cards are assigned in a fixed order, so each world has exactly one generating path and its proposal density $q(w)$ is the product of the per-card choice probabilities. The proposal is quota-proportional — the natural approximation to uniform — multiplied by a boost for choices that would satisfy an as-yet-unsatisfied clause, and forced when a clause has a single live card left. Self-normalised importance weights

$$\tilde{w}_i \propto \frac{L(w_i)}{q(w_i)}, \qquad \widehat{\mathbb{E}}[\varphi] = \frac{\sum_i \tilde{w}_i \varphi(w_i)}{\sum_i \tilde{w}_i} \tag{4}$$

are then consistent for any target $\propto L$; with $L \equiv 1$ the target is the uniform posterior of Proposition 1.

Two properties are required and both are verified against exhaustive enumeration rather than argued:

- **Support completeness.** Every world in Ω must have $q(w) > 0$, since a weight cannot repair mass the sampler never reaches. We check this *analytically* for every world of every small system — random hit-testing cannot distinguish "unreachable" from "rare" — over 55 systems and 1,797 worlds, all reachable.
- **Density correctness.** The density used to build the weights must be the density the sampler realises. Analytic and empirical path densities agree to a worst gap of 0.0097 over 20 systems.

Weighted marginals then matched exhaustive enumeration to a worst error of 0.029 at 4,000 draws, while the *unweighted* estimator over the same draws was measurably biased — a control without which the test would not be known to discriminate. The Kish effective sample size is $\mathrm{ESS}/N = 0.52$ on real positions (Section 14.3), so the cost of steering the proposal is substantial and, more importantly, is reported rather than assumed. An earlier draft put it at 0.81 here while Section 14.3 put it at 0.52; the 0.52 is what `results/ess_probe.json` measures over 641 real decisions, and the two sentences had disagreed with each other for as long as both existed.

Drawing the batch one card at a time across all draws in lockstep, rather than one draw at a time, makes this 6.8× faster at the median for identical output (11.5× on average, over 60 positions); before that change the scalar loop was 67% of the policy's total runtime.

6.2 A correction: option (c) is in fact tractable

Our estimate above that folding the clauses into the dynamic program is unaffordable was measured on the wrong decomposition, and a separate study overturned it. Refining by *half-suit* is coarser than necessary: what actually needs to stay together is a connected component of clauses that share cards, and OR-free cards can be processed one at a time. With that decomposition and a dynamic program over the residual-quota lattice, an exact backend refused none of 349 real positions at a work budget of 10^7, costing a median of 7.5 ms and a p90 of 17.5 ms.

That result is worth stating in its own right, because it inverts the usual shape of an accuracy-versus-compute study: *the frontier here is not a curve, it is a point.* The exact backend sits below and to the left of every sampled one, and there is no compute budget above roughly 8 ms at which a sampler is the right answer. Below that budget the best available option is a hybrid — exact where it fits, sampled where it does not — which still beats every pure sampler at matched cost (L_1 error 0.133 at 5.5 ms against 0.258 for v0.3's sampler at 3.2 ms).

A mode that promised exactness it could not deliver. The shipped posterior takes the counting DP where no OR clause is active and the importance sampler everywhere else, and reports which it used. It also accepted `mode="exact"`, which forced the DP *whatever the clause set looked like* and then set `Posterior.exact = True` for the result. The DP does not represent OR clauses — that is the entire reason this module contains a sampler — so its draws come from a strict superset of the feasible worlds. At the first position of one seeded game where a clause bites, 30% of those draws are worlds the belief state has already *proved* impossible. The mode a caller would reach for when they wanted a guaranteed-correct answer was the one mode that could quietly return a wrong one under a correct label. It now refuses. Nothing in the shipped agent used it, which is why it survived: a trap with no current caller is invisible to every test that exercises the paths that are taken.

We nevertheless ship the importance sampler, and the reason is the subject of Section 7: the exact backend computes the *uniform* posterior, and the uniform posterior is not the target we want. What the measurements in Section 14.9 add is that the difference between an exact posterior and a well-sampled one is worth nothing measurable in play, while the difference between the uniform target and an opponent-model target is worth 1.9 sets per deal-pair. Exactness was the cheaper thing to buy and the less valuable one.

The natural synthesis is available and untried: the choice model (6) is a product of per-(player, half-suit) depth factors, and a depth is a function of one half-suit's count vector. So if the items of the dynamic program are taken to be whole half-suits rather than clause components, the likelihood factorises over exactly the same structure as the constraints do, and exact inference *under the opponent model* costs the same dynamic program with reweighted item tables. We did not build it, because the evidence above says it would buy approximately nothing in strength; we state it because it is the right thing to build if that evidence ever changes.

7 Opponent modelling: relaxing the false hypothesis

Proposition 1 rests on a hypothesis that is plainly false: that a player's choice of action depends on their hand only through legality. It does not. A player who asks in a half-suit is more likely to be deep in it.

Under the simplest defensible choice model — a player picks which half-suit to ask in in proportion to how many cards of it they hold —

$$\mathbb{P}(\text{ask in } H \mid \text{hand } h) \;=\; \frac{d_H(h)}{|h|}, \tag{5}$$

and the denominator is public, hence identical across worlds. The likelihood of the whole observed ask sequence is therefore proportional to a product of depths, giving the importance weight

$$\log L(w) \;=\; \gamma \sum_{(a,H)} n_{aH} \log\big(d_{aH}(w)\big), \tag{6}$$

where n_{aH} counts asks by player a in half-suit H and $d_{aH}(w)$ is a's depth in H under world w. The single parameter γ controls how sharply the model is believed; $\gamma = 0$ recovers Proposition 1 exactly, so the entire idea is one ablatable number. Because (6) decomposes over cards, it costs one integer increment per card inside a draw.

Remark 4 (The obvious refinement does not help)**.** *Depth in* (6) *is evaluated on the initial deal, whereas the choice model is really about what the player held* at the moment they asked*. Cards move, so the two differ. We implemented the at-ask-time version — which needs only one replay of the public log per decision, not per candidate world, because a card the propagator has not pinned was never located at any time and so is exactly the card the sampler draws — and it is not better. Posterior negative log-likelihood is* 1.3536 *against* 1.3522 *for the initial-deal version at matched settings, and* 1.3391 *against* 1.3373 *at the best settings of each. In duplicate-deal play at matched* $\gamma = 0.35$ *the two are likewise indistinguishable (*-0.050*,* $[-0.377, +0.277]$ *over 400 pairs).*

That is not the end of the story, and Section 14.5 is where it goes: at $\gamma = 1.0$ *the at-ask covariate* is *demonstrated,* $+0.102$ $[+0.006, +0.197]$ *over 6000 pre-registered pairs. The two statements are compatible and the difference between them is the finding. What is demonstrated is the* pair *— at-ask depth together with a sharper* γ *— and neither component is claimed on its own, because the pre-registration bound them together on the argument that a covariate measuring the right thing deserves to be believed more sharply. Fixing* γ *at the incumbent* 0.35 *and swapping only the covariate is the comparison this remark makes, and it buys nothing. The theoretically-correct quantity buys nothing over the simpler one, so the simpler one ships.*

We flag one methodological point here because it nearly became a false finding. Our first at-ask-time implementation counted only cards that had visibly *changed hands, silently omitting every card the propagator had* deduced *to be someone's — the large majority. Depths came out far too small, and the variant measured as catastrophic on both instruments (posterior NLL* 2.22 *against* 1.35*;* -1.54 *sets per deal-pair in play). Two independent instruments agreeing on a large effect is normally reassuring; here they were agreeing about a bug. What caught it was that the magnitude was implausible for a refinement, not that either measurement looked wrong.*

8 Scoring a posterior against the truth

Every belief metric in v0.3 was indirect: sampler cost, agreement with an exact endgame solver, or downstream win rate. In simulation the hidden hands are available, so a posterior can be scored on the question it is actually trying to answer: how much probability does it place on where the cards really are?

We score over cards that are still genuinely uncertain for the acting seat — a card the propagator has already pinned is scored perfectly by every configuration and would only dilute the comparison — using mean negative log-likelihood of the true holder, the six-way Brier score, top-1 accuracy, and a reliability curve in deciles. Reliability must bin the probability assigned to a *fixed* outcome; binning by the probability assigned to the true holder would make every entry a hit by construction and report perfect calibration for any model whatsoever.

9 Exact solving, and a closed form

The exact solver is the project's only source of *absolute* truth: in a position small enough to solve, there is a right answer, so an engine that disagrees has a defect rather than a preference. v0.3's solver enumerated card placements and stopped at roughly seven live cards. Widening it produced three results that change how the previous version's absolute benchmark should be read.

9.1 There is no such thing as a seven-card position

Claims strip all six cards of a half-suit from every hand, so

$$\text{live cards} \;=\; 6 \times (\text{unresolved half-suits}) \qquad \text{exactly, always.} \tag{7}$$

Every "maximum live cards" threshold in the previous codebase was therefore a binary switch: any value from 6 to 11 selects the same policy. v0.3 solved the one-half-suit layer and refused everything else, and the two-half-suit layer is $6^{12} \times 6 \approx 1.3 \times 10^{10}$ card placements, which is why.

9.2 The count abstraction

Proposition 2 (Card identity is irrelevant under perfect information)**.** *Let σ be any permutation of the deck that maps each half-suit onto itself. Then σ is an automorphism of the perfect-information game, and consequently the value of a position depends only on how many cards of each live half-suit each player holds.*

Proof sketch. σ acts on states by relabelling hands. $\textsc{Ask}(t, c)$ is legal in s iff $\textsc{Ask}(t, \sigma(c))$ is legal in $\sigma(s)$: the target and all hand counts are unchanged, "the asker holds a card of this half-suit" is preserved because σ fixes half-suits setwise, and "the asker does not hold c" is preserved because σ is a bijection of each hand. The ask succeeds iff the target holds c iff σ(target) holds $\sigma(c)$, and the transfer commutes with σ. $\textsc{Claim}(h, a)$ maps to $\textsc{Claim}(h, a \circ \pi^{-1})$ with π the restriction of σ to h; exactness, the "some card sits with an opponent" case and the wrong-distribution case are all preserved, and both states lose exactly the six cards of h. $\textsc{Pass}$ and turn normalisation depend only on which hands are empty. Hence $V(\sigma(s)) = V(s)$, and two states have equal per-(half-suit, player) counts iff some such σ relates them. □

A half-suit's counts are a composition of 6 into 6 ordered parts, of which there are 462, so a layer with m unresolved half-suits has at most $462^m \times 6$ abstract states: 2,772 for $m = 1$ and 1,280,664 for $m = 2$, against 279,936 and 1.3×10^{10} card placements. Both are solved once, in full, as dense tables — a genuine Fish tablebase, after which a query is an array lookup. Measured on the same 300 one-half-suit positions, and this is one cross-check rather than two: the old solver agreed with the new one on both the value and the complete optimal-action set in 300/300 cases, with no refusals, at 2,240 CPU-seconds against 0.052. That is **43,000**× by total CPU time and **10,900**× by median per position (990 ms against 91 μs). An earlier draft reported "131 further positions" at 4,205×; there is no second corpus, and 4,205 is not a ratio anything in `results/exact2_cross_check.json` produces.

9.3 A closed form for the perfect-information value

Both full tables match, exactly, the formula

$$V \;=\; \text{sign(mover's team)} \times (2f - m), \tag{8}$$

where m is the number of live half-suits and f the number of them in which the team on move holds at least one card.

This is not only an empirical match at the layers we can enumerate. It holds at every m, and the reason is that it turns on three properties of the *rules* rather than of any position, none of which needs a table:

A. Footholds never grow. `check_legal` refuses an ask unless the asker already holds a card of that half-suit, so a team holding none of h can never receive one. Its foothold set only shrinks, and the mover's team can therefore claim at most the f it has *now*.

B. A team with no foothold in h cannot deny h. All six cards of h sit with the opponents, so every assignment it could submit reveals an opponent holder and `_apply_claim` awards the half-suit to the opponents. There is no spite-null available: a certain -1 cannot be converted into a 0. Hence the opponents take all $m - f$ whatever the mover does.

C. The mover takes all f without surrendering the turn. A hit retains the turn, a claim never moves it, and a claim that empties the claimant passes within the team. With perfect information every ask can be made to hit, so the mover drains and claims each of its f half-suits, and only when its team is cardless does the turn normalise to an opponent — who by symmetry takes the rest.

A and B bound V above by $2f - m$; C bounds it below. Each is checked against the engine rather than against a reading of it, and the optimal perfect-information policy that falls out is one line: *take a hitting ask if one exists; otherwise claim a half-suit your team wholly owns; otherwise pass to a teammate with cards.* Played from 360 random positions spanning $m = 1$ to $m = 9$ it realises $2f - m$ every time, and at $m = 1$ and $m = 2$ it agrees with the exact solver on every position tested — the one step in the argument anchored to a solver rather than to internal consistency.

9.4 The perfect-information game is trivial

The closed form is worth stating as a value because the value is degenerate. A team lacks a foothold in a half-suit only if all six of its cards landed in the opponents' twenty-seven, with probability $\binom{27}{6}/\binom{54}{6} = 1.15\%$, so at a fresh deal

$$\mathbb{E}[V] \;=\; 2\left(9 - 9\binom{27}{6}/\binom{54}{6}\right) - 9 \;=\; +8.794 \tag{9}$$

of the nine available: **whoever moves first takes everything**.

Premise C is the constructive step and the one an adversary could attack, so it is worth testing against opponents that are trying to win rather than against the greedy rule itself. The premise is stronger than the proof needs: the mover never surrenders the turn, so the opponents do not act at all until its team is cardless, and only the *mover's* team needs to know the deal. The oracle duel reported in §22 is that experiment. Across 80 games against the champion, in every one of the 40 where an oracle seat moves first its team's realised set count equals its foothold count at the deal — 40/40 exactly, not on average. Mean footholds were 8.94 of nine against the 8.897 that (9) implies.

The other forty games are the sharper result. There a *champion* moves first, holding the advantage that (9) prices at +8.79 of nine, and converts essentially none of it: the oracle team still finishes with 8.95 of its 8.97 footholds. A first move that is decisive under perfect information is worth nothing at all to a player who cannot see the deal.

Two things follow for the project. First, extending the tablebase to larger m would buy no ground truth about Fish, because the closed form already answers every layer and the answer is a different game's answer. Second, the inference ceiling reported in §22 should be read accordingly: its +17.483 per deal-pair was framed as what perfect card-reading is worth, and it is closer to (8) being played out — seven of its ten per-deal differentials are exactly $18 = 2 \times 9$.

This has an uncomfortable consequence for the previous version's headline absolute result. v0.3 reported that its belief agents choose a provably optimal move in 100% of information-resolved positions — but every one of those positions had a single live half-suit, where (8) reduces to "the team on move wins it". The score was real, and it is reproduced here, but it is a shallower certificate than it reads. The two-half-suit slice is the one with teeth, and the corpus used in Section 14.7 is 79% two-half-suit for that reason.

9.5 Bellman-optimal is not optimal in a loopy game

Fish's state graph is cyclic, so the Bellman operator has many fixpoints and value-preservation is not optimality: if a side can force +1, every action keeping the value at +1 looks optimal, including one that walks a cycle forever and banks nothing. Measured on the full tables, in **28.6%** of decided two-half-suit states some value-preserving action makes no progress. At one half-suit the figure is 0%, which is why v0.3 never noticed. The solver therefore carries an attractor rank and the agent asks for a *progress-optimal* action, and Section 14.7 reports agreement against that stricter criterion.

Two further corrections to v0.3. First, although the state graph is cyclic, *optimal play always terminates* under perfect information: cycles exist, but no optimal line needs one. Second, the fixpoint reached by value iteration was checked to be unique in real Fish layers — eight different initialisations (zeros, $\pm m$, ± 5, three random) and three sweep orders at $m = 1$ and $m = 2$ all land on the attractor value — so v0.3's zero-initialisation and dictionary ordering were never load-bearing, even though in general they could have been.

10 The ask objective

10.1 A wider basis

v0.3 scored an ask as $\mathbb{P}(\text{success}) + 0.06 \cdot \text{depth}$, later adding a turn-risk and a scarcity term. Those terms are in incommensurable units — a probability, a card count, a normalised hand-size difference, a share — so the weights combining them have no meaning beyond "what won a sweep". That is consistent with what v0.3 measured: a clean inverted-U in each weight, which is the signature of a *tie-breaker* rather than an objective.

v0.4 keeps the additive form, because it is ablatable and because the previous version's evidence says the success probability really is dominant, but widens the basis to eleven terms and computes all of them from the posterior in a single vectorised pass. Writing π for $\mathbb{P}(\text{success})$ and H for the asked card's half-suit, the score is

$$\text{score}(a) \;=\; \pi \;+\; \sum_j w_j\, f_j(a), \tag{10}$$

with the terms in Table 1. Two deserve comment because they are, as far as the surveyed literature shows, unstudied.

Exposure. v0.3's turn-risk term penalised handing the turn to an opponent with many cards. Hand size is a proxy; the quantity that actually matters is how much that opponent can *immediately take from us*, which we can compute: cards of ours whose location is already *public* and that sit in a half-suit the opponent can legally ask in. Both parts are public knowledge, so using them is inference, not leakage.

Signalling. An ask publicly certifies that the asker holds a card of that half-suit. In a game with no legal communication channel, that is the only channel there is — and it is simultaneously read by the opposition. v0.3 modelled only the cost (its `reveal` term, worth a marginal +0.29).

term	origin	feature value
`suit`	v0.3	own depth in H (raw card count)
`turn`	v0.3	$(1-\pi)\cdot$ negated normalised hand-size excess of the target
`scarce`	v0.3	$2(\bar{s}_H - \frac{1}{2})$, with $\bar{s}_H$ our team's expected share of H
`reveal`	v0.3	-1 if H has not previously been shown by us
`deplete`	v0.3	$\pi \cdot (1 - \text{target hand}/q)$
`expose`	new	$-(1-\pi)\cdot$ our publicly located cards the target could take
`claim`	new	$\pi \cdot \big(\prod_{c \in H} \mathbb{P}(c \text{ with our team})\big)^{1/6}$
`info`	new	$4\pi(1-\pi)\cdot$ holder entropy of H
`certain`	new	1 if $\pi = 1$
`concent`	new	largest single teammate share of our team's holding in H
`signal`	new	$2(\bar{s}_H - \frac{1}{2})$ if H has not previously been shown by us

Table 1: The ask objective basis. Every term is zero by default, so setting a weight to zero ablates exactly that idea and nothing else — a property enforced by test, for the reason given in Section 13.

The benefit is that a teammate learns where a card of a half-suit our team is winning must be, which is exactly the localisation that lets a set be declared. So the sign should depend on who gains more, and the natural proxy is our team's share of the half-suit. Cost and benefit are carried as independent weights so that the data, not the story, decides.

10.2 An alternative in the units that are actually paid out

The additive form has a principled competitor. What the game pays is half-suits, so define, for each unresolved half-suit,

$$V(H) \;=\; \mathbb{P}(\text{our team ends up scoring } H) - \mathbb{P}(\text{they do}), \tag{11}$$

which is measured in sets, is bounded in $[-1, 1]$, and whose sum over half-suits is the final differential. An ask is then worth the expected change it produces,

$$\text{score}(a) \;=\; \pi\,\big[V(H \mid \text{success}) - V(H)\big] \;+\; (1-\pi)\,\big[V(H \mid \text{failure}) - V(H)\big], \tag{12}$$

with every quantity in the same unit. The substantive difference is that this is a *marginal* value rather than a level: "our team is winning this half-suit" is a level, and what should drive a decision is how much one more card moves the outcome.

V is obtained by supervised learning rather than by guessing a functional form: self-play games supply, for every unresolved half-suit at every decision, a feature vector computable from the acting seat's posterior alone, labelled with how that half-suit eventually resolved. That is roughly nine labels per decision, which is abundant and cheap, in contrast to the rollout regression that v0.3's variance analysis made expensive. The counterfactual posteriors in (12) are local updates on the half-suit's marginal block: a success pins the asked card to us, a failure removes the target from that card's candidate set.

11 Claiming

v0.3 swept the claim-confidence threshold directly and found two things. First, claiming eagerly is a real, measurable mistake: dropping the bar from 0.97 to 0.70 costs 0.45 sets per deal-pair.

Second, and more usefully, thresholds from 0.85 to near-certainty play *identically*: over 150 duplicate deals the 0.97 and 0.999 policies never once diverged.

The mechanism is worth stating, because it justifies the design rather than leaving 0.97 as a magic number. Suppose our team genuinely holds all six cards of a half-suit but we cannot place them within the team. What does waiting cost? An opponent cannot take the set by claiming it — a claim by a player whose team does not hold all six *gives* the set to us. So the only cost of waiting is running out of opportunity, and the benefit is that continued play localises teammate holdings. Waiting is close to free, which is exactly what the sweep measured and what v0.3's expected-value model missed by treating the resolution of a split as a coin flip.

The corollary is that the leverage in claiming is not the threshold. It is (i) getting the declared distribution right, since nulls cost roughly 0.6 sets per deal-pair and a null is a bookkeeping failure rather than an unlucky gamble; and (ii) playing the forced claims well, where no threshold is involved at all. v0.4 therefore keeps a high threshold and changes what happens underneath it:

- The declared distribution is the posterior MAP, found by shortlisting on the per-card marginals and scoring the shortlist with the *joint* posterior. The product of marginals is not the joint — cards compete for the same quota slots, so per-card modes can be jointly impossible.

- Where the clause set is empty the joint probability is computed exactly by the dynamic program of Section 5; where it is not, deduction still settles the extremes exactly, because a zero under the OR-free system implies a zero under the smaller OR-constrained one, and likewise for one.

- Forced claims maximise expected sets, $\mathbb{P}(\text{exact}) - \mathbb{P}(\text{an opponent holds one})$, rather than confidence.

That second probability was, until recently, an independence product, and the first bullet is the sentence that condemns it. The claim evaluator returned the *joint* for "this split is right" and $\prod_c \sum_{p\in\text{team}} \mathbb{P}(p \text{ holds } c)$ for "the half-suit is ours at all", in the same tuple, from the same method, three lines below its own docstring saying the product of marginals is not the joint. Under the baseline null rule the forced-claim ranking is $\mathbb{P}(\text{exact}) + \mathbb{P}(\text{ours}) - 1$, so the two carry equal weight; and their difference is read as "ours but wrongly split", a quantity subtracted across two different distributions and therefore able to come out negative — which it did, and a clamp hid it.

Measured over 29406 half-suit queries in 60 games of real play, the two disagree on 27902, and on 86% of those the product *overstates* the joint — by a median of 0.008 and by as much as 0.313. Overstating "our team holds it all" is the direction that makes a declaration look safer than it is. Over the 5885 positions in the same games where a forced claim could arise, the product produced a negative "wrongly split" 7 times and the correction changes which half-suit is declared at 31 of them. The decision counts are small, and neither is what makes the case: an expected value assembled from two different distributions is wrong at any magnitude, and the magnitude is reported so that the correction is not mistaken for a result.

The screening tier is unaffected and deliberately so. It returns a product for *both* elements, so the pair is internally consistent; what could not stand was one of each.

But the screen itself carries a correctness claim, so we measured that too. The middle tier skips the joint query when the product of the per-card MAP marginals falls below 0.35, and returns the product as the half-suit's probability. Its justification is the comment "most half-suits are nowhere near claimable" — an assertion about a distribution, inside a tier whose entire premise is that the product and the joint differ. If the product understates by enough, a half-suit whose true probability clears the 0.97 threshold is returned below 0.35, no claim is made, and nothing records that a certain set was left on the table.

Over 11687 screened half-suits in 25 games, the number that turn out to be claimable is **0**: the largest true joint among them is 0.581, against a threshold of 0.97. The shortcut is sound. What makes it sound is the margin and not agreement between the two quantities — the gap reaches +0.255 on individual half-suits — so the tier is safe at these settings and would not survive a much lower threshold. That is now a stored measurement (`results/claim_screen_check.json`) with a test that fails if a screened half-suit ever turns out to be claimable, rather than a comment.

Finally, a hard architectural constraint taken from the literature: a claim must never be selected by determinized search. Inside any determinization the searcher knows the true layout, so every claim evaluates as a certain gain and the declared assignment differs between sampled worlds — textbook strategy fusion (Frank and Basin, 1998). Claims in v0.4 are always decided from the posterior.

12 Search over the belief

Every search this project has tried — PIMC, Information-Set MCTS, and two value-network variants — evaluates a candidate move by sampling hidden layouts and solving each one, and every one of them lost to its own prior. v0.3 did not leave that as a mystery: it measured the standard deviation of a position's value across possible layouts at about 2.4× the gap between the best and the worst candidate move, so a searcher scoring different candidates against different sampled worlds is ranking sampling noise. Common random numbers removed the deficit and never produced a surplus.

That diagnosis rules out a family of designs, not search itself. What it rules out is sampling. So the design it leaves open is a lookahead whose state is the *posterior* rather than a sample from it: the marginal matrix $M[c, p] = \mathbb{P}(\text{player } p \text{ holds card } c)$, with transitions that are local edits of that matrix. Given a belief such a search is a deterministic function — run it twice on one position and it returns the same vector — so the variance that killed the previous attempts is structurally absent rather than merely reduced.

12.1 What it computes

A Literature turn is a run of asks that ends the instant one misses. Greedy scoring maximises the immediate success probability; what a possession is actually worth is how many cards it banks before the turn flips. Define

$$W(B, d) \;=\; \max_{a \in \mathcal{A}(B)} \; p_a \left[\, 1 + W(B \mid a \text{ succeeded}, \, d-1) \,\right], \qquad W(B, 0) = 0, \tag{13}$$

where $\mathcal{A}(B)$ is the legal-ask set implied by the belief and $p_a = M[\text{card}(a), \text{target}(a)]$. The failure branch contributes nothing to W because a miss ends the possession; what a miss costs positionally is already priced by the incumbent objective's turn-risk and exposure terms, and duplicating it here would double-count.

W is in units of cards expected to be banked in the rest of this possession. It enters the policy as an additive bonus, not as a replacement:

$$\text{score}(a) \;=\; \underbrace{\pi_a + \textstyle\sum_j w_j f_j(a)}_{\text{the incumbent objective, unchanged}} \;+\; w_{\text{look}} \cdot p_a \, W(B \mid a \text{ succeeded}, \, d-1). \tag{14}$$

The reason it is additive is Section 14.9: replacing $\mathbb{P}(\text{success})$ with a learned half-suit value — correct units, a calibrated model — costs 7.4 sets per deal-pair. Whatever else this game rewards, the probability that the ask lands is the objective and not a term in one. At $d \leq 1$ the bonus is identically zero, so the policy reproduces the incumbent decision for decision; that is enforced by test, for the reason Section 13 gives.

12.2 Why the obvious version is provably a no-op

The natural transition is to collapse the asked card's row to a point mass on us. That is exact, and on its own it is *inert*, which is worth stating because it determines whether the design can do anything at all.

Proposition 3 (Greedy optimality under an inert transition)**.** *Suppose the transition edits only the asked card's row, so that $p_{a'}$ is unchanged for every $a' \neq a$, and suppose every candidate remains available after any other is taken. Then (13) is maximised by taking candidates in descending order of p, and its value equals that of the greedy policy at every depth.*

Proof sketch. Under the hypotheses, a depth-d chain is a choice of an ordered d-subset of a *fixed* multiset of probabilities, with value $p_1(1+p_2(1+\dots)) = \sum_{k=1}^{d} \prod_{i \le k} p_i$. Exchanging an adjacent pair (p_i, p_{i+1}) with $p_i < p_{i+1}$ changes the value by $\left(\prod_{j<i} p_j\right)(p_{i+1} - p_i) \ge 0$, so descending order is optimal; and the greedy policy selects exactly that order. $\square$

Proposition 3 is the null this section is tested against, and it is why a lookahead built only on the point-mass update would be worth measuring only as a bug-check. Its hypotheses fail in three places, of which the third is the substantive one:

1. a card taken cannot be asked for again, and one card is askable from several targets, so candidates do disappear in correlated groups;

2. a target can run out of cards, which removes every candidate aimed at them;

3. **the quota system couples the candidates.** Learning that the target held this card makes it less likely they hold the others — they are now known to hold one card fewer, and that probability mass has to go somewhere.

The third is handled by scaling the target's column down to the count they now have and renormalising each affected row across the remaining players: one sweep of the same proportional fitting the exact posterior does globally. A row whose whole mass sits on the shrunk column renormalises straight back to where it started, which is correct — a publicly located card cannot be made less certain by a counting argument. The sweep costs one column scale and one row normalisation per node, and it is separately ablatable, so Proposition 3 is not merely argued but measured: the no-coupling cell in Table 8 is the experiment.

12.3 Cost

The branching factor of a real mid-game position is large: measured over 615 decisions from six self-play games at the champion configuration, the number of legal asks has mean 45.0, median 45, p90 81 and maximum 102, and exceeds 8 in 94.6% of decisions. (An earlier draft of this section said the branching factor "averages 8". That figure is the mean legal-action count of the *exact-solver corpus* in `results/exact_positions_v2.json`, which consists of endgame positions with one or two live half-suits — an endgame statistic read as a general one.)

Continuations are beamed to the best 4 candidates by immediate p. The root ply is *not* beamed, because the policy needs a bonus for every candidate it is ranking, so a naive depth-3 tree costs $n\,(1+b+b^2) = 21n$ expansions — about 945 at the median $n = 45$ — rather than the $b + b^2 + b^3 = 84$ a uniformly-beamed tree would give.

Three quarters of those expansions were doing no work. At the last ply the continuation is $W(\cdot, 0) = 0$ by definition, so $q = p\,(1+0) = p$ and the maximum over the beam is simply the largest probability — a number the sort key has already computed. The implementation nonetheless performed up to b full transition/undo cycles there, each copying the marginal matrix and running a column scale and a 54-row renormalisation, to rediscover it. Short-circuiting that ply, and fusing the row renormalisation to avoid a boolean-index round trip, leaves the returned vector bit-identical (maximum absolute difference 0.0 over 24 real positions and every candidate) and costs $n\,(1+b) = 225$ expansions instead of 945. Measured median cost falls from about 14 ms per decision to 6.7 ms (mean 8.3, p90 18.1, over 40 positions, minimum of three repetitions each) on the contended host of Section 5.

Against the configuration the lookahead runs actually used — 160 draws, the `fish4/agent4.py` default, since no lookahead cell in `results/v04_duels.jsonl` overrides `n_draws` — the relevant posterior cost is the 31.9 ms row of Table 2, not the ∼100 ms rows, which are all 512-draw. So the bonus now adds roughly a fifth to a decision. An earlier draft of this section claimed 15% by comparing the pre-optimisation cost against a 512-draw budget these runs never used, which was wrong twice in opposite directions; the duelling results in Table 8 were all produced at the slower 945-expansion cost, since the optimisation is exact and postdates them.

13 Evaluation methodology

13.1 Duplicate deals

Raw win rates in a card game are dominated by the deal. Every result here uses **duplicate (paired) deals**: each deal is played twice with the teams swapped, on identical cards, an identical rotated starting seat, and identical agent randomness. Only the policy assignment differs, so per-pair set differentials are i.i.d. across deals and confidence intervals mean what they say.

13.2 How much can an experiment of size n resolve?

The per-pair standard deviation of the set differential between two policies of the strength we compare (v0.3's champion against its own baseline) is **3.869 sets**, measured over 300 pairs. The distribution is wide: the theoretical bound is ±18 and observed pairs stayed within ±11, but a single pair says almost nothing. The consequences are stark:

effect (sets/pair)	pairs @ 80% power	pairs @ 90%	hours @ 80%
0.10	11 751	15 731	3.9
0.20	2 938	3 933	0.98
0.30	1 306	1 748	0.44
0.50	471	630	0.16
1.00	118	158	0.04

Read the other way: the 600-pair cells of the v0.3 champion search could resolve effects down to about 0.44 sets per pair, so every "no significant difference" it reported below that size was *uninformative rather than negative.* We therefore report, alongside every interval, the smallest effect the run could have detected.

That standard deviation is not a property of the game. The table above, and every power calculation in this paper up to this point, treats 3.8 sets per pair as a constant of the deal population. It is not. It is a property of a *pair of policies* that disagree constantly, and the experiments most in need of careful sizing are the ones whose arms barely disagree at all. We noticed this only when a screening cell measured a per-pair standard deviation of 2.328 — seven standard errors from the figure it had been sized against.

The mechanism is immediate once stated. Under common random numbers, a deal-pair on which the two arms play *identically* contributes a difference of exactly zero. Writing s for the share of pairs on which they diverge at all,

$$\mathrm{Var}(D) \;\approx\; s \cdot \mathrm{Var}(D \mid \text{diverge}), \qquad \text{so} \qquad \mathrm{sd} \;\approx\; \sqrt{s} \,\cdot\, \mathrm{sd}(D \mid \text{diverge}).$$

That much is arithmetic and follows from the design. The empirical content is what the second factor does. Across the 46 cells of this study that store their per-pair differentials it moves little in absolute terms — 3.86 ± 0.29, a relative spread of 7.6% against 15.5% for the raw standard deviation — and our first draft stopped there and called that the finding. It does not survive being looked at properly.

Three things weaken it, and the third matters for use. *Most of the residual spread is not structure*: the sampling noise of an sd estimate at these sample sizes is alone 2.8%. *The comparison is carried by a handful of cells*: restricted to the 26 cells with divergence share above 0.83 — where most of this study lives — the raw spread is 3.38% and the conditional spread is 3.45%, the same number to within noise, and the correlation between share and sd collapses from +0.95 to −0.03. Where share barely varies the decomposition has nothing to explain, and it explains nothing. And *the conditional term is not constant*: it drifts with share, corr = +0.79, fitted as $2.53 + 1.66\,s$.

That drift matters precisely where the rule gets used, because the rule is only useful when extrapolated to a low-divergence contrast. At $s = 0.31$ the flat model predicts sd = 2.18 and the fitted line predicts 1.68 — a 30% over-estimate. The direction is the saving grace: over-predicting noise asks for more pairs than are needed, never fewer, so a run sized this way is over-powered rather than under-powered. But a rule invoked to justify a *smaller* experiment than the A/A figure implies should not be quoted as though it were calibrated at the point where it is used, and it is not.

One further caveat on what s means. It is measured as $\mathbb{P}(D \neq 0)$, and D is zero on many pairs where the two arms genuinely diverged and the deal still ended level — zero is not an isolated atom in these histograms. So the cheap substitute the rule advertises, counting disagreements at the level of decisions rather than of finished pairs, is a strictly larger quantity than the one the constant was calibrated against, and using it directly would inflate the predicted noise further.

What survives is narrower than we first wrote and still worth having: the per-pair standard deviation is a property of the comparison rather than of the game, the $\sqrt{s}$ scaling is exact by construction, and sizing every experiment against a figure measured between two maximally-different policies is systematically conservative for the near-identical comparisons that most need care.

The extrapolation was untestable, and then it was tested. When the above was written this study had no cells below $s = 0.44$, so the calibration of the constant at low share — the only place the rule is ever *used* — was not established, and the paragraph said so. The gated re-take run has since landed two blocks at $s = 0.233$ and $s = 0.236$, and the prediction that the flat model over-states low-share noise by roughly 30% was recorded in `jobs/PREREGISTRATION_retake_gate.md` *before any pair of that run was played.* Scoring it out of sample, over the three cells now below $s = 0.5$:

cell	s	measured sd	flat	drift-corrected
re-take gate, block 0	0.233	1.443	1.863	1.409
re-take gate, block 1	0.236	1.440	1.875	1.420
ungated re-take penalty	0.440	2.328	2.560	2.163
mean relative error			**+23.1%**	**−3.6%**

The flat constant over-predicts by 23% where it is used; the drift-corrected line lands within 4%. This is the only out-of-sample prediction in the paper that was written down before the data existed and then checked against it, and it is worth more than the several in-sample fits around it. Sizing at low s should use sd $= (2.53 + 1.66\,s)\sqrt{s}$, and the flat form remains the conservative choice — it asks for more pairs than are needed, never fewer.

The correlation between s and the raw standard deviation is $+0.952$, and we report it only to say that it is *not* evidence. If sd really is $c\sqrt{s}$ for roughly constant c, that correlation is the correlation of s with $\sqrt{s}$, which is near 1 over any range — it is implied by the identity rather than a test of it. A reader will compute it anyway, so it is better stated and defused than omitted.

Sizing therefore reduces to predicting one quantity, s, which can be counted from *decisions* rather than from finished pairs and is correspondingly cheap. The consequence is not cosmetic. A candidate that perturbs 10% of pairs has sd ≈ 1.23, not 3.80: sized on the constant it would need $9.5\times$ the pairs, which is the difference between an experiment that looks impossible and one that is routine. We used exactly this to decline a run — a sixth cell in a family of five failures, whose largest possible effect we could show in advance was inside what the cell would resolve, so that it would have returned a null whatever the truth was — and to size its honest replacement at 2000 pairs rather than the 6200 the constant demanded.

The direction of the error is worth naming: sizing on the A/A figure is *conservative*, so nothing already published here is too narrow. What it cost was experiments never run, and nulls recorded from cells that could not have found anything. Those are the expensive kind of mistake precisely because they leave no trace in any interval.

13.3 Anytime-valid stopping

Screening many candidate terms invites peeking. A fixed-n 95% t-interval, monitored after every deal, rejects a *true null* 68.8% of the time on an empirical null and 92.1% on a skewed one, with a median false stop after 11 and 3 deals respectively. We therefore use anytime-valid procedures: an empirical-Bernstein confidence sequence and a betting test supermartingale (Waudby-Smith and Ramdas, 2024). Measured type-I error under continuous monitoring over 4000 null experiments each monitored at every one of 4000 deals: 2.1–2.4% for the betting test and 0.3–0.5% for the confidence sequence, against a nominal 5%.

13.4 Two corrections to the previous harness

Timed-out games must be scored, not dropped. Nothing in Literature forces progress, so a harness needs an action cap. v0.3 dropped any deal-pair containing a capped game. Dropping conditions on the outcome, which is precisely what a paired design exists to avoid, and an audit conducted for this paper found it biases rather than merely blunts: replaying the same 60 deals at reduced caps, the dropped pairs had a true mean differential of -7.14 against -10.13 for the kept ones, and at one cap the reported 95% interval *excluded* the true value. v0.4 scores a capped game on the half-suits actually resolved, with the unresolved ones counting for nobody — the same semantics the exact solver gives an unbroken cycle.

An ablation must reduce to its baseline. v0.3's most instructive methodological failure was a confounded ablation: the ablated agent silently normalised a second term differently, so every cell compared two changes at once and produced tight, confident, and completely wrong intervals. The confound was caught only by noticing an implausible monotonic pattern in the magnitudes. We therefore enforce by test that a zero weight reproduces the baseline *decision for decision* over real positions, and separately that a non-zero weight demonstrably changes some decision. Every term in Table 1 passes both.

14 Results

14.1 Exact inference alone changes nothing

The first experiment is the one the research programme predicted would pay. Holding the ask objective at v0.3's champion weights and changing *only* the inference — exact counting

inference	NLL ↓	Brier ↓	top-1 ↑	ms/decision
exact DP, clauses ignored	1.4776	0.7568	0.287	50.4
v0.3 sampler, 32 draws	1.4399	0.7065	0.386	5.9
v0.4 importance sampler, 160	1.3731	0.7006	0.396	29.5
v0.4 importance sampler, 512	1.3618	0.6955	0.403	92.3
v0.3 sampler, 96 draws	1.3552	0.6891	0.406	16.6
v0.3 sampler, 512 draws	**1.3408**	0.6833	0.416	87.8
v0.4 + opponent model, 160, γ=0.45	1.3476	0.6856	0.401	31.9
v0.4 + opponent model, 512, γ=0.30	1.3332	0.6791	0.415	99.8
v0.4 + opponent model, 512, γ=0.45	**1.3251**	**0.6747**	0.413	98.6
v0.4 + opponent model, 512, γ=0.60	1.3248	0.6744	0.411	100.8

Table 2: How much probability each inference configuration places on where the cards actually are. Lower NLL and Brier are better. The row that matters is the comparison between the two bold entries: the *exactly correct* uniform posterior at 512 draws (1.3618) is worse than v0.3's *biased* sampler at the same budget (1.3408).

where the clause set is empty, unbiased importance sampling where it is not, in place of a sampler documented as non-uniform — gives a set differential of −0.22 per deal-pair, 95% CI $[-0.96, +0.52]$, over 100 duplicate deals. Statistically indistinguishable, and the interval is wide enough that only effects above about 1.1 sets per pair could have been detected.

This is a genuine negative result about a change that is provably correct. The explanation is in the next subsection.

14.2 Posterior quality against ground truth

Scoring posteriors against the true hidden hands over 517 decisions and 13,640 uncertain-card predictions (Table 2) shows what happened.

Two things stand out.

First, the exactly-uniform posterior is *worse* at predicting reality than the heuristic sampler it replaced: 1.3618 against 1.3408 at a matched draw budget. That is not a contradiction. Proposition 1 is a correct theorem about a false hypothesis. v0.3's sampler seeded the OR-clauses by picking a live card of the asked half-suit for the asker, which systematically over-weights worlds in which askers are deep in the suits they asked in — an accidental opponent model. Removing the bias removed the model with it.

Second, making the model explicit and tunable recovers the loss and passes both: 1.3251 at $\gamma = 0.45$. And the improvement is not a large-budget artefact. At 160 draws the opponent model takes the sampler from 1.3731 to 1.3476, which is better than v0.3's deployed 32-draw configuration (1.4399) by a wide margin.

Third, and separately: ignoring the OR-clauses is by far the worst option tested, confirming the design decision of Section 6 from the other direction.

14.3 How much of the sampling budget survives reweighting

Table 2 scores the posterior. This asks what it costs to get there, and finds a diagnostic that has been computed on every decision this engine has ever made and reported on none of them.

The posterior is drawn by importance sampling and self-normalised. Self-normalisation is consistent, not free: if the proposal is a poor match for the target, a few draws carry most of the weight and the batch behaves like far fewer than n independent worlds. Kish's effective sample size measures that, and the sampler already computes it.

At the champion configuration, 160 draws are worth 83.7 effective — 52% efficiency — with a p10 of 64.8, a median of 87.0 and a p90 of 100.0 over 641 decisions across 36 seat-deals. The exact DP is almost never the path taken: 9 decisions in 641. So the precision of every $\mathbb{P}$(success) in this engine is the precision of that one sampled batch.

A control this paragraph used to claim, and did not have. Earlier drafts read "83.7 effective ... against 99.9 with the opponent model disabled", and reported 14 exact decisions in 646 for that arm. No such arm was ever run. `results/ess_probe.json` holds four rows, all at $\gamma = 0.35$, differing only in the proposal tilt; the 99.9 is the *p90 of the champion row itself* (99.986), and the 14 in 646 has no source at all. A percentile of one arm had been written up as the mean of another, and the paragraph that followed then explained a gap between them.

That is worth more than a correction, because the argument it supported was sound and the number under it was not. The efficiency loss from steering the proposal is real and is measured — 52% — and the remedy discussed below was tested and did make the marginals worse. What was never measured is how much of the loss the opponent model accounts for, and with $\gamma > 0$ changing the target the exact DP is built to count, the comparison is not a straightforward one to run. It is left open rather than asserted.

The gap has an obvious explanation and an obvious remedy, and the remedy is a trap. The proposal is quota-proportional and knows nothing about the choice model, so the whole tilt of the target lands in the weights; folding one step of the likelihood into the proposal is textbook, and it does exactly what the textbook says — effective sample size rises to 105.8, past even the untilted $\gamma = 0$ figure. Partial strengths are worse than either end, the signature of a proposal distorted but not matched.

Measured against a high-precision reference, the twist makes the marginals 3.4× worse.

Effective sample size measures how *flat* the weights are, and a proposal that covers one corner of the target well and the rest not at all has beautifully flat weights over the corner it covers. The one-step twist overshoots because the likelihood's depth-zero term is a numerical floor of $\log 10^{-9}$: the astronomical $0 \to 1$ ratio makes the proposal read "this slot is empty now" as "this slot ends empty" and drives nearly every draw into one configuration. The parameter ships at zero, verified bit-identical to the untwisted sampler, and the tests pin both halves of the result so that the next person to notice the efficiency gap does not have to rediscover why closing it fails.

Is precision still worth buying? v0.3 found belief precision unsaturated, but with a biased sampler and a different estimator, so it does not transfer. Measured here against a 16,000-draw reference, L1 error per card falls as $n^{-0.475}$ from 40 draws to 1280 with no bias floor anywhere:

draws	40	80	160	320	640	1280
L1 per card	0.248	0.174	0.128	0.091	0.066	0.048

Two things follow. The exponent is the pure Monte Carlo law, which independently confirms the sampler is unbiased — a bias floor would show as the curve flattening, and it does not. And precision at the operating point of 160 draws is nowhere near saturated: the prize is real and can be bought directly with draws rather than with a cleverer proposal.

The policy can feel it, and it is the largest single change here apart from the opponent model. The duel that paragraph declined to claim was pre-registered before its screening cell finished — deliberately, because the belief-space search had already cost four rounds partly by being sized against a number a screen had inflated, and the only reliable defence is to fix the design before the number exists. It was sized against a *minimum interesting effect* of +0.15 sets per deal-pair rather than against any estimate, on the ground that an effect smaller than the already-shipped search delivers, for triple the sampling budget, is not one this engine should adopt.

Six blocks of 1000 duplicate deal-pairs, `n_draws` = 480 against 160, everything else identical:

$$+0.340 \text{ sets per deal-pair,} \quad 95\% \ [+0.243, +0.436].$$

All six blocks are positive, Cochran's $Q = 2.92$ on 5 degrees of freedom ($p = 0.71$), $I^2 = 0\%$, $\hat{\tau} = 0$. The entire interval clears the threshold fixed in advance. For scale, this is more than three times the belief-space search, which took four rounds and 6000 pre-registered pairs to establish at +0.104.

The 400-pair screen of this same cell had read +0.160. The confirmatory run reads +0.340 — *higher*. That is what an unselected screen looks like, and it is worth stating only because every screen this project *selected* on inflated instead.

What it costs, and why that is a separate decision. The pre-registration committed in advance that a demonstrated effect would not automatically move the default, because `n_draws` trades strength against latency and the public table has a request budget. Measured on 90 positions, a decision costs

$$3.57\,\text{ms} \;+\; 6.34\,\mu\text{s per draw},$$

a fit whose residuals reach 0.21 ms across 40 to 1920 draws, and are worst at the low end — −0.21 at 40 draws and +0.20 at 160, against +0.03 and −0.02 at 480 and 1920. So the straight line is a good description of the expensive end and a rough one exactly where the shipped budget sits, which is worth knowing before using it to price a change at 160. The ratio quoted next does not depend on it: it is measured per decision, not read off the fit. At the shipped 160 the sampler is only a fifth of a decision, so tripling it costs +1.9 ms — 1.38×, not 3×. That headroom is finite and the same fit says so: by 1920 draws a decision is already 3.3× the shipped one. With the effect and the price both measured, the default moved to 480.

That fit is not an explanation, and the difference matters: if the fixed term were setup, it could be hoisted and draws would become nearly free. Timing the sampler alone, on one already-built sampler across 1 to 1440 draws, gives $2.50\,\text{ms} + 8.0\,\mu\text{s}$ per draw — about $60\,\mu\text{s}$ of fixed cost for each of the 42 free cards it must place. That is per-card numpy dispatch. Sequential importance sampling walks the free cards one at a time, so batching makes each *step* wider without making the number of steps smaller, and the per-card constants are already cached. There is nothing there to hoist.

The consequence is a design rule with a number attached: going from 160 to 480 draws costs 2.2 ms inside the sampler, while going from one decision to two costs 2.5 ms before a single draw is taken. Draws are cheap and decisions are expensive, so under a latency budget a better posterior at one decision beats more decisions with worse ones. The belief-space search of Section 12 already obeys this without having been designed to: it never samples, and operates on the single posterior marginal matrix it is handed.

One caution on those two fits. The per-draw slope of the sampler alone ($8.0\,\mu$s) exceeds that of the whole decision containing it ($6.34\,\mu$s), which is impossible until the position mix is accounted for: per-draw work scales with the number of free cards, and the two runs drew positions averaging 42 and 28 of them. Scaling by that ratio predicts $5.4\,\mu$s against the 6.34 measured, which restores the ordering and accounts for most of the gap but not all of it — the remainder is the rest of a decision that also grows with the draw count. We report it as a partial reconciliation rather than a clean one.

Two of this paper's four documented winner's curses were paid for by sizing an experiment against an inflated screen. This is the case where pre-registering against a threshold instead cost nothing and returned the study's largest result.

14.4 The opponent model in play

Posterior accuracy is not strength. Table 3 measures the model by duplicate-deal play against the otherwise-identical $\gamma = 0$ policy.

γ	pair score	set diff / pair	95% CI
0.10	0.613	$+0.993$	$[+0.470, +1.517]$
0.30	0.667	$+1.847$	$[+1.230, +2.463]$
0.45	0.700	$+1.860$	$[+1.336, +2.384]$
0.60	0.657	$+1.300$	$[+0.795, +1.805]$
0.90	0.723	$+1.827$	$[+1.199, +2.455]$

Table 3: Opponent model against the identical policy with the model disabled, 150 duplicate deal-pairs per cell. Every cell excludes zero. The dose-response rises to a plateau by $\gamma = 0.30$ and stays there over a threefold range, which is what one wants from a model with a single free parameter — and unlike the ask-objective terms of v0.3, which showed a sharp inverted-U, there is no weight at which this becomes destructive within the range tested.

The cleanest attribution. Tables 3 measures the model against copies of itself. The sharper experiment holds the *opponent* fixed at the v0.3 champion and varies only whether the model is on, at identical deal seeds and identical agent seeds, 250 duplicate deal-pairs each:

v0.4 configuration	pair score	set diff / pair	95% CI
exact inference, $\gamma = 0$	0.502	-0.008	$[-0.482, +0.466]$
exact inference, $\gamma = 0.30$	0.684	$\mathbf{+1.920}$	$[+1.479, +2.361]$

The first row is as close to exactly zero as a 250-pair experiment can report. Provably correct inference, replacing a sampler its own authors documented as biased, moved the result by eight thousandths of a set per deal-pair. The second row differs from the first in one parameter.

14.5 The choice model, measured rather than assumed

Section 14.4 establishes that the model is worth +1.9 sets per deal-pair, which is the largest single effect in this engine. It rests on one sentence: *a player asks in a half-suit in proportion to how many cards of it they hold.* Proportional means an exponent of one, and that exponent was never measured. It was chosen because it is the one value whose denominator is public — the depths of a hand sum to its size — which is an excellent reason to try it and no reason whatever to believe it.

There is a specific reason to doubt it. Under an objective dominated by $\mathbb{P}(\text{success})$, what makes a half-suit worth asking in is how many cards it still offers: holding five of six offers exactly one card to ask for, holding one of six offers five. Depth and opportunity pull in opposite directions, so linear-in-depth is not obviously right even in sign.

The design. The policy is available, so the propensity can be measured instead of argued. Over 200 self-play games we record, at every ask, which half-suits were legal, the asker's depth in each *on the initial deal*, and which was chosen. That is a conditional-logit design in which the alternatives are known from the rules rather than inferred, and it yields 17,005 asks where at least two half-suits were legal — 72,091 alternatives in all. Fitting $\mathbb{P}(\text{ask in } H) \propto \text{depth}_H^{\alpha}$ by exact conditional likelihood gives α directly, with $\alpha = 1$ the shipped model and $\alpha = 0$ legality carrying all the signal and depth none.

Standard errors are block-bootstrapped over games. This is not a refinement: the likelihood's own curvature treats every decision as an independent draw, and 17,005 decisions come from 200 deals whose members share a layout, six hands and one set of policies. The correction doubles every interval, and the symptom was visible beforehand — a smooth curve through the per-band estimates returned a χ^2 several times its degrees of freedom.

half-suits resolved	0	1	2	3	4	5	6–8
α	2.00	1.28	1.20	0.68	0.30	0.41	−0.02
clustered SE	0.08	0.08	0.12	0.12	0.13	0.18	0.16
decisions	4410	2881	2527	2437	1958	1541	1251

Table 4: The fitted exponent of the choice model against how far the deal has run. The shipped model is a constant 1. It is wrong at both ends, in opposite directions: at the opening the true exponent is about 2 and the model understates its own strongest signal, and by the endgame it is 0 and the model applies a tilt that is not there.

What it says. Pooled over the whole game the best constant is $\alpha = 1.207 \pm 0.046$ — close enough to 1 that a constant never looked broken, and far enough from either end of Table 4 that it could not be right. Split at the halfway point, $\alpha = 1.331 \pm 0.049$ early against 0.232 ± 0.118 late. The likelihood prefers the fitted constant to $\alpha = 0$ by 1402 nats and to $\alpha = 1$ by 40.

And most of the decay is the covariate, not the player. The obvious reading of Table 4 is behavioural: a player asks where they are deep only if they had a choice, and late, holding two cards with most of the deck resolved, legality binds. That reading is largely wrong, and the same 17,005 decisions say so.

The fit conditions on depth *on the initial deal*, because that is what the sampler assigns. Cards move, so as a game runs on that covariate drifts from the hand the asker actually held — mean absolute disagreement rises from 0.16 cards at the opening to 0.66 by the endgame, with correlation falling from 0.77 to 0.44. Noise in a covariate attenuates its coefficient, and noise that grows with the game manufactures precisely this decay. Refitting on depth at the moment of the ask:

covariate	opening	1–2	3–4	5–8	pooled
initial-deal depth	2.00	1.25	0.52	0.23	1.207
depth at the ask	2.90	2.32	1.72	1.34	2.195

The decay survives and shrinks, and never approaches zero. Late asks do carry depth signal; what decays is how well the initial deal still describes the hand it is about.

The better covariate is free. Depth at the moment of the ask looked expensive — it appeared to need the public history replayed against every candidate world — and it is not, because of a fact about the rules. Cards move only when an ask succeeds, and a successful ask is entirely public: the card, the asker and the target are all in the log. So $\text{depth}_{\text{at ask}}(p, H) = \text{depth}_{\text{initial}}(p, H) + \delta(p, H, t)$ with δ *identical in every hypothesised world.* The sampler goes on assigning initial deals and the likelihood adds a constant known before any drawing. Verified against the engine on 23,268 (player, half-suit, time) triples across six games, with no mismatches.

Measured as a description, the difference is not marginal: at-ask-time depth fits the same choices 4,654 nats better, reaching 6,057 nats above uniform where the initial deal reaches 1,403. The shipped covariate captures under a quarter of what its own one-parameter family can.

This is not the `attime` mode of Section 14.9, which computes belief-*located* depth and adds a present-tense count to free cards drawn over the initial deal; that mode measured −1.544 sets per deal-pair over 250 pairs. Whether the exact version plays better is a duel, and is not claimed here.

Is the power law the right shape? Fitting one functional form well is not evidence that it is the right one. The saturated alternative gives every depth value a free coefficient; it nests the power law, so it can never fit worse, and it is the best that anything conditioning on depth alone can do. On the same decisions the power law gives up 110 nats to it on initial-deal depth and 269 on depth at the ask.

Set against the 4,654 nats that changing the covariate is worth, the shape is not what limits this model — the covariate matters roughly seventeen times more. That is the useful bound: further gains in the choice model need a different conditioning variable or the normaliser below, not a better curve. (The initial-deal figure is a lower bound; that fit had not converged, and its depth-5 coefficient rests on five alternatives in the whole corpus.)

What the engine drops, and where it sits. The likelihood is written as proportional to a product of depths, and the proportionality hides a denominator: a proper choice model divides by the sum of depth$^{\alpha}$ over the half-suits the player could legally ask in. At $\alpha = 1$ that sum is the player's hand count — public, identical in every world, and it cancels. That is the original argument for an exponent near 1, and it holds for every covariate here: the depths over live half-suits sum to the hand count whether depth is read at the deal, now, or at the ask.

The cost of being elsewhere had never been measured. Across worlds drawn from the posterior at 30 real positions, the coefficient of variation of the dropped denominator is

α	0.35	0.60	0.80	1.00	1.207	1.60	2.195
mean CV	0.100	0.063	0.032	**0.000**	0.035	0.107	0.226

a V centred exactly on 1, with the $\alpha = 1$ column serving as the control that the measurement is reading the right quantity. Two readings follow. The shipped $\gamma = 0.35$ pays more for the omission (0.100) than the fitted exponent on its own covariate would (0.035 at 1.207) — the approximation that motivated an exponent near 1 is being violated more at the setting in use than at the one the data prefer. And $\gamma = 1.0$ is not just another point on the sweep of Table 3: it is the unique exponent at which this likelihood is correctly specified. That sweep never tested it; 0.90 and 1.50 bracket it at 150 pairs each, an MDE of 0.87, which separates nothing.

And in play, it is real and not worth buying. Everything above is an argument from likelihood and from a normaliser, not from a duel, so the configuration it points at — at-ask-time depth at $\gamma = 1.0$, the better covariate at the exponent where the model is correctly specified — was pre-registered and run. Six blocks of 1000 duplicate deal-pairs against the champion, sized against a minimum interesting effect of +0.15 fixed in advance, with the run committed to proceed whatever its five screening cells said. They said −0.05, +0.08, +0.32, −0.13 and +0.01, every one of them uninformative at ±0.53, which is why they were never load-bearing.

$$+0.1015, \quad 95\% \ [+0.0063, +0.1967].$$

The interval excludes zero, so by the test fixed in advance the effect is demonstrated. Three things belong next to that, and all three were decided before the data:

It is *below the bar it set itself.* The pre-registration named +0.15 as the smallest effect worth adopting — roughly what the belief-space search delivers — and this is +0.10, with the interval's upper end barely reaching it. Real, and not worth buying. The threshold was chosen before the number, so that judgement is not a reaction to it.

The blocks *disagree.* Cochran's $Q = 12.9$ on 5 dof ($p = 0.024$), $I^2 = 61.4\%$, $\hat{\tau} = 0.150$ — the same size as the pooled effect. Two explanations fit that on its own: the harness understates its uncertainty, or this particular effect varies with the deal population. They are distinguishable, because the identical six-block design has now been run three times and the A/A null twenty-four times:

run	pooled	Q	p	$\hat{\tau}$
lookahead settling	+0.104	4.08	0.538	0.000
precision, 480 vs 160	+0.340	2.92	0.713	0.000
at-ask depth, $\gamma = 1.0$	+0.102	12.94	**0.024**	**0.150**
A/A null, 24 blocks	0.000	18.41	0.735	0.000

Every other run gives $\hat{\tau}$ *exactly* zero, including two real effects of comparable and larger size, and so does the null. So the harness is not understating its uncertainty by default, and a deal-population-dependent effect is the better explanation here. We state the multiplicity rather than leave it implicit: four homogeneity tests at the 5% level give roughly a one-in-five chance of at least one significant result under a complete null, so one significant Q out of four points at where to look and does not settle it.

The verdict rests on *two of the six blocks.* Leaving out block 0 gives $[-0.041, +0.168]$ and leaving out block 4 gives $[-0.060, +0.148]$; dropping either one alone puts zero back inside. No block may be dropped for its result and none is — the pre-registered pool is the pool — but under $I^2 = 61\%$, with the interval clearing zero by 6.7% of its own half-width, saying so costs nothing and omitting it would be a choice.

The random-effects interval is $[-0.052, +0.255]$ and *includes zero.* We report it and it decides nothing. The fixed-effect pool was fixed as primary before any of this was known, and adopting a wider estimator once the narrow one is inconvenient is the failure this whole protocol exists to prevent — it is the winner's curse run backwards, choosing the analysis that gives the answer one now prefers. Both numbers are printed; the pre-registered one governs.

So the normaliser argument survives contact with play, and the engine keeps $\gamma = 0.35$ and initial-deal depth anyway. The two parameters were argued for jointly and the screens cannot separate them, so the pre-registration binds them to move together or not at all; they do not move.

The competing story is refuted by the same data. If asks went where the opportunity was rather than where the depth was, propensity would fall as cards held rises. Observed over expected runs 0.46, 1.05, 2.26, 2.64, 2.90 for one through five cards held — monotonically the other way. Whatever else is wrong with the model, its sign is right.

A rule, not a regularity. In 72,091 legal alternatives, not one had an initial-deal depth of zero. That is a theorem rather than a coincidence: asking in a half-suit requires holding a card of it, and cards move only to askers who already hold one, so a player's first card of any half-suit cannot have arrived by play. The consequence matters for the implementation, because the likelihood floors a zero depth at $\log 10^{-9}$ and that floor is therefore excluding worlds the *rules* forbid rather than softening an awkward case. The engine's `allow_bluff_asks` variant does not undo it; that flag widens which cards of a half-suit may be asked for and leaves the half-suit guard alone.

What was changed, and what was not. `gamma_schedule` interpolates between the constant and a weighted quadratic through Table 4 ($\chi^2 = 8.3$ on 4 dof), clamped flat past its vertex because past there the parabola turns up and the measurements do not. Given the paragraph above it is best read as a *dilution correction*: the engine conditions on initial-deal depth, that covariate degrades as the game runs on, and a degrading covariate should be believed less. That is a more modest claim than the behavioural one this was first written up as, and a better supported one — and it predicts the term should become unnecessary under the at-ask-time covariate, where nothing drifts. It is normalised by the profile's mean over the observed distribution of asks, so full strength redistributes the model's belief across the game without changing how much of it there is in total. That separation is deliberate: γ was already tuned by duels to the broad

v0.4 champion versus	pairs	pair score	set diff / pair	95% CI
v0.3 champion (`tuned`)	**500**	**0.705**	**+1.854**	**[+1.542, +2.166]**
— replication	300	0.758	+2.440	[+2.052, +2.828]
— replication, $\gamma = 0.30$	250	0.684	+1.920	[+1.479, +2.361]
v0.3 `probabilistic`	200	0.792	+2.785	[+2.274, +3.296]

Table 5: The v0.4 champion is `fishbot4(`γ`=0.35)`: exact/unbiased inference, the opponent choice model, and v0.3's own ask weights. No new objective term is enabled.

plateau of Table 3, and a term that moved shape and strength together would leave any positive result unattributable. Zero remains the incumbent, decision for decision.

Whether the reshaped model plays better is a duel, and is not claimed here.

Limitation. This measures the champion against copies of itself, which is exactly the situation the opponent model is used in throughout this paper, and exactly not the situation of a human table. The profile is a fact about this policy's behaviour, not about Fish.

14.6 The strength ladder

Against the two weakest rungs the champion wins every pair: 120/120 against the public-information heuristic (+14.09 sets per pair) and 80/80 against random (+16.30). Those are reported for completeness rather than as evidence; a shutout means the rating gap is a bound, not a measurement.

The headline comparison was run three times at different seeds, and the three estimates span +1.85 to +2.44 — wider than their individual intervals suggest, which is a useful reminder that a 95% interval covers the mean of the deals actually played, and that three such intervals will occasionally fail to overlap. We take the largest experiment as the headline and report the others rather than the most flattering one.

What the strongest configuration is worth, and what nobody has played. Two changes in this version clear a pre-registered bar against the champion: the tripled posterior sampling budget at +0.340 and the belief-space search at +0.104. Each was measured against the champion *alone*, so neither says what their combination is worth — and the combination is not obviously additive, since the search's whole mechanism is to search a belief the sampler draws.

The stacking run lets the two be chained, because it measured the search *on top of* 480 draws. The two runs use disjoint deal seeds, so they are independent and their standard errors add in quadrature:

link	estimate	pairs
480 draws vs. champion	$+0.340 \pm 0.049$	6000
search on top of 480 draws	$+0.072 \pm 0.042$	6000
chained: search + 480 draws vs. champion	$\mathbf{+0.411 \pm 0.064}$	12000

95% CI [+0.285, +0.538] — larger than either change alone and, apart from the opponent model itself, the largest figure in this paper. It is not the largest *single* change: that is the sampler's +0.340, and this number is two changes at once.

And it has now been played directly, which is the only reason the number above can stand unqualified. A chained estimate inherits both links' assumptions — in particular that each change's effect does not depend on the deal population the other was measured over. That was plausible, since both pools were drawn the same way, and it was *unmeasured*; this paper's recurring failure is exactly a plausible unmeasured step carrying a published number. There was also a cheaper reason to run it: **nobody had ever played the shipped configuration against the reference in a single duel**, so every "vs. champion" figure in this document was for a spec that is not the one on the website.

`jobs/PREREGISTRATION_combined.md` fixes, before any pair: two blocks of 1000 fresh pairs; the A/A 3.796 as the per-pair standard deviation rather than the divergence model, deliberately conservative because this arm differs from the champion by *two* knobs and the model was fitted on arms differing by one; a fixed-effect pool as the only thing that decides; and three outcomes named in advance — agrees, lower, higher — so none could be chosen afterwards. It also states, in advance, that at 2000 pairs against the chain's 12000 this run *cannot* make the estimate more precise. Its job is to make it honest, and a wider interval is the design working rather than a disappointment.

	estimate	
chained, 12000 pairs	$+0.411$ $[+0.285, +0.538]$	indirect
direct, 2000 pairs	$+\mathbf{0.357}$ $[+0.191, +0.524]$	**excludes zero**
difference	-0.054 ± 0.107	does not resolve

The chain agrees. The intervals overlap and the difference is half a standard error, so the assumption the chain rests on is now measured and holds, and the indirect caveat can leave this section. Worth noting separately: the design's sizing assumption held almost exactly — a planned per-pair standard deviation of 3.796 against a realised 3.801 — which is unusual enough in this study to be worth one clause, given that the claim-threshold run's equivalent came in 21% low.

What is *not* settled by this is whether the combination beats the sampler *alone*: that is the stacking run's own estimate, whose interval contains zero at 68% power. So the defensible ordering remains

$$\text{search} + \text{sampler} \;\geq\; \text{sampler} \;>\; \text{champion},$$

with the first inequality undemonstrated — and the leftmost quantity now measured directly rather than assembled. The engine names this configuration `V04_COMBINED` and the public table has been playing it all along; until this run reported, no figure in this paper described what the website was actually running.

And it stops paying: precision's returns diminish. If tripling the sampling budget is worth +0.340, the obvious question is whether tripling it again is worth as much. Pre-registered in `jobs/PREREGISTRATION_precision2.md` with a *minimum interesting effect* of 0.150 fixed before any pair was played — so that a small positive would be reported as a small positive rather than promoted:

rung	set diff	95% CI
160 → 480 draws, 6000 pairs	$+\mathbf{0.340}$	$[+0.244, +0.436]$
480 → 1440 draws, 6000 pairs	$+0.094$	$[-0.002, +0.191]$
difference	$-\mathbf{0.245} \pm \mathbf{0.069}$	$-\mathbf{3.5}$ SE

Cochran's $Q = 4.01$ on 5 df ($p = 0.55$), $I^2 = 0\%$, $\tau = 0$: the six blocks of rung 2 agree with each other completely, and with zero.

Rung 2 on its own is **not demonstrated** — its interval includes zero and its point estimate sits below the 0.150 threshold fixed in advance. But the *contrast* is demonstrated, and it is the more interesting quantity. Both rungs multiply the budget by exactly three, so under a log-linear response they would buy the same amount. They do not: the difference between the rungs is

-0.245 ± 0.069, which excludes zero at 3.5 standard errors. Precision's returns diminish, and they do so somewhere between 480 and 1440 draws.

That is worth putting beside the earlier claim it repairs. The failed-experiment section reported "precision has saturated" on the strength of 200-pair screens around 160 draws, and Section 14.3 refutes that: saturation had not begun at 160. It has begun by 480. The screens reached the right word for the wrong rung, and getting from one to the other took 12,000 pre-registered pairs across two runs.

The default stays at 480. Rung 1 is demonstrated and bought; rung 2 is not demonstrated, costs a further 1.4× on the sampler, and its own pre-registration says a point estimate below 0.150 does not move a default whatever its sign.

A demonstrated effect that is deliberately not shipped. At-ask-time depth at $\gamma = 1.0$ measures +0.102 over 6000 pre-registered pairs and its interval excludes zero, so the effect is real. It is also below the 0.15 that `jobs/PREREGISTRATION_at_ask.md` fixed *in advance* as the smallest effect worth adopting, and the upper end of its interval, +0.197, barely reaches that. It therefore does not ship, and `depth_mode` stays `"initial"` in every named configuration. A real effect below a threshold chosen before the data is not a reason to move a default; deciding otherwise after seeing +0.102 is exactly what fixing the threshold in advance prevents. A test asserts the default, so the decision cannot quietly reverse itself.

Do v0.3's objective terms survive better inference? A reasonable worry about the whole result is that v0.3's hand-tuned tie-breakers were compensating for a poor posterior and would become redundant once it improved. They do not become redundant, but they do become *worth less*. Removing both — setting w_{turn} and w_{scarce} to zero and changing nothing else — costs, over 250 duplicate deals each:

inference	cost of removing both terms	95% CI
without the opponent model	1.660 sets/pair	[+1.240, +2.080]
with the opponent model	1.084 sets/pair	[+0.611, +1.557]

v0.3 measured the same pair at +1.41 against its own inference, which sits between the two. So the terms and the opponent model are partially substitutable: about a third of what the tie-breakers were contributing is absorbed once the posterior knows more about opponents' holdings, and the remaining two-thirds is positional judgement that no amount of card-reading supplies. That is the expected shape if the tie-breakers were doing two jobs — compensating for a blurry posterior, and encoding something genuinely strategic — and it is worth noting that the gains do *not* simply add: 1.85 (the headline) is less than 1.92 (the model alone against the v0.3 champion) plus anything.

14.7 Absolute strength: agreement with provably optimal play

Every number above is relative. Table 6 is not. It reports, over 988 exactly solved positions harvested from 260 real games by four different generator policies, how often each agent chooses a *progress-optimal* action — the stricter criterion of Section 9, which excludes value-preserving moves that make no progress.

Four things are worth drawing out.

Every belief-tracking agent is progress-optimal in every information-resolved position, on a corpus four times the size of v0.3's and 79% of it two-half-suit, and against the stricter criterion. This reproduces v0.3's strongest correctness claim on harder ground — while Section 9 explains why the one-half-suit version of that claim was shallower than it read.

The absolute metric confirms the relative one, independently. Exact inference without the opponent model scores 67.5% under uncertainty against the v0.3 champion's 67.6%:

agent	resolved	uncertain	m=1	m=2
random	60.8%	32.7%	28.3%	43.9%
heuristic	71.6%	46.8%	34.4%	59.0%
memory	**100.0%**	61.7%	81.1%	70.1%
probabilistic	**100.0%**	66.9%	86.8%	73.3%
v0.3 champion	**100.0%**	67.6%	86.8%	74.0%
v0.4, no opponent model	**100.0%**	67.5%	85.4%	74.2%
v0.4 champion	**100.0%**	**72.5%**	85.8%	**78.7%**
v0.4 champion, tablebase disabled	**100.0%**	72.5%	85.8%	78.7%

Table 6: Agreement with progress-optimal play over 988 solved positions, 278 of them information-resolved and 776 with two live half-suits. "resolved" and "uncertain" mean what they did in v0.3; m is the number of live half-suits.

no change, exactly as the duplicate-deal experiment found and exactly as the ground-truth posterior scores predicted. Adding the opponent model moves it to 72.5%. Three different instruments — paired play, posterior likelihood against the true hands, and agreement with an exact solver — agree on which component does the work.

The tablebase contributes nothing here, and we report that rather than claiming the speedup as strength. Disabling it reproduces the champion's counts in all six columns of Table 6 exactly — 278 resolved-optimal, 585 uncertain-optimal, 515 uncertain-progress, 182 at m=1, 611 at m=2. That is consistent with changing no decision and is not a proof of it: the comparison is of aggregates, and two different decision sets can produce the same totals. The per-decision check would be stronger and was not run. In an information-resolved position the marginals are one-hot and the ordinary policy already plays it correctly; in an uncertain position the tablebase refuses by construction. Its value is a 10,900× reduction in the median cost of the endgame path and insurance against a class of error v0.3 demonstrably suffered, not extra strength.

Where the remaining gap is. At m=2 under uncertainty the champion agrees with optimal play 78.7% of the time. The missing fifth is not in the endgame in the sense v0.3 meant — these positions *are* the endgame — it is in choosing correctly when the hidden information genuinely has not resolved.

14.8 How much the ask objective leaves on the table

Table 6 is absolute but narrow: it needs a position the exact solver can reach, and it scores agreement with a *perfect-information* optimum rather than value. This measures the objective where it actually operates — positions with hidden information and no solvable endgame — by running one step of policy iteration and reporting the gap.

At a decision with belief b and policy π, the value of ask a is $Q(a) = \mathbb{E}_{w\sim b}[V^{\pi}(w, a)]$: draw worlds from the posterior, take a, and play the deal to the end with π in all six seats. The one-step regret is $\max_a Q(a) - Q(\pi(\text{root}))$, and it is zero exactly when the objective already picks the best ask available *with respect to its own continuation.* It bounds what any better tie-break could buy.

This is not strategy fusion. Every seat inside a rollout sees only the public history plus the hand the sampled world gives it; nothing consults the world it is being played in. Determinization becomes strategy fusion when the searcher conditions its plan on the sampled world, and here the world only generates observations.

We harvest 40 on-policy positions with at least five half-suits resolved, where the action set is small enough to evaluate *exhaustively* — 18.4 legal asks on average, all of them, rather than a shortlist the objective proposed. Thirty-seven yielded an ask decision rather than a claim.

The estimator cannot be the obvious one. $\max_a \widehat{Q}(a)$ over eighteen noisy estimates is biased upward by about two standard errors, so a policy that already played perfectly would measure a regret near +1.8. On synthetic rollouts with twenty-five *truly equal* actions the naive estimator reports +1.83 sets of regret that does not exist. This is the same selection bias Section 14.11 is about, arriving at a fourth level: inside the tool built to measure the objective.

Cross-fitting removes it. Choose the challenger on half the sampled worlds, score it on the other half, both ways round; the half that scores it played no part in choosing it, so the estimate is unbiased for the value of the action actually selected — which is at most the best available, making the result a lower bound and never an inflated one. On the same synthetic null it returns -0.004 ± 0.012 over five runs of 4000 trials, and given one action truly worth +2.0 it recovers +0.88.

The result.

cross-fitted one-step regret	-0.097 ± 0.105, 95% $[-0.302, +0.109]$
naive maximum over actions	+0.579
selection bias, measured	+**0.676** sets

The selection bias is the solid finding. Two thirds of a set per decision is the entire difference between reporting that the objective throws away most of a set and reporting that it throws away nothing detectable.

What the interval can and cannot say. Its resolution is set by the noise, and getting that right changed the reading. A first calibration modelled each action's rollout noise as independent, which is not the design: every action runs through the same worlds with the same seat seeds. Measured properly the paired spread across actions is 1.253 sets against a raw per-world spread of 1.437, so common random numbers barely help — changing the root ask sends the deal down a genuinely different path, and there is little shared variation left to cancel. At that noise the estimator recovers 21% of a true half-set edge and 57% of a full one.

So the honest reading is not that the objective is optimal. It is that its one-step regret is under roughly half a set per decision and this design cannot resolve anything smaller. The same procedure applied to a *random* legal ask returns $+0.128 \pm 0.130$, indistinguishable from zero, which confirms the limit is the design and not the policy: at this budget one-step lookahead cannot reliably beat a coin flip either.

Ranking. Over the same rollouts, the objective's ordering of asks correlates $+0.164 \pm 0.072$ with what those asks turn out to be worth, against $+0.133 \pm 0.072$ for $\mathbb{P}(\text{success})$ alone. The full objective ranks slightly better than its dominant term, by an amount well inside noise — which is what Section 14.6's tie-break story predicts, and is not on this evidence more than that.

What this says about where to spend effort. Three measurements point the same way, and they are worth stating together because the conclusion is not visible in any of them alone. The objective ranks barely better than $\mathbb{P}(\text{success})$ by itself. Almost every term added *beside* $\mathbb{P}(\text{success})$ — Table 7 — is a null. And the largest single effect in this study, +0.340 sets per deal-pair, comes from tripling the sampling budget, which does not touch the objective at all: it makes the $\mathbb{P}(\text{success})$ the objective is already dominated by more accurate.

The reading we take is that in this game the leverage is in the belief rather than in what is computed from it. That is a claim about where the remaining strength is, and it is the reason the next experiments in this line buy posterior quality rather than richer ask features.

We are explicit that the regret bound above contributes *nothing* to this argument. It is per decision, while the play results are per deal-pair, and the two are not commensurable by any factor we can defend — a deal contains dozens of decisions, and one-step improvements do not add. A regret under half a set per decision is compatible with enormous headroom. The argument rests only on the three measurements that are in the same units as each other.

14.9 Everything else we tried

Table 7 is the rest of the study. It is mostly nulls, and it is reported in full because a table of only the things that worked would misrepresent how much was tried. Each cell is a duplicate-deal duel against the champion with exactly one thing changed, and the last column is the smallest effect that cell could have resolved at 80% power given a per-pair standard deviation of 3.869 sets.

That column uses one standard deviation for every row, and we now know it is a property of the comparison rather than of the game. Since the correction is conservative — the constant is the noisiest case — it can only have made these cells look *less* informative than they were, which raises the question of whether any of them was in fact resolvable. Rescoring each cell that stored its per-pair differentials against its own measured noise moves the resolvable effect by a median factor of 0.92, and by 0.60 in the quietest, and changes no verdict at all.

That last clause was where we stopped, and it was the wrong check. "$|\text{est}| > \text{MDE}$" is a test at $\alpha \approx 0.005$, because the MDE is $(z_{0.975} + z_{0.80})\,\text{se} = 2.80\,\text{se}$; everywhere else in this paper a cell is said to have resolved something when its *95% interval excludes zero*, which is 1.96 se. We had been auditing the table against a bar 43% higher than our own definition, and since the gap between the two MDEs is a median 5% of the MDE, finding no change was close to arithmetically guaranteed.

Under the definition the rest of the paper uses, **five cells change**: the retake penalty at -0.340 $[-0.663, -0.017]$, a decisive lookahead cell at $+0.307$, and three blocks of pre-registered runs. All five sit below the MDE their row quotes and have intervals excluding zero. The MDE column says what a cell could *detect* at 80% power, which is a stricter and different thing from what it did *resolve*, and reading the column as "this cell found nothing" understates them. The retake cell is the sharpest case: another pre-registration in this study treats that -0.340 as an established effect and designs a follow-up against it, while this table files it as a null. Both readings cannot be right, and the interval is the one that governs.

None of this widens a published interval — each cell's interval was always computed from its own sample variance, and the constant only ever fed the MDE column. What it corrects is how that column should be read. The cells that recorded only a mean and an interval cannot be re-examined at all, which is itself the argument for storing per-pair data, and why v0.4 does.

Four of these deserve comment.

† **These two rows measured nothing, and we found out afterwards.** The no-declaration term was assembled into the importance weights and then overwritten: the batch sampler built it first and the opponent-model branch that follows did `logl =` rather than `logl +=` , so whenever any non-self player had already asked — 632 of 641 decisions in `results/ess_probe.json` — the term was discarded before use. In the handful of decisions where it survived, it addressed the half-suit by column indices built against the caller's free-card order and applied to a matrix whose columns are in the sampler's own sort; six of eight columns differ on a typical position, so the "did the opponents take this whole half-suit" test read a mixture of cards from other half-suits. And the scalar reference sampler stored the term and never applied it at all, so the batch/scalar agreement test could not see any of this.

Both cells are therefore reported as *invalid rather than null*: they are what a disabled feature scores, which is the champion plus sampling noise, and the intervals above are consistent with exactly that. All three bugs are fixed and pinned by tests that fail against the previous code; the feature has not been re-measured, and until it is there is no result here in either direction. Nothing else in this paper is affected, because `opp_lambda` defaults to zero and no other cell sets it.

Precision appeared to have saturated, and it had not. v0.3 found that raising the sampled-world count from 32 to 96 was worth +0.54 sets per pair and concluded that belief

what changed	setting	pairs	set diff	95% CI
inference precision				
draws per decision	96 (from 160)	160	−0.300	[−0.852, +0.252]
draws per decision	320 (from 160)	200	−0.035	[−0.578, +0.508]
count mode	$\sqrt{\cdot}$, γ=0.70, 160	200	+0.480	[−0.042, +1.002]
count mode	$\sqrt{\cdot}$, γ=0.70, 320	200	−0.165	[−0.701, +0.371]
no-declaration signal†	$\lambda = 0.9$	200	+0.205	[−0.301, +0.711]
no-declaration signal†	$\lambda = 1.8$	200	+0.190	[−0.313, +0.693]
new ask-objective terms				
exposure	0.40	160	−0.113	[−0.660, +0.435]
claim progress	0.30	160	−0.206	[−0.807, +0.395]
signalling	+0.25	160	−0.300	[−0.890, +0.290]
signalling	−0.25 (conceal)	160	+0.056	[−0.530, +0.642]
information gain	0.10	160	−0.394	[−0.932, +0.144]
concentration	0.15	160	−0.037	[−0.653, +0.578]
turn-risk retune	0.35 (from 0.60)	160	−0.075	[−0.670, +0.520]
scarcity retune	0.35 (from 0.20)	160	+0.125	[−0.381, +0.631]
claiming and the endgame				
claim threshold	0.99 (from 0.97)	200	+0.010	[−0.010, +0.030]
informed pass choice	on	160	+0.013	[−0.012, +0.037]
tablebase	one half-suit only	200	+0.005	[−0.005, +0.015]
tablebase	disabled entirely	200	+0.005	[−0.005, +0.015]
the learned half-suit value objective				
blended	$w_{\text{value}} = 0.5$	200	+0.410	[−0.086, +0.906]
blended	$w_{\text{value}} = 1.5$	200	+0.040	[−0.502, +0.582]
replacing the objective	pure ΔV	200	**−7.355**	[−7.875, −6.835]

Table 7: Everything else. Only the last row resolves, and it resolves against us. Every other interval contains zero at a minimum detectable effect of roughly 0.8 sets per pair, so these are "not shown to matter at this budget", not "shown not to matter".

precision had not saturated. Under v0.4's inference these screens said it plainly had: halving or doubling the draw budget around 160 changed nothing measurable, and the natural reading was that v0.3 had been buying accuracy rather than precision — more draws partially averaging out a *bias* — so that once the bias was gone, sampling noise was no longer binding.

That reading is refuted, by a pre-registered 6000-pair run reported in Section 14.3: going from 160 draws to 480 is worth +**0.340** sets per deal-pair, 95% CI [+0.243, +0.436], all six blocks positive, $I^2 = 0\%$. Precision had not saturated; the screens that said so were 200-pair cells resolving about ±0.53, which cannot distinguish +0.340 from zero and returned exactly the null they were bound to return. Their agreement with each other was not corroboration — underpowered cells agree on nothing in particular, and four of them agreeing on a null is what four underpowered cells do. The paragraph is kept in its original form above because the inference it drew was reasonable on the evidence available and wrong anyway, which is this paper's most frequently repeated lesson and is more useful stated against a concrete case than in the abstract.

The word was right and the rung was wrong, which is the more interesting correction. A second pre-registered run at 480 → 1440 draws returns +0.094 [−0.002, +0.191], and the contrast against rung 1's +0.340 is -0.245 ± 0.069 — so precision's returns *do* diminish, just not where the screens said. Saturation has not begun at 160 and has begun by 480. See Section 14.3.

One cell technically resolves, and should still be ignored. Lowering the claim bar from 0.97 to 0.90 gives +0.035 [+0.007, +0.063]: an interval excluding zero, and an effect of one fortieth of a set per deal-pair. After forty-odd cells at a nominal 95% level, an interval that narrowly excludes zero at a magnitude this small is not evidence of anything, and we report it rather than promoting it — particularly having already watched a larger apparent effect fail to replicate twice (Section 14.10). The neighbouring cell in the other direction (0.99) gives +0.010 [−0.010, +0.030], which is the shape of a policy that barely changes its decisions at all. This one has since been run properly, and Section 14.16 reports what an unselected 2000 pairs did to it.

The claim threshold and the tablebase are exact no-ops. The intervals [−0.010, +0.030] and [−0.005, +0.015] are not small effects; they are almost no divergence at all. Raising the claim bar from 0.97 to 0.99 changes essentially no decisions, reproducing v0.3's bimodality finding on a much better posterior. Disabling the exact endgame solver entirely changes essentially nothing, which agrees exactly with the absolute benchmark (Section 14.7), where it changed not one of 988 decisions.

Replacing the objective with the learned value is catastrophic. The half-suit value model is good — held-out log-loss 0.727 against 0.821 for the class prior and 0.743 for the best single feature, and a mean calibration bias of −0.006. That last figure flatters it, and the flattery is worth one sentence because "well-calibrated" is doing work in the argument below. Five of its ten deciles are miscalibrated at the 95% level: it is over-confident at both ends (decile 0 says 7.6% and delivers 0.5%; decile 9 says 92.7% and delivers 90.0%) and under-confident in the middle (deciles 4 and 5 both run about 3 points high). The mean bias is near zero because those errors cancel, not because each decile is right. Scoring asks by the expected change in it, in units of sets, loses 7.4 sets per deal-pair. Blending it in at low weight is harmless and does nothing. This is the sharpest confirmation of v0.3's central claim about this game that we have: the probability that an ask lands is not one term among several, it is the objective, and a decently-calibrated model of what a half-suit is worth is no substitute for it.

The eleven-term basis bought nothing. Neither the two terms motivated by the literature survey (exposure, signalling) nor the four motivated by inspection improved on v0.3's three. The signalling term is worth singling out because its sign was the interesting question: +0.25 (advertise a holding in a half-suit our team is winning) scored −0.300 and −0.25 (conceal it) scored +0.056, so if anything the concealment direction is favoured, but neither interval comes close to excluding zero.

14.10 The near-misses, and the one that resolved

Two cells in Table 7 had positive point estimates whose intervals only just included zero, which is exactly the situation in which a screening study manufactures a false positive if the best cell is promoted without re-measurement. We reran both at 500 pairs on fresh seeds — minimum detectable effect 0.48 rather than 0.77 — together with their combination:

candidate	screen (200 pairs)	retest (500 pairs)	retest 95% CI
blended value objective, 0.5	+0.410	−0.208	[−0.534, +0.118]
$\sqrt{\cdot}$ counting, $\gamma = 0.70$	+0.480	+0.212	[−0.092, +0.516]
both together	—	**+0.490**	[**+0.184**, **+0.796**]
replication, fresh seeds	—	+0.068	[−0.254, +0.390]
replication again	—	−0.066	[−0.395, +0.263]

When this was written the combination was the only cell in the study — out of more than twenty-five — whose interval excluded zero in the improvement direction, which is exactly why it looked like something. On a fresh set of 500 deals it returned +0.068, and on a second fresh set −0.066. It was noise.

That framing no longer holds, and the reason it stopped holding is the point. Three cells in this paper now exclude zero upward — the pre-registered lookahead at +0.104, posterior precision at +0.340 and at-ask depth at +0.102 — and every one of them was *designed* to be able to, at thirty times this cell's sample size with the analysis fixed in advance. The distinction that matters is not how many intervals clear zero, it is whether clearing zero was a possible outcome of a design or the best of twenty-five looks.

An earlier draft of that sentence said *four*, and counted the claim threshold's +0.035 among them. That was the error the sentence is about, committed inside the sentence about it: the claim-threshold figure is a screening cell selected out of 103 for excluding zero, which is the *other* category entirely. It is the eighth appearance of the pattern this paper keeps documenting, the fourth of them one level above where the error was last caught, and the only one caught by a *run* rather than by rereading — the confirmatory 2000 pairs of Section 14.16 put the effect at +0.002.

We report this at length because it is the single most useful methodological observation the study produced. Twenty-five screening cells at a nominal 95% level will produce roughly one apparent winner under a complete null, and this one arrived with a respectable-looking interval, a plausible story (two independently-motivated changes that stack, exactly as v0.3's two winning terms had) and a point estimate twice the sum of its parts — which should have been the tell. Promoting it would have shipped a change worth nothing and, worse, would have anchored the next version's baseline on it.

14.11 Belief-space search: four rounds to one answer

Section 12 describes the one search design the variance diagnosis does not rule out, and predicts, in Proposition 3, that a version of it without the quota coupling would collapse to greedy. Both halves were measured. Table 8 is the whole record, and it took 9,700 deal-pairs across four rounds — 800 in screens, 1500 in replications, 1400 decisive, 6000 in the pre-registered settling run — to answer a question a 200-pair cell appeared to answer at the start.

The answer is that the search helps, by about a tenth of a set per deal-pair. The section is long because the route to that number is the more useful part: the first cell overstated it by a factor of five, the next three rounds disagreed with each other, and the disagreement turned out to be a fact about selection rather than about the harness or the game.

Three runs of one configuration disagree. The screening cell, at 200 pairs, returned +0.570 and excluded zero. Following the replication policy of Section 14.10 we reran the identical configuration twice more at 500 pairs, changing nothing but the seed streams. The two retests returned -0.044, 95% CI $[-0.321, +0.233]$, and $+0.386$, 95% CI $[+0.086, +0.686]$. The second of those excludes zero. The first does not, and its interval excludes the value the second reports.

That is not a result about the lookahead. It is a result about the experiment. Three estimates of one quantity, differing only in seeds, are mutually inconsistent: Cochran's $Q = 7.03$ on 2 degrees of freedom, $p = 0.030$, with $I^2 = 71.5\%$ — so roughly seven-tenths of the spread between these runs is *not* the sampling variation their own intervals model.

The within-run intervals are not the problem. The per-pair standard deviation recovered from each cell separately is 3.21, 3.17 and 3.42 sets. Those agree with each other and with the 3.869 figure used throughout this paper, so each interval correctly describes the deals that run actually played. The natural inference is that the variance the intervals miss sits *between* runs; the DerSimonian–Laird estimate of such a component from these three cells is $\tau = 0.268$ sets per deal-pair. *We measured that inference directly and it is wrong* — see below — but it is worth following through first, because the two pooled answers it produces are the reason the question mattered enough to measure.

Pooling all 1200 pairs on that assumption gives two answers, and only one of them is usable:

pooling	estimate	
fixed effect	$+0.226$ $[+0.041, +0.411]$	excludes zero
random effects	$+0.278$ $[-0.083, +0.638]$	includes zero

cell	pairs	set diff	95% CI	MDE
depth 3, $w = 0.25$	200	$+$**0.570**	$[+0.125, +1.015]$	0.75
depth 2, $w = 0.25$	200	$+0.345$	$[-0.083, +0.773]$	0.75
depth 3, $w = 0.60$	200	-0.080	$[-0.583, +0.423]$	0.75
depth 3, *no quota coupling*	200	$+0.070$	$[-0.369, +0.509]$	0.75
depth 3, $w = 0.25$ — retest	500	-0.044	$[-0.321, +0.233]$	0.48
— retest again	500	$+$**0.386**	$[+0.086, +0.686]$	0.48
no coupling — retest	500	$+0.112$	$[-0.164, +0.388]$	0.48
depth 3, $w = 0.25$ — decisive A	700	$+$**0.307**	$[+0.063, +0.551]$	0.40
— decisive B	700	$+0.001$	$[-0.240, +0.243]$	0.40
depth 3, $w = 0.25$ — settle, block 1	1000	$+0.179$	$[-0.034, +0.392]$	0.34
— block 2	1000	-0.009	$[-0.217, +0.199]$	0.34
— block 3	1000	$+0.118$	$[-0.089, +0.325]$	0.34
— block 4	1000	$+$**0.241**	$[+0.040, +0.442]$	0.34
— block 5	1000	$+0.064$	$[-0.147, +0.275]$	0.34
— block 6	1000	$+0.028$	$[-0.175, +0.231]$	0.34

Table 8: Belief-space lookahead against the v0.4 champion, which is the identical policy with the bonus disabled. MDE is the smallest effect that cell could have resolved at 80% power, so a null row says the run did not detect an effect of that size, not that there is none. This table uses the per-pair standard deviation of 3.796 measured over the 4800 A/A pairs reported later in this section, rather than the 3.869 used elsewhere in this paper: the settling run was powered against 3.796, and a table reporting a run against a different resolution from the one it was designed for would misstate what it could see. The two differ by under 2%.

A fixed-effect pool assumes the runs are three samples of one true value, which is precisely what Q appears to reject; the random-effects interval, which adds τ^2 to every cell, would then be the one reflecting where a fourth run might land, and it includes zero. Read that way the effect is unresolved and could be reported neither as working nor as a null — which is where we left it, and which is what made the size of τ worth establishing rather than assuming.

The three cells prompted a question about the harness, and the answer is that the harness is fine. The reading above — that a duplicate-deal cell has a variance component its own interval does not model — would, if true, make the minimum-detectable-effect table of Section 13 optimistic for every single-cell result in this paper. That is too consequential to leave inferred from three points, so we measured it.

The instrument is an **A/A**: one policy against a copy of itself, where the true differential is exactly 0. That required a change to the harness, and the reason is worth recording, because it means this question could not previously have been asked. Under the default seat-based seeding an A/A plays a bit-identical game in both halves of a duplicate pair, so its differential is $(a - b) + (b - a) = 0$ for every deal — a point mass, not a small effect correctly measured as zero. The harness could not observe its own null. Giving each side an independent randomness stream (`independent_seeds`, off by default so the archive stays comparable) makes the null a real random variable.

Twenty-four blocks of 200 pairs, each with its own deal seeds and its own agent seeds — differing in exactly the way two separate runs differ:

nominal 95% intervals covering the truth	$23/24 = 95.8\%$ $[79.8, 99.3]$
Cochran's Q	18.41 on 23 df, $p = 0.735$
I^2	0.0%
τ, between-run standard deviation	**0.0000**
grand mean of the 24 estimates	-0.0215 (truth: 0)
per-pair standard deviation, 4800 pairs	3.796 (this paper uses 3.869)

Coverage is nominal. The observed spread of the block estimates, 0.2454, is *smaller* than their own standard errors predict, 0.2701; there is no excess variance to attribute. A duplicate-deal interval means what it says.

So we retract the between-run variance. An earlier version of this section reported $\tau = 0.268$ sets per pair from the three cells above and argued that it made every small result in this paper more fragile than it looks. That was a Type I error. The $Q = 7.03$, $p = 0.030$ which produced it is exactly the one-in-thirty event, and we promoted it to a structural claim about the harness on the strength of three points — which is the error this paper devotes Section 14.10 and two of its four lessons to warning against, committed inside the section doing the warning. The minimum-detectable-effect table stands as written.

Which leaves the lookahead genuinely open, on better terms. If the cells are homogeneous — and a harness that produces no excess heterogeneity under the null is strong evidence that they are — then fixed-effect pooling is the valid analysis, and over all 1200 pairs the estimate is +0.226, 95% CI $[+0.041, +0.411]$.

We did not report that as a result. It excludes zero by 0.04; this configuration had already produced one screening cell that failed to replicate; and reaching for the pooling that happens to resolve, immediately after the pooling that happened not to, is the same error in the opposite direction. So we fixed a size in advance and ran it: two further cells of 700 pairs each, on fresh seeds, pre-registered as decisive.

They returned +0.307, 95% CI $[+0.063, +0.551]$, and +0.001, 95% CI $[-0.240, +0.243]$.

The verdict, on the test we committed to. Pooled, those two cells give +0.153, 95% CI $[-0.019, +0.324]$ — which *includes zero*. The experiment designed in advance to settle this does not settle it, and we report that rather than the alternative below, because a post-hoc pool does not get to overrule a pre-registered one.

What the fuller picture says, and why we still do not claim it. Dropping the original screening cell — it was *selected* for resolving, so including it in any pooled estimate is winner's curse — and pooling the four unselected cells over 2400 pairs gives +0.153, 95% CI $[+0.022, +0.285]$. The point estimate is identical to three decimal places across both analyses, which is fairly strong evidence that a small real effect exists, of order a seventh of a set per deal-pair. It is not evidence that we have demonstrated one.

Dropping that cell also disposes of the heterogeneity this section opened with: Q falls to $p = 0.063$ over the four unselected cells and $p = 0.081$ over the two decisive ones, neither significant. Combined with the A/A measurement above, the story is simply that one cell was selected for being large and the rest is sampling noise — no between-run variance, and nothing wrong with the harness.

And the sizing was wrong, by the error this section is about. We chose $n = 2600$ on the grounds that its minimum detectable effect of 0.21 would separate +0.23 from zero. But +0.23 was the pool *including* the selected cell — the inflated estimate. Against the unbiased +0.153, a correctly powered test needs roughly 5000 pairs. We committed the winner's curse in the power calculation for the experiment whose purpose was to correct for the winner's curse, which is the third time in this study the same error has appeared one level up from where it was last caught. It appears a fourth time in Section 14.8, inside the estimator built to measure the ask objective, where taking a maximum over eighteen noisy action values would have reported two thirds of a set of regret that does not exist. The pattern is worth naming: each occurrence is not a repetition of the last but a level above it, and none was caught by being more careful — each was caught by measuring the thing that had been reasoned about.

The pre-registered run, and the answer. So we wrote the analysis down before running it. `jobs/PREREGISTRATION_lookahead.md` fixes, in advance: six blocks of 1000 fresh pairs; sizing against the *unbiased* +0.153 rather than any figure containing the selected cell; a per-pair standard deviation of 3.796 from the A/A rather than the older 3.869; a fixed-effect pool of the six new blocks as the only thing that decides; no cell droppable for its result; and no further run added to chase significance. The last six rows of Table 8 are that run.

	estimate	
primary — the six new blocks, 6000 pairs	**+0.104** [+0.020, +0.189]	**excludes zero**
secondary — all ten unselected cells, 8400 pairs	+0.119 [+0.048, +0.190]	excludes zero

The effect is demonstrated. Three things belong beside that sentence.

The estimate has fallen at every stage — +0.570 at the screening cell, +0.153 pooled over the unselected cells, +0.104 now. That is the winner's-curse decay in its textbook shape, and the pre-registered figure is the one with no selection in it.

The run was sized against +0.153, whose 80% power at 6000 pairs needs a true effect of 0.137. Against a true +0.104 it was under-powered, and the lower bound of +0.020 says how narrowly it cleared. A replication would be worth more here than another decimal place.

And the heterogeneity that opened this section is gone: $Q = 4.08$ on 5 degrees of freedom, $p = 0.54$, $I^2 = 0\%$, $\tau = 0$. Six blocks of one configuration agree within their own intervals, which is what the A/A predicted and what the original three cells did not. The disagreement really was one selected outlier and sampling noise.

The implementation shipped disabled while the effect was undemonstrated. On the strength of the primary test that default can now change — for a gain of about a tenth of a set per deal-pair, against the cost measured in Section 12.3, which is a trade rather than a free win.

What survives regardless. Proposition 3 does not depend on any of this. A possession-chain search over a belief whose transition edits only the asked card's row is exactly greedy at every depth, so the obvious version of this idea is a no-op by construction rather than by measurement. That is a structural statement about why search keeps failing in this game, it removes a whole branch of the design space rather than a point in it, and it is the part of Section 12 we would keep. Its empirical shadow is the no-coupling cell, and the shadow is worth stating carefully, because the proposition entails greedy equivalence *at a node* and the step from there to a null set differential is a modelling inference rather than a theorem. Setting `couple=False` also does not restore the proposition's hypotheses outright — the transition still zeroes the asked row and decrements the target's count. What is demonstrated is narrower and still to the point. Instrumenting the search itself over 60,277 multi-branch nodes on champion trajectories (`scripts4/greedy_shadow.py`), the coupled arm departs from the highest-probability branch at 452 of them, 0.750%; the uncoupled arm departs at **1**, 0.0017% — a factor of 452. Its set differential is a null at both 200 and 500 pairs (+0.070 and +0.112, both intervals comfortably containing zero), which is consistent with that.

The single uncoupled departure is worth a sentence, because it is real — relative gap in q of 1.9×10^{-3}, three orders of magnitude above floating point — and Proposition 3 predicts exactly zero. It is not a contradiction. `couple=False` does not restore the proposition's hypotheses, as the module says: the transition still decrements the target's count, so a player reduced to zero cards drops out of `legal_asks` and the available branch *set* changes downstream. That is a channel the exchange argument does not cover. The empirical shadow is therefore a near-null with a mechanism rather than a null, and an earlier draft reporting it as "53,273 of 53,273" overstated it — those counts were in the paper and in no results file, which is how a number stays unchallenged. We would also withdraw a remark from an earlier draft that the coupling "changed no decision at all": over 1,110 champion-trajectory positions the coupled and uncoupled arms disagree on 0.54% of decisions, so observing no disagreement in a sample of 59 is the modal outcome rather than evidence of equality.

Does it still pay once precision is bought? The two demonstrated features of this version — the belief-space search and the larger posterior sampling budget — were each measured against the champion *alone*, so neither says anything about the other. They could add, or the sharper belief could already be earning what the search was earning. Pre-registered in `jobs/PREREGISTRATION_stack.md` and written at one of six blocks, which is the only moment at which writing such a thing proves anything:

	set diff	95% CI
lookahead on top of 480 draws, 6000 pairs	+0.072	$[-0.010, +0.154]$

Cochran's $Q = 1.94$ on 5 df ($p = 0.86$), $I^2 = 0\%$, $\tau = 0$ — the six blocks agree completely.

The interval contains zero, and the pre-registration fixed in advance how that must be read. At 6000 pairs this run has 68% power against the +0.104 it was sized against, so failing to exclude zero is a *failure to resolve an effect of the size assumed* and is not evidence that the search stops paying. The script that prints this verdict refuses to print the second sentence. It is a genuine limit of the design rather than a finding, and the honest summary is that we do not know.

What the run does bear on is additivity, and only weakly. The search on its own measured +0.104; on top of the sampler it measures +0.072, a difference of −0.032 with the two intervals overlapping. As far as this study can tell the two features add rather than overlap — with one point on each side of the comparison, which is exactly as weak a statement as it sounds. Neither `WEB_DRAWS` nor `V04_STRONGEST` moves on this result in either direction: both features are independently demonstrated, and this run is about what may be *claimed*, not about what ships.

`w_lookahead` still defaults to 0, and that is now a choice about cost rather than about evidence. The effect is demonstrated; it is worth about a tenth of a set per deal-pair for roughly 6.7 ms per decision (Section 12.3), so a caller under a latency budget should decide deliberately instead of inheriting. The engine exposes the enabled configuration under its own name rather than by moving the default, because every "vs champion" cell in this paper names the default spec and moving it would redefine what those runs measured.

14.12 Is the opponent model just fitting its own kind?

The model is a hypothesis about how opponents choose, and the opponents it was measured against in Table 3 are copies of the same policy. Two tests separate a real effect from self-play overfitting.

Against `memory`, a policy with no probabilistic component whatsoever, the model is still worth about +0.93 sets per pair. Against an opponent built specifically to violate its assumption it is still worth about +0.84. The second number deserves care: the exploiter is a weak policy in absolute terms (inverting the depth term costs it a great deal), so what the comparison shows is that the model is not *harmed* by an opponent designed to mislead it, not that it cannot be exploited by a stronger adversary. A genuinely coordinated best-responder is future work.

opponent	agent	pairs	set diff / pair	95% CI
`memory` (off-distribution)	$\gamma = 0.30$	150	+4.727	[+4.119, +5.335]
	$\gamma = 0$	150	+3.793	[+3.247, +4.340]
anti-model exploiter	$\gamma = 0.30$	200	+6.570	[+6.113, +7.027]
	$\gamma = 0$	200	+5.735	[+5.237, +6.233]

Table 9: The opponent model against opponents it was not shaped by. `memory` acts only on facts it can prove and has no probabilistic component at all. The anti-model exploiter is the v0.4 policy with the depth term *inverted* ($w_{\text{suit}} = -0.35$), so it deliberately asks in half-suits it is shallow in — the exact behaviour the model assumes does not happen.

rule	comparison	pairs	set diff	95% CI
declare at any time	v0.4 vs v0.3 champion	200	+2.175	[+1.647, +2.703]
declare at any time	γ vs no γ	200	+1.935	[+1.450, +2.420]
48-card deck	v0.4 vs v0.3 champion	200	+1.080	[+0.557, +1.603]
wrong split → opponents	v0.4 vs v0.3 champion	200	+1.690	[+1.191, +2.189]

Table 10: The result holds under every rule variant tested, including the declare-at-any-time rule as the game is usually written down. Note the 48-card row: with eight half-suits instead of nine, the margin roughly halves, which is what one would expect if the advantage is inferential — a smaller deck leaves less to infer.

14.13 Does it survive the rule variants?

The engine supports several rule options, and a result that only holds under one configuration is a weaker result. Table 10 repeats the headline comparison under three of them.

The declare-at-any-time rows required an engine change. v0.3's game loop only ever asked the player on move, so although the rule was accepted by the configuration it could not be exercised by any policy — and it matters most precisely where it is unreachable, for a seat that has run out of cards and therefore never gets a turn of its own while a teammate still has some. v0.4 polls off-turn seats for a claim they can make *by deduction alone*, which costs a belief update rather than a posterior per seat, and offers it identically to both generations so that the comparison stays a comparison of one thing.

14.14 Learning the ask objective

v0.3's stated top priority for this version was to learn the ask objective rather than hand-weight it. We did, by common-random-numbers paired rollout regression: 874 positions, 83,168 games played to the end, all pairs of candidate asks within a position regressed on the difference of their feature vectors, with standard errors clustered by position (the naive standard errors are on average 2.00× too small, since pairs from one position share candidates, worlds and rollout seeds).

The machinery works: common random numbers cut the variance of a pairwise comparison to 0.64 of independent worlds, bringing the noise-to-signal ratio from v0.3's 2.4 to 1.19; a permutation test over position-level sign flips gives $p = 0.003$ against a null R^2 of 0.005; and a torch MLP on the same paired target improves validation R^2 from 0.234 to 0.249, so the objective's *shape* is not the limitation either.

The learned weights lose. Played against the incumbent over 120 duplicate deal-pairs, the best of three variants scored −2.183 sets per pair [−2.746, −1.621].

The diagnosis is specific, and it is a calibration failure of the target rather than of the fit. Three of the eleven coefficients have answers already established by *play*: v0.3's 600-pair sweeps put `turn` at 0.60 and `scarce` at 0.20 and measured the pair as worth +1.41 sets per deal-pair.

The regression returns `turn` $= 0.157$ with a clustered standard error of 0.058 — so an effect the size play has already measured would sit ten standard errors from zero, and the dataset does not see it. More starkly, position-centred rollout value rises by only +0.101 sets across the entire range of $\mathbb{P}$(success), and an unpinned fit puts that coefficient at 0.063.

The likeliest mechanism is the continuation policy. A rollout has to be finished by a policy that can attach to a determinized mid-game position, and the belief tracker cannot: it is anchored on the initial deal and refuses. That leaves a public-information heuristic, which throws away most of the value of a marginal card, so a card won by a good ask is largely squandered before the game ends and the ask stops mattering to the final differential. The binding constraint is not precision — pairing already fixed that — it is that the continuation is too weak for the difference between two asks to survive to the end of the game.

This is the same shape as v0.3's search failure, one level up. There, the problem was that the evaluation target could not rank asks; here, a much more careful attempt to build a better target hits the same wall, and locates it in the rollout policy rather than in the statistics.

The mechanism was right and the wall was not there. Everything above stands except one clause, and that clause carried the whole line. "The belief tracker cannot" is false. `BeliefState` is anchored on the initial deal, but it does not have to be *handed* one: given the current hand and the real public history, `initial_hand()` back-computes a deal consistent with both, and the tracker attaches to that. The stored positions did not carry the public history — they recorded its *length* — so the strong policy had nothing to attach to, and a property of the dataset was read as a property of the engine.

Repeating the paper's own measurement with the full v0.4 policy finishing every rollout, over 110 positions and 3056 scored candidate asks: position-centred rollout value rises by $+0.681 \pm 0.142$ across the range of $\mathbb{P}$(success), against the +0.101 above.

Those two numbers must not be quoted against each other, and we nearly did. The +0.101 was measured over the LEARNING harvest, whose positions span the whole deal — median two half-suits resolved, only 29% with four or more — while this measurement harvests late positions, four or more by construction. Attributing the gap to the continuation, when the position distribution also differs, is a two-factor change read as one: the error this section is about, committed inside the correction to it.

The control is cheap and settles it. Running the public-information heuristic on the SAME 110 positions, the same determinized worlds, the same seeds and the same 3056 root asks gives a slope of $\mathbf{+0.040 \pm 0.045}$ — consistent with the published +0.101 and indistinguishable from zero. Because the two arms score identical asks at identical positions, $\mathbb{P}$(success) is bit-identical between them, and the difference of the two within-slopes is exactly the within-slope of the differenced outcome. That paired contrast is

$$+0.641 \pm 0.145 \quad \text{(4.4 standard errors, clustered by position).}$$

So the continuation is the cause, and the earlier comparison was right for a reason rather than by a coincidence of position mix.

One thing that contrast changes twice, and it had to be checked rather than argued. The engine arm finishes the game *and* is handed the real public log; the heuristic arm finishes the game *and* starts blind to every card the table has already watched change hands, because an empty log is the knowledge set `PublicInfoHeuristic` was defined and audited against and is what the original learning rollouts gave it. So the two arms differ in information as well as in policy, and reading the whole +0.641 as "the continuation policy" would be this section's own two-factor error committed one level further in — inside the correction to the correction.

A third arm settles it: the same weak heuristic, on the same 110 positions and 3056 root asks, with the public log seeded. It scores $\mathbf{+0.084 \pm 0.071}$, beside the unseeded heuristic's +0.040

and nowhere near the engine's +0.681. Because all three arms score identical asks at identical positions, the contrasts are exact within-slopes of differences on the same x and therefore add:

$$\underbrace{+0.597 \pm 0.145}_{\text{policy alone}} + \underbrace{+0.044 \pm 0.069}_{\text{the log alone}} = +0.641.$$

Handing the weak policy the whole public history moves the target by an amount this design cannot distinguish from zero. The asymmetry was real; it is not what the contrast measures.

This arm exists because of an audit note about something else entirely — that seeding the history also shortens the stall rule's effective window, since `stalled` scans the last 80 actions and the seeded prefix eats a mean of 10.2 of them. That particular asymmetry turns out to be inert: over 13,290 engine-continuation decisions the rule fired *zero* times, because the prefix eats ten actions and the engine finishes in twenty-six more, so nothing approaches eighty without a claim. It fires at 1.02% of the heuristic arm's decisions instead, which is the same fact as the 181 plies — but chasing it is what surfaced the information asymmetry, which was the one that mattered.

One caution on that pairing, because it briefly said the opposite. Averaging the per-position slope *differences* without weights gives $+0.020 \pm 0.246$, apparently a null. That is a different estimand, not a more careful version of the same one: an unweighted average is dominated by positions carrying almost no contrast in $\mathbb{P}(\text{success})$, whereas the reported slopes weight each position by its own sum of squared centred $\mathbb{P}(\text{success})$. One position with four scored asks contributes a slope difference of -21.7, because a slope fitted to four points is not an estimate of anything. The robust summaries agree with the weighted one: the *median* per-position difference is +0.299 and the engine's slope is the larger at 63% of positions. A pairing that changes the weighting is not a pairing.

We first computed that unweighted figure as $+0.194 \pm 0.160$, which is what a silent bug produced. Positions were being kept only when their clustered standard error came out positive — ordinary-looking defensive code, and wrong, because within a single cluster the CR0 numerator is $(x'e)^2$ and $x'e$ is the OLS normal equation, hence exactly zero in exact arithmetic. The filter kept the 97 of 110 positions where floating-point roundoff happened to leave a nonzero residual dot product, and among the 13 it discarded was the -21.7 outlier. A defensive filter on a quantity that is analytically zero selects on roundoff, and it flattered the comparison it was printed to stress-test. No single position carries it (the largest leave-one-out move is 0.06), and the run's first 40 positions, which had been measured on their own first, agree with the remaining 80 within 1.7 standard errors, so it is one slope rather than an average of two regimes. Per-position slopes nonetheless run from -6.4 to $+3.7$: it is that between-position spread, and not the number of asks scored, that sets the interval, which is why tripling the run barely narrowed it.

The mechanism is measurable too, and turns out to be almost embarrassingly concrete. Handed the same position, the same world and the same root ask, the heuristic needs 181 plies on average to reach a terminal state where the engine needs 26 — a factor of seven. The extra plies are exchanges that undo each other: nothing stops the heuristic trading a card back and forth until its stall rule fires, and every one of those plies is a chance for the root ask to stop mattering to the final differential. Neither policy comes near the 400-action cap (324 and 56 at worst), so no part of this is capped games being scored on partial boards.

So the diagnosis was correct about the mechanism — a weak continuation does flatten the target — and wrong about it being a wall. The target the regression could not learn from was flat because of what finished the game, and finishing the game properly un-flattens it by a factor of nearly seven. The learning line is re-opened rather than abandoned, at a cost: with six belief trackers in the loop a rollout takes 240–500 ms against 17–23 ms for the heuristic, roughly 20×.

Two further things had to be true and only one of them was obvious. The stored worlds must satisfy everything the public log implies — not merely preserve hand counts, which is all the public heuristic ever needed. A successful ask locates a card; a *failed* ask asserts, under the no-bluff rule, that the asker holds at least one card of that half-suit, and a count-preserving shuffle violates that constraint routinely. Worlds must therefore come from the belief sampler. And the failure had to stay loud: seeding a rollout with an empty history does not error, it anchors six trackers on a determinized mid-game hand as though it were the deal and returns a number computed from a game that could not have happened. That configuration is refused rather than run.

A second obstacle, which the continuation fix does not touch. With the target no longer flat, the obvious next step is to ask which of the eleven objective terms it responds to, and the answer is that this design cannot say. Fitted one at a time over the 3076 rows of that run — twenty more than the 3056 the slope above uses, because a slope needs two scored asks at a position and twenty rows sit alone at theirs — several features look more informative than $\mathbb{P}$(success) itself: `expose` $+1.807 \pm 0.359$, `deplete` $+1.387 \pm 0.224$, `certain` $+0.919 \pm 0.169$, against $\mathbb{P}$(success)'s $+0.681 \pm 0.142$. Fitted together, $\mathbb{P}$(success) turns *negative*, at -0.329 ± 0.264. Neither column should be read, and the diagnostic that says so belongs beside them: $\mathbb{P}$(success) carries a variance inflation factor of **13.5** here, meaning 92.6% of its within-position variation is reproduced by the other ten terms. It is not an independent regressor; it is computed from the same posterior the features are, and correlates with `deplete` at 0.854 and `certain` at 0.738. A coefficient that is not identified does not become identified by being reported, and the negative sign is an artefact of the basis rather than a finding about play.

This matters for what happens next. The diagnosis above was right — a weak continuation does flatten the target, and finishing rollouts properly un-flattens it sevenfold — but it was not the only thing standing in the way. Near collinearity of the feature basis is a *second* obstacle, independent of the first, and fixing the continuation does nothing about it: the design matrix is nearly singular whatever finishes the game, so the fitted weights are unstable however much signal the target carries. The whole eleven-term fit explains 11.7% of within-position variance, so there is signal to work with; what there is not is a way to attribute it to individual terms as they currently stand.

The two most-quoted figures in this paper were the two nothing could check. `check_paper_numbers.py` watches over a hundred figures against the results files that produce them. It could not watch the abstract's headline — the +1.85 margin over the previous champion — because that lives in the JSONL duel pool while every other watched figure lives in a JSON file, and the loader only read JSON. It could not watch Table 6 either, because that table's *rates* were computed in the LaTeX from counts the pipeline stored, so the number the paper prints was never a number the pipeline held. Both are correct; both sat outside every drift check for reasons as dull as a file format and a division done in the wrong place. This is the third time the same shape has appeared — after `settle_verdict.py` printing the lookahead headline and persisting nothing, and the deadlock quartet existing in no file at all — and each time the unwatched figure was among the most load-bearing in the document, for the same reason: the numbers a paper repeats most are the ones nobody re-derives.

The manifest cannot find its own gaps, so the check is now inverted as well. `scripts4/unwatched_claims.py` extracts every number the paper asserts in **bold** — its strongest claims, the ones a reader takes as measured — and requires each to be either pinned by the manifest or exempted with a *stated reason*; an exemption with no reason is indistinguishable from an oversight, which is the failure being guarded against. Running it the first time returned nineteen unpinned figures. Six were version numbers, row labels or a saturated column; five belonged to the two figures above; and the remaining *eight* were ordinary measurements nobody had thought to watch — among them the A/A per-pair standard deviation that every power statement in this paper divides by, the stalling fraction that motivates the progress-optimal criterion the absolute-strength table is scored against, and the champion posterior's own NLL and Brier scores.

A supervisor a bystander could switch off. The rollout pass is restarted by a shell supervisor whose liveness check was `pgrep -f "learn_ask_objective.py rollout -run v2"`. That matches the *full command line of every process*, so a one-line loop written to *watch* the pass — one whose own command line contained that string — was itself a match. The supervisor concluded the job was alive and stopped restarting it; the pass stayed dead for as long as the watcher ran, and was found only because a separate progress check reported zero workers. The widening script used the same pattern to choose a PID to `kill`.

This is the mirror image of a failure this paper already records, in which the same function returning `False` on any exception made a guard fail *open*. Both come from reading "this string appears somewhere in the process table" as "this job is running". The check now walks `/proc` and counts a process only when it is a Python interpreter whose own `argv` carries every token, excluding the caller and its ancestors; its test asserts both directions, including that `pgrep` really does match the bystander it is given.

One term of the eleven will not be fitted, and the reason is worth one sentence. The `claim` feature's formula was corrected after this harvest was taken — it multiplied by the product over all *six* cards of the half-suit including the one being asked for, so it scored exactly zero on provably certain steals, the asks it exists to reward. Its stored column therefore describes a feature the engine no longer computes. The loader refuses such a column by default, and the fit stage names it, zeroes it, and records that it was not fitted — because a ridge coefficient of exactly 0.0 for `claim` reads identically to "fitted and came out at zero", and those are different statements. The other ten terms are unaffected; re-harvesting to recover the eleventh is a day of rollouts for a term the champion already weights at zero, and the paragraph below is the reason that is not the bottleneck.

"So use a different basis" was reasoning, and the measurement says no. The natural conclusion from the paragraph above — and the one we wrote first — is that the re-run needs a different basis, or a penalty, or terms dropped, rather than more rollouts. It is cheap to check, because the completed run already holds every scored ask, so we checked it before spending anything. Eighteen candidate bases, each scored by *leave-one-position-out* cross-validated within-position R^2 (position-level folds, because asks inside a position share worlds, seeds and a value level):

basis	CV R^2	vs. the full basis
full, eleven terms and $\mathbb{P}$(success)	0.0959	—
strongest three, selected *within* each fold	**0.1030**	$+0.0070 \pm 0.0038$
drop terms until every VIF < 5	0.0964	$+0.0005 \pm 0.0017$
ridge, $\lambda = 1$	0.0968	$+0.0009 \pm 0.0015$
$\mathbb{P}$(success) alone	0.0430	-0.0530 ± 0.0234
principal components, $k = 4$	0.0150	-0.0809 ± 0.0306

The best of them beats the basis in use by $+0.0070 \pm 0.0038$, an interval that *contains zero*. Ranked by point estimate a three-term basis wins; measured against the spread across positions, nothing here is distinguishable from what the objective already uses, and picking one off this table would be selecting on noise.

That is the textbook consequence of collinearity and it is worth stating plainly, because it is easy to get backwards: near-collinearity makes individual *coefficients* unidentified without making *predictions* worse. A reduced basis is therefore not the fix, and neither is a penalty — ridge does nothing at any λ that does not also destroy the fit. Two smaller things fall out of the same table, and the first has to be stated carefully because it reads, at a glance, like the opposite of this paper's central advice.

Dropping terms by variance inflation removes $\mathbb{P}$(success) *first*, and costs nothing measurable ($+0.0005 \pm 0.0017$). That is **redundancy, not unimportance**, and the two are easy to confuse. The variance inflation factor of 13.5 says precisely that 92.6% of $\mathbb{P}$(success)'s within-position variation is already reproduced by the other ten terms — which is unsurprising, since all eleven are computed from the same posterior over the same worlds. A regression cannot miss a quantity its other regressors have already encoded, and it does not follow that the quantity does not matter. Section 20's finding that the probability the ask lands *is* the objective rests on *play* — scoring asks by anything else loses by -7.4 sets per deal-pair — and nothing in this table bears on it. What the table says is narrower: given ten features already carrying that information, an eleventh copy of it adds nothing.

And every principal-component basis is far worse, so the predictive directions in this design are not its high-variance directions — rotating to the biggest components discards exactly the signal.

We record this as the seventh appearance of the pattern, and the shortest-lived: the sentence it refutes was written earlier the same day, in the paragraph immediately above, by the same argument the paper keeps warning about. A plausible mechanism that generates no experiment is not a finding, even when the mechanism is real — and here the mechanism *is* real. The variance inflation factor of 13.5 stands. What does not stand is the inference we drew from it about what to do next.

The first version of this comparison also got its own selection wrong, which is worth one sentence because it is the same error at a third level. It ranked terms by $|\beta \cdot \mathrm{sd}|$ on a fit to *all* the data and then cross-validated the winner, so the selection had already seen every held-out position. Selecting inside each fold instead is what the table above reports.

Collinearity is not a licence to explain away every disagreement, and it does not cover the one that started this section. `turn` is the term play has measured most directly, and it is precisely the term collinearity cannot be blamed for: its variance inflation factor is 1.4, and it varies between candidate asks at 111 of the 113 positions the feature fit retains (the slope's 110, plus three the within-position fit keeps and the slope does not). It comes back at $+0.010 \pm 0.154$ all the same, where duplicate-deal play puts it at 0.60. That gap is real but not yet a finding in either direction: these positions are late by construction (four or more half-suits resolved), and holding the turn may simply be worth less when most of the board is already decided, so a coefficient near zero here is not in itself evidence against a weight of 0.60 across a whole game. What can be said is that the explanation is not the one immediately to hand. Reaching for the nearest available mechanism is how this paper's recurring error starts, and it would have started again here.

We record this as the fifth appearance of the pattern this paper keeps documenting, and the first in which reasoning about a quantity did not merely inflate it but retired an entire research direction. The diagnosis was well-argued, specific, mechanistic, and supported by a number. It was also never tested, because testing it required exactly the thing it asserted was impossible.

And the weights were played, which settles the line. Everything above is estimation. `jobs/PREREGISTRATION_learned_weights.md` fixes, before the fit was run and before any weight vector had been looked at, that the *validation duel is the test*: two blocks of 1000 fresh pairs, the pinned-p vector only, the shipped champion as the baseline rather than the weaker `fishbot4()` defaults v0.4 validated against, and a fixed-effect pool as the only thing that decides.

	estimate	
v0.4's attempt, 120 pairs *(different design)*	-2.183	best of three variants
this run, 2000 pairs	$-\mathbf{0.745}$ $[-0.914, -0.576]$	**entirely below zero**

The learned weights lose, and this is the stronger negative. The two blocks agree exactly ($I^2 = 0\%$, Q at $p = 0.33$), the realised per-pair standard deviation of 3.857 matches the

3.8 the design assumed, and the arms diverge on 87.6% of pairs — an ask-weight change alters decisions constantly, which is why the A/A figure is the right one here and why this run resolves what it was built to resolve.

Read against the diagnosis it was built to test, the result is unambiguous. v0.4 blamed the weak rollout continuation; that diagnosis was measured, found right about the mechanism, and the continuation was replaced — the target's slope on $\mathbb{P}$(success) went from +0.101 to +0.681 on late positions and +0.355 across the whole harvest. *And the weights still lose.* So the continuation was a real obstacle and not the binding one. The binding one is the basis, which this section measured before the duel was played and which the table above says a smaller basis does not fix.

What the vector is not. The fit moves several terms with permutation $p \leq 0.023$ — `expose` to +0.825 from an incumbent of zero, `certain` to −0.502, `signal` to −0.249 — and the individual signs are not findings. `certain` correlates 0.738 with $\mathbb{P}$(success), which carries weight 1.0 by convention, and the within-position feature fit reported earlier in this section gives the same term the *opposite* sign on a different population (`results/target_feature_fit.json`, +0.649 against this −0.502). That is the collinearity this section already documented for $\mathbb{P}$(success)'s own coefficient, arriving a second time.

The single most informative row is `turn`: incumbent +0.600, learned +0.108. The regression cuts by more than 80% the weight on the term duplicate-deal play values most highly and the term collinearity cannot be blamed for, and then loses by three quarters of a set per pair. Whatever the regression is fitting, it is not what wins games.

One term of the eleven carries 0.0 by construction rather than by measurement: `claim`'s stored column predates the correction to its formula, so it was named, zeroed and excluded from the fit at every stage. The incumbent weights it at zero too, so the duel is unaffected — but the fit is a ten-term fit and is reported as one.

A cut-off resolved before it could be chosen. The pre-registration fixed the fit at "every position the pass has completed at the moment the duel queue drains". The queue drained at 743 of 1023 and the pass did not stop there, because the supervisor that widens it exists precisely to make draining the queue speed the pass up. Worse, that timing had been moved earlier the same session by a repair to the supervisor's liveness check — so the trigger named a number we had influenced. The amendment resolving it to the full 1023 was written at 919, before any fit had been run, and committed to computing the 743 fit anyway as a robustness check. That check came back with *no sign flips*, a largest single move of +0.135 and total L_1 movement of 0.394 across eleven terms: the vector is stable, and the ambiguity turned out not to matter. Learning that afterwards is the only order in which it is worth anything.

14.15 The split posterior is badly under-confident

Section 11 sets the claim threshold at 0.97 and argues for it from a specific mechanism: while our team genuinely holds all six cards of a half-suit an opponent cannot take it — a claim by a team that does not hold all six *gives* the set away — so the only cost of waiting is running out of opportunity. Waiting is close to free. v0.3 measured the consequence and could not explain it: thresholds from 0.85 to 0.999 play identically, and over 150 duplicate deals 0.97 and 0.999 never once diverged.

Scoring the posterior against ground truth explains it. Over 40 games we recorded, at every position where a half-suit was in fact entirely within one team, the posterior probability the engine assigned to its own MAP split and whether that split was the true one — 2719 decisions on 359 distinct half-suits, ground truth used for analysis only:

posterior says	decisions	actually the true split
$[0.000, 0.500)$	2062	66.5%
$[0.500, 0.700)$	203	93.6%
$[0.700, 0.850)$	75	97.3%
$[0.850, 0.970)$	24	100.0%
$[0.970, 1.000]$	355	100.0%

The interior cut points used to be 0.700/0.825/0.970, where 0.825 was 1 minus the null rate on stuck half-suits — a decision bar imported from Section 14.17. Re-measuring that null rate therefore moved two rows of a table about the *sampler*, for a reason having nothing to do with the sampler. How well a stated probability is calibrated is a property of the posterior alone, so the bins are now fixed, the bar is reported separately where it belongs, and the whole table is stored in `results/stuck_claim_value.json` rather than printed — which is the failure `scripts4/check_verdicts.py` exists to catch, found here in a table rather than a verdict.

Two things follow. The distribution is **bimodal** — 76% of decisions sit below 0.5 and 13% above 0.97, with almost nothing between — which is why the threshold is insensitive across the whole range v0.3 swept. There is nothing there to be sensitive to. That is the explanation the sweep lacked, and it is a fact about the posterior rather than about the claim rule.

And the posterior is **badly under-confident**. It says "below one half" and is right two thirds of the time; it says "between a half and 0.7" and is right 93.6% of the time. The claim threshold is the one place in this engine where a split probability is consumed as a *number* rather than as a ranking, and a threshold set on a miscalibrated number does not mean what it says. This is a finding about the sampler, not about the claim policy, and it is where we would look next.

What we did not compute, and why. The obvious next step is to price the change: declaring on splits the engine currently waits on, against waiting and accepting the measured 23.4% null rate on stuck half-suits, would beat waiting wherever the MAP is right more than 76.6% of the time — which the table says it is, everywhere above 0.5. Carried through, that arithmetic returns a number of the size this paper calls demonstrated.

It is not reported, because the two numbers are measured over different populations. The 23.4% is the null rate for half-suits a team is *stuck* on in Section 14.17's sense: it can *prove* it holds all six and still cannot place the split. The population in the table above is every half-suit that *is* entirely within one team, provable or not — 8.97 per game out of nine, so very nearly all of them. Subtracting one from the other is a difference between two different things, and it produces a publishable-looking number by doing so. This is the same error as quoting the learning harvest's +0.101 against a late-position +0.681 (Section 14.14), and it is recorded here because we made it, saw the size of the answer, and went looking for why it was too good. Pricing this properly needs the null rate re-measured over this population, and until then there is no estimate and no queued run.

14.16 The claim threshold: the one screen that resolved, re-run

Section 14.9 reports a cell that lowers the claim bar from 0.97 to 0.90 and returns +0.035 $[+0.007, +0.063]$ over 400 pairs. It is the only small, cheap, unexplained positive in a table the paper describes as mostly nulls, and it contradicts a documented conclusion: `fish4/claim4.py` argues that waiting to claim is close to free, and reports that over 150 duplicate deals the 0.97 and 0.999 policies never once diverged.

That cell was selected out of 103 for excluding zero, so it is inadmissible as evidence and `jobs/SCREENING_DISCIPLINE.md` forbids it from sizing its own confirmation. `jobs/PREREGISTRATION_claim_thr` fixes, in advance: two blocks of 1000 fresh pairs; a minimum interesting effect of +0.02, deliberately an order of magnitude below the +0.15 used elsewhere, because this change is one constant and costs nothing to run; a fixed-effect pool of the two blocks as the only thing that decides; and no block droppable for its result.

	estimate	
screening cell, 400 pairs *(selected)*	+0.035 [+0.007, +0.063]	excludes zero
primary — two new blocks, 2000 pairs	+**0.002** [−0.013, +0.017]	**includes zero**

The effect does not survive. The confirmatory estimate is a twentieth of the screen's, and the decay of -0.033 ± 0.016 is itself 2.1 standard errors — selection inflation measured rather than inferred. The two blocks agree ($Q = 1.35$ on 1 df, $p = 0.25$).

This is a null with a mechanism. A behavioural measurement taken before the run, over 998 decisions carrying a claim candidate, found that raising the bar from 0.97 to 0.90 admits *one extra claim in 998*. The run then diverged on 2.6% of pairs. Two arms that make one different decision in a thousand and finish 97% of deals identically are not a null from thin data; they are the same policy, and `claim4.py`'s argument survives its one piece of counter-evidence.

The design missed its own power target, and says so. The pre-registration derived a per-pair standard deviation of 0.286 from the screening cell's recorded interval, giving a minimum detectable effect of 0.018 — just under the +0.02 it was built to detect. The run's realised standard deviation is 0.346, 21% higher, so the realised MDE is 0.022 and sits *above* the threshold. An effect of exactly +0.02 would have been missed more often than the design intended, and the null is reported against 0.022 rather than against the number that was planned. The interval-derived standard deviation, which reproduces a cell's measured value to within 1% everywhere else in this study, is the one input that failed here, and it failed in the direction that flatters the design.

A prediction recorded in advance, and scored. The divergence model of Section 14.9 was deliberately *not* used to size this run, because every cell it was fitted on diverges on 23–88% of pairs and a rarely-binding threshold does not. The pre-registration records what the model would have implied and predicts it will be wrong in a stated direction: at this per-pair standard deviation the model implies a divergence share of 0.8%, about sixteen pairs in 2000, and the prediction is that the true share is substantially higher *because* a threshold change alters one decision rather than the whole line of play, so its conditional differences must be smaller than the model's constant 3.88. Both halves hold: the measured share is 2.6%, and the measured conditional standard deviation is 2.17.

That divergence share is a ninth of the low end of the range the model was fitted on, which makes this the most extreme out-of-sample point available for the drift correction of Section 14.9 — and the correction was fitted before this run existed. The flat conditional term of 3.859 overstates the measured 2.166 by 78%. The drift-corrected form $2.53 + 1.66\,s$ predicts 2.576, an overstatement of 19%: still wrong, four times less wrong, and wrong in the same direction the correction was built to fix. A model extrapolated an order of magnitude past its data has no business being right, and it is not; what is worth recording is that the term added to it keeps earning its place where it cannot have been tuned.

14.17 Perpetual states, measured

Two hundred games per column, with ground truth used for analysis only.

Repetition is universal, not exotic. A position repeated in *every one* of 200 games, and some position recurred eight times in a single game. v0.3 established cyclicity by search, as a possibility; it is in fact a routine feature of ordinary play.

	normal	no progress rule	signalling on
actions per game	104.4	104.4	104.8
games hitting the 4,000-action cap	**0**	**0**	**0**
games in which a position repeated	**200**	**200**	**200**
most repeats of a single position	8	8	8
games reaching a *dead* position	**200**	**200**	**200**
share of plies fully dead	4.24%	4.24%	4.56%
actions played after going dead	4.4	4.4	4.8
games where a team held an unplaceable set	110	110	116
dead half-suits at peak	3.13	3.13	3.15
nulls per game	0.275	0.275	**0.220**

Table 11: Repetition and dead positions in 200 games of champion self-play.

Every game dies before it ends. All 200 games reached a position in which no ask anywhere on the board could succeed. It happens late — about 4.4 actions from the end, roughly 4.2% of plies — but it is universal. The last phase of a game of Fish is not a contest at all; it is a bookkeeping exercise in which the sets are already decided and only the declarations remain.

Termination is not the progress rule's doing. With the agents' stalling rule fully disabled, not one game in 200 hit the action cap, and the mean game length was identical to four significant figures. So although the rules genuinely permit non-termination, and although positions genuinely repeat, in practice a strong policy walks out of every cycle on its own. The convention is insurance, not the mechanism. Consistent with this, widening the stall window from 80 to 200 actions produced 367 exact ties in 400 duplicate deals: the rule almost never fires.

The deadlock is real, and it is expensive. In 110 of 200 games (55%) a team reached a position where it could prove it held an entire half-suit and still could not place the split. Over the 1800 half-suits of those 200 games, 171 (9.5%) reach that state, and they are nulled **23.4%** of the time (± 3.2) against **0.92%** (± 0.24) for half-suits that never get stuck — a factor of 25 — so they account for **73% of all nulls** while being under a tenth of half-suits.

Four figures in this paragraph were wrong in every draft before this one, and the way they were wrong is the reason the paragraph now names its denominators. They read 17.5%, 2.8%, a factor of 6.3, and 27% of all nulls. Those numbers appear in no results file: `perpetual_study.py` tracked which *teams* were stuck and recorded nothing per half-suit, so nothing in the pipeline could have checked them and nothing did. They were carried from draft to draft as though measured. The direction survives re-measurement — it survives it more strongly, since the true ratio is four times the one claimed — but 27% against a measured 73% is not a rounding difference, and one of these numbers had already been used downstream as a threshold (Section 14.15). Every figure here now comes from `results/perpetual_study.json` and is checked by `scripts4/check_paper_numbers.py` against it.

The free-signalling escape works, and does not pay. Turning on the protocol of Section 16 cuts nulls from 0.275 to 0.220 per game, a 20% reduction, which confirms the mechanism does what the theory says. It buys no wins: against the identical policy with signalling off, over 500 duplicate deals, $+0.002$ sets per pair $[-0.086, +0.090]$; restricted to strictly dead positions, $+0.012$ $[-0.000, +0.024]$. Those intervals are far tighter than the usual ± 0.48 because the change is a near-no-op — it alters so few decisions that most pairs come out identical.

Two things are going on, and both are worth stating. The strictly-free version of the protocol applies only when the *whole board* is dead, which the seat can prove on about 0.3% of plies — too rare to matter. The relaxed version fires more often but is no longer free: it spends a turn to send one bit, and the tempo it concedes cancels the sets it saves. So the escape is real theory with a null in practice, and the honest summary is that Fish's stalemate is a genuine structural feature of the game that a strong engine is already handling adequately by other means.

14.18 Game statistics, and a statistic that stops working

Homogeneous tables of each champion, 100 games each, on the measures v0.3 used to characterise skill:

	v0.3 champion	v0.4 champion
cards gained per possession	1.419	1.295
possessions gaining nothing	44.5%	48.8%
failed-ask share	40.9%	43.2%
asks per game	100.3	96.5
ask accuracy, thirds of the game	54.1 / 59.7 / 64.2%	52.2 / 58.0 / 60.8%
null rate (natural half-suits)	4.4%	3.1%
claim delay, median / p90 (actions)	0 / 58	0 / 49
first-move advantage	$+0.23$ $[-0.27, +0.73]$	$+0.32$ $[-0.22, +0.86]$

Three of these reproduce v0.3 and one contradicts it.

Reproduced: there is still no measurable first-move or seat advantage, both intervals comfortably containing zero with the starting seat rotated across deals; the synthetic "eights and jokers" half-suit still behaves like any other (mean resolution rank 4.08 against 3.99, null rate 3.0% against 3.1%); and ask accuracy still rises from the opening into the later game, so the opening remains the least-informed phase.

Contradicted, and worth stating carefully: v0.3 named cards-per-possession "the single best summary statistic of skill", on the evidence that it doubled from the `memory` tier to the `probabilistic` tier. The stronger engine here scores *lower* on it. The mechanism is visible in the other rows: v0.4 nulls fewer sets (3.1% against 4.4%) and claims sooner (p90 49 against 58), so half-suits resolve faster and the game contains fewer, harder asks — 96.5 per game against 100.3. Cards-per-possession compares two tables on how easy their asks were, and how easy the asks are depends on how quickly the claiming policy is removing the resolved material. Within a tier that shares a claim policy it tracks skill; across engines whose claiming differs, it does not. A statistic that inverts when the thing it is measuring improves is not a measure of skill, and we would retract the v0.3 claim rather than qualify it.

15 The engine as an instrument

A strong policy that can only be measured in aggregate is a weak research instrument, and an unusable one. The three things a chess engine gives a user all exist here; they just mean something different in a game whose hidden state is the entire problem.

Evaluation, in sets. The banked differential plus, for every unresolved half-suit, the learned probability that our team scores it minus the probability that theirs does (11). So +1.4 reads as "expect to finish 1.4 sets ahead of the current score". Because the underlying model is fitted to outcomes and checked for calibration — held-out log-loss 0.727 against 0.821 for the class prior, mean bias -0.006, reliable in every decile — the number means what it says. This is the one component that is a good *predictor* and a disastrous *objective* (Section 14.9), which is itself worth knowing: an evaluation function and an objective function are not the same object.

A ranked move list with attribution. Every legal ask, with the probability it lands, the evaluation on each branch, and a per-term breakdown of the policy's score. The breakdown matters for a research instrument rather than a toy: when a human disagrees with the engine, the disagreement can be attributed to a named term instead of left as an oracle's verdict.

A principal variation. In chess the PV is the expected line. The Fish analogue is the *possession*: the chain of asks the engine would make if each landed, since a turn in Fish is a run of successful asks. It is computed by pinning each card as it is won and re-ranking, so it is the engine's actual plan rather than a static list.

Two things a chess engine never has to show, and this one must. The **card table** is the posterior over who holds every unresolved card — where the strength of this version actually lives, so it is displayed rather than hidden. The **declaration table** gives, per live half-suit, the probability our team holds all six, the most likely split, and the exact probability that split is right, which is the quantity the claim decision turns on.

The information boundary survives the user interface. Every panel is built from the seat's own `Observation`; the posterior shown is an inference from the public record, not a look behind it. The only exception is the reveal after the last card is public anyway.

Mixed tables. The same server seats any mixture of people and engines at one table, which is what makes the engine usable as an instrument on human play rather than only on self-play. The design point worth recording is that this is not a cosmetic choice: teams in Fish are the alternating seats $\{0, 2, 4\}$ against $\{1, 3, 5\}$, so a request as ordinary as "three of us against three bots" is a constraint on *which* seats the people take, not on how many. A table therefore takes a seat *arrangement* rather than a bot count — one that fills a whole side before crossing, and one that interleaves — and the 3-versus-3 configuration that a group of friends actually wants is the former at $n = 3$. Engine seats move on a clock rather than instantly, because a decision costs about 4 ms and an unpaced possession appears to a human as a single discontinuous jump. Pacing alone is not enough: a longer gap is only a longer gap unless there is something to read in it, so the table states each move in words together with what it *proved*. That second part is free, and is the one place where the no-bluff rule pays the observer directly: an ask is a truthful statement about the asker's hand as well as a question about the target's, so a failed ask establishes that the target does not hold the card, that the asker does not either, and that the asker holds some other card of that half-suit. The default gap is 12 s and is adjustable in play. Freezing the table turns the same control into a step-through — one engine move per request, still frozen afterwards — which is what makes an engine's whole possession readable, since a possession is a run of successful asks and it is the run, not any single ask, that carries the plan. Each seat is served only its own `Observation`, so the information boundary of Appendix C is enforced by the transport, not merely respected by the client.

Analysis costs 130–270 ms, which is interactive, and the same objects drive the terminal client, the browser client and the batch experiments, so what a user sees is what the experiments measured.

16 Perpetual states: does Fish have a stalemate?

Chess has stalemate and threefold repetition. The natural question is whether Literature has an analogue — a position the game cannot get out of — and the answer comes in three layers, of which only the third is interesting.

16.1 Layer 1: the state graph is cyclic

Nothing in the rules forces progress. Two opponents can trade a card back and forth and return to a bit-identical position, so positions can repeat. This is a property of the rules, not of any implementation. It follows immediately that *termination is a property of the players*: any claim that "every game of Fish ends" describes a convention rather than the game. Our policies carry an explicit progress rule and our harness a cap, and Section 14.17 measures how much of "games end" those conventions are actually doing.

16.2 Layer 2: under perfect information it does not matter

Cycles exist, but the exact solver of Section 9 shows that no *optimal* line needs one: with all cards visible, optimal play always terminates. Cyclicity on its own is therefore not a strategic phenomenon. The reason is the closed form (8) — footholds only shrink, so the mover simply collects everything reachable and the position marches forward.

16.3 Layer 3: dead positions

This layer has no chess counterpart, and it is the one worth having.

Proposition 4 (Dead half-suits)**.** *Call a live half-suit* dead *when all six of its cards sit with one team. Then no ask in that half-suit can ever succeed.*

Proof. A legal ask in half-suit H requires the asker to hold a card of H. If one team holds all six, no member of the other team holds any card of H, so no opponent of the holding team may ask in it at all. And a member of the holding team may only ask an *opponent*, who by hypothesis holds no card of H, so such an ask fails. No card of H therefore ever changes hands again. □

Call a position *dead* when every live half-suit is dead. By Proposition 4 the material is then frozen for the remainder of the game: every legal ask is guaranteed to fail, the turn merely circulates, and the only state changes left are declarations. This is as close to a stalemate as Fish gets — and, unlike a chess stalemate, it is not a draw. The sets are already decided; they have simply not been claimed.

16.4 The apparent deadlock, and why it is not one

A dead position looks like a genuine trap for a team that holds a half-suit but cannot place the split among its own members. No card will move again, so — the argument runs — no new information can arrive, so the team must either gamble on a declaration or stall forever.

That argument is wrong, and the way it is wrong is a strategy.

Information does not only arrive when cards move. Under the no-bluff rule a player may not ask for a card they hold, so **asking for a card proves publicly that you do not hold it**. In a dead position that ask is guaranteed to fail, which means it costs no material whatsoever: there is no card to lose, and the turn it concedes is worthless because no opponent can win a card either. The ask is a *free one-bit broadcast to your partners*.

A team stuck in a dead position can therefore talk its way out. Each member asks for cards of the contested half-suit that they do not hold; after a few such asks the split is common knowledge within the team and the declaration is safe. Literature's only communication channel is normally costly and read by the opposition; in a dead position it becomes free, and the opposition can do nothing with what it hears, because the half-suit is already beyond their reach.

Remark 5 (A definition that made the phenomenon invisible)**.** *Our first implementation defined "our team provably holds this half-suit" as "every card's holder is pinned and all six are teammates". That makes "we hold it but cannot place the split" a contradiction in terms, since pinning the* *holders* is *placing the split, and the condition never fired once in 1,047 measured plies. The correct condition is weaker and is the whole point: every* possible *holder of every card is a teammate, which does not require knowing which teammate. Under that definition the situation occurs on 2.6% of plies.*

17 The price of a turn, and why it is conditional

Every result in this paper is denominated in one unit: the information exchange rate of about 0.45 sets per hidden card revealed. Half the ask objective prices *tempo* — turn_risk, depth, the retained turn a hit buys — and none of those weights was ever fitted against a measured scale, because none existed. This section supplies one, and the answer is not a constant.

17.1 Measuring it

The design is a duplicate deal with one intervention. Both arms play the identical deck with identical agent seeds; at one pre-chosen decision of one pre-chosen seat the treatment substitutes an ask that provably cannot land, taken from the true state so the donation is certain rather than a modelled mistake, and both arms then play on honestly. The difference in the donating team's differential is the price of exactly one turn. Substituting the agent's *own* action instead gives exactly 0.000 on every deal, which is the control that makes the number mean anything.

Two runs of 1500 pairs each put the price at +0.271, 95% CI $[+0.092, +0.450]$ when the donation falls at decisions 0–4, and +0.497, 95% CI $[+0.324, +0.671]$ at decisions 14–21. Tempo value rises through a game.

17.2 It depends entirely on whether you have an ask worth making

Bucketing 1464 donations by p_{best}, the success probability of the ask the seat would otherwise have made:

p_{best}	pairs	price	se
$[0.00, 0.25)$	374	−0.043	0.169
$[0.25, 0.50)$	511	+0.004	0.143
$[0.50, 0.75)$	179	**+0.508**	0.258
$[0.75, 1.01)$	400	**+0.415**	0.164

Below $p_{\text{best}} = 0.5$ a turn is worth nothing; above it, about +0.45 sets. **A turn is worth nothing unless you have something to do with it.** Tempo in Fish is not a resource in itself.

This is not the timing effect in disguise, and the direction rules that out: a late turn prices *higher* than an early one, so if low-p_{best} positions were merely late positions they would be worth more rather than nothing.

17.3 What it explains, and what it cost to find out

The measurement began as a critique of the engine. On 1.5% of decisions (229 of 15,542) the champion plays an ask that cannot land while one with median success probability 0.385 was available — it surrenders the turn for certain. That looked like a defect, and `avoid_doomed_asks`, which restricts the choice to asks that can land and ranks them by the same objective, was pre-registered against it with a predicted +0.05 to +0.15.

Over 8000 duplicate deal-pairs it scored +0.017, 95% CI $[-0.024, +0.059]$: at most +0.06 with 95% confidence, under a fifth of the largest gain this project has demonstrated, and plausibly zero.

Reconciling that with the turn price took four rounds, and three of the four found errors in the bridge rather than in the game. Treating the measured price as the value of a turn understated tempo by a factor of 1.7, since the donation displaces the agent's normal action, which itself retains the turn only with probability 0.583. Counting firings across all six seats double-counted, since the arm changes only the three seats it plays. Neither helped. The band table did: the branch fires when $p_{\text{best}} = 0$ *by definition*, at the floor of the worthless range, and the substitute's median $p = 0.385$ sits inside it too. The arm retains turns worth approximately nothing, 38.5% of the time, and its predicted effect is approximately zero.

So the champion's objective was right and the critique was wrong. Those 1.5% of decisions throw away a *worthless* turn, which is exactly the circumstance in which the objective's other terms should dominate the probability term — and they do. The discrepancy was a selection effect in the instrument: turns priced at positions sampled without regard to option quality and spent at positions where $p_{\text{best}} = 0$ by construction.

17.4 A note on public information

One candidate explanation, tested and set aside, produced a fact about the game worth keeping. Forking each firing and playing the champion's doomed ask in one branch and the substitute in the other, the substitute sheds 0.344 more card-equivalents of a teammate's uncertainty — and 0.269 more of each *opponent's*. An ask is public, and it reaches two partners and three opponents, so the team-weighted balance is

$$2 \times (-0.344) - 3 \times (-0.269) = +0.120 \text{ card-equivalents per firing, favouring the quieter ask.}$$

In six-player Fish a public disclosure is net-negative for the discloser's team whatever it says, because the opposition outnumbers the partnership three to two. This is the arithmetic behind the no-bluff structure noted in Section 10: signal is welded to move, and what you tell your partner you tell your opponents more of.

18 The endgame solved under imperfect information

These figures were recomputed. The solver's transposition table keyed nodes on hands, turn, half-suit winners and belief weights, and *not* on the history. The opponents in it are the champion, whose action is a function of its whole observation, so two nodes identical in every key field but reached by different move orders have different continuations and different values. Merging them returned one branch's value for the other, and the maximisation read those values — on the position that exposed it, the search reported +0.2500 for a move a rollout of the same strategy scores at +0.7500, and visited 154 nodes where the corrected search visits 12,086. The fault was found by an invariant that holds with no ground truth at all: a best response may copy the champion at every information set, so it cannot score below it. Five $m = 2$ positions did.

On the 304 $m = 1$ positions that both the broken and the corrected run solve, the aggregate barely moved — the gain went +0.4403 to +0.4392. That is not because the bug was harmless. It corrupted 43 of those 304 positions individually, and the errors went *both ways*: the fix raised the optimum in 24 and lowered it in 19, so they very nearly cancelled in the mean. The champion figure is identical position for position, as it must be — it is a rollout that never consulted the table — and that identity is what licenses the comparison at all. Everything below is the corrected run.

Every "exact" result above is a statement about the perfect-information game, and Section 17's companion result showed that game is degenerate. The v0.3 headline — belief agents choosing a provably optimal move in 100% of information-resolved positions — was scored against that same perfect-information optimum, and every one of those positions had a single live half-suit. This section measures what the champion does at $m = 1$ when it genuinely cannot see, which is a different question with a different answer.

18.1 An exact best response over information sets

`fish4/exact_ii` enumerates the deals consistent with the public record and optimises over the deviator's *information sets* — the same action at every deal it cannot tell apart — by backward induction against champion opponents. There is no sampling anywhere in it.

Two assumptions, stated rather than buried. The prior is uniform over consistent deals, which is what the constraint system alone licenses; tilting it by the champion's own opponent model would make the answer a statement about that model. And the opponents are a *deterministic realisation* of the champion, seeded from a hash of the observation, because the champion samples and a distribution over policies is not a strategy one can best-respond to. That second choice rests on a property that was checked rather than assumed: over 629 real decisions a freshly constructed agent given the same observation computes a bit-identical belief to one attached from the deal.

Two controls run first and gate the rest. Where the belief pins every card there is nothing hidden left, so the exact value must equal the closed form $2f - m$. Over 200 games it does, 344/344. And where the support is small enough to afford it, the champion's value is computed twice — once by rolling each deal forward, once by walking the same tree the best response walks and playing the champion at the deviator's nodes. Same strategy, two code paths, so they must agree, and they do: **601/601**. The first control cannot see a fault that needs several deals to show, which is exactly the fault that was there.

18.2 What the endgame is worth, and what the champion gets

Across 305 genuinely hidden positions solved exactly (13 skipped for a belief support above 24 deals and 3 over the 300,000-node budget, so 95% coverage):

exact optimum in the seat on move	+0.5084
the champion in the same seat	+0.0679
exact gain from deviating	**+0.4405**

with median gain +0.3333, maximum +1.5556, and — as the invariant requires, since a best response can always copy the champion — *no* negative entry in 305 positions. That invariant is now a hard failure rather than a printed warning: the run that violated it wrote its results file anyway.

Two readings. First, hidden information is expensive: the closed form values the last half-suit at +1 to the team on move, and played optimally without sight of the cards it is worth +0.51. Some positions are worth −0.5 — won under perfect information, lost under optimal play without it. Second, the v0.3 certificate is not wrong but certifies a much easier thing: where the champion cannot see, it captures +0.068 of the +0.508 available.

The caveat, measured. Best-responding to a realisation one can predict is easier than beating the mixture it came from, so this figure likely overstates exploitability against the champion as it plays. How much it overstates is not settled here, but its relevance is: over 41 hidden $m = 1$ information sets at 12 seeds each, the champion repeats its move every time in only 32% of them, averaging 2.41 distinct actions. It mixes at most of these decisions.

18.3 Where the disagreements are, and one that is provable

Comparing the champion's move to the exact optimum's at each hidden decision, 84 of 273 agree. Of the 189 disagreements, 79 cost *exactly* zero — ties broken differently, not errors — and the rest cost +0.5267 on average. Their shape does not implicate one component: of 96 costly ask-versus-ask disagreements, 49 share the target only, 12 the card only, and 35 neither. No repair is proposed from that.

Thirteen could not be priced at all, because the declared split was in no candidate the solver enumerates — and that set contains every assignment true in at least one consistent deal. Those claims were true in *no* deal the record allows. Checked directly, and the weak version of the check refuted the reading it was built to confirm: testing each card's own holder mask gives 0/1080. A declaration can satisfy every card's mask separately and match no complete deal, because the hand *counts* do not add up — the failure `claim4` names as "per-card modes can be jointly impossible".

	claims	jointly impossible
$m = 1$	60	**7 (11.7%)**
$m = 2 \dots 9$	480	0

Only at $m = 1$, and structurally: there the hand counts are tight enough that a mis-split is detectably infeasible, while above it there is slack enough that almost any team-only declaration fits some deal. Every such claim nulled: +0.000 sets each against +0.968 for the rest.

18.4 The repair, and why it is not adopted

A max-flow feasibility test over the live cards decides the question exactly at any layer, and agrees 40/40 with brute-force enumeration where enumeration is available. Supplying the most likely *feasible* declaration instead — filtering alone changes nothing, because `forced_claim` rebuilds a declaration from the masks when no candidate survives — gives, over 6000 pre-registered pairs:

$$+0.028, \quad 95\% \text{ CI } [+0.024, +0.033].$$

Every block positive, every block excluding zero, and 27 wins against 973 ties and 0 losses in a representative block: the arm alters play about 0.09 times a game, so most deals are bit-identical and repairing a claim that could not have won cannot cost anything.

The bar fixed before the arm existed was +0.05. **It is not adopted**, and the champion is unchanged. A provable error, a free repair, and an effect smaller than the smallest one this project agreed in advance to care about — all three are the finding.

19 The all-champion profile is not an equilibrium

Computed with the corrected solver, and therefore not covered by the withdrawal in Section 18.

Section 18's figures average over *positions*, and a game can contain several $m = 1$ decisions. That is the right unit for asking how well the champion plays the endgame and the wrong one for asking how exploitable it is: summing gains across positions in one game double-counts a continuation the deviator only gets to play once. So this takes exactly one position per game — the first $m = 1$ decision with genuinely hidden cards — and lets the deviator play champion-identically before it and optimally from it.

The value is already a team differential: +1 for a half-suit the deviator's team takes and −1 for one the opponents take, so a half-suit changing hands moves it by 2, exactly as it moves the set differential `scripts4/exploitability.py` reports. Over 60 games:

	per game	95% CI
sampled rollout best response, whole game	-0.4875	$[-1.2442, +0.2692]$
exact, **one** endgame decision only	$+\mathbf{0.1111}$	$[+0.0460, +0.1761]$

The interval excludes zero, so a single seat deviating at a single endgame decision takes a positive amount — and if the all-champion profile were an equilibrium, no single seat could gain by deviating at all. The rollout responder, deviating for a *whole* game, measured -0.4875: it does not merely fail to find an exploit, it loses ground, which is what `fish4/bestresponse.py` warns strategy fusion can do to a PIMC responder.

Why the per-game figure is the smaller one, and why it is the right one. Of the 60 games, only 22 contain a first $m = 1$ decision with anything hidden. In the other 36 the belief has *pinned every card* by the time the endgame arrives — not a wide support, not a missing position: all 36 are pinned — so the closed form already answers them and the restricted deviator gains exactly zero. Conditional on a hidden decision existing the gain is $+0.2928$; per game it is $+0.1111$. Quoting the conditional figure beside a whole-game bound would compare a mean over a favourable subset against a mean over everything. The 2 games whose first hidden decision exceeded the node budget are excluded from the denominator rather than scored zero: unsolved is not worth nothing.

That split is the better description of the champion. Its inference finishes the endgame outright in 60% of games; where it does not, it leaves about a third of a set on the table.

What this does not bound. The opponents are a *deterministic realisation* of the champion, seeded from a hash of the observation, and best-responding to a realisation one can predict is easier than beating the mixture it came from. The champion mixes at most of these decisions — over 41 hidden $m = 1$ information sets at 12 seeds each it repeats its move every time in only 32% of them. So this bounds the realisation, not the champion as it plays, and the claim is correspondingly narrow: *that* profile is not an equilibrium.

20 What this says about playing Fish

The results have readings that do not require an engine, and we state them because a study of a game people play ought to say something to the people playing it.

Watch where opponents ask, not just what they ask for. This is the finding of the paper, and it is actionable at the table. A player who asks in a half-suit is probably deep in it, in rough proportion to how many of its cards they hold, and that single inference is worth more than every other improvement we made combined. Concretely: when an opponent asks in hearts, do not only note the hard fact that they hold at least one heart — shift weight toward their holding several. The engine's version of this is one parameter, and it is worth about 1.9 sets per duplicate deal-pair against a strong opponent that ignores it.

An exact rule for the perfect-information endgame. Once every remaining card is public, the game is solved in closed form: the team on move scores every live half-suit in which it holds at least one card, and the opponents score the rest. Footholds only shrink and a well-informed asker never misses. The practical corollary is that in a fully-known endgame the only thing that matters is *how many live half-suits your team has a foothold in when it is your turn* — not how many cards, not which cards.

Do not agonise over when to declare. Reproducing v0.3's finding on a much better posterior: raising the claim bar from 0.97 to 0.99 changes essentially no decisions. The reason is structural rather than statistical. If your team genuinely holds all six cards of a half-suit, no opponent holds any card of it, so no opponent can legally ask in it — and a claim by a team that does not hold all six gives the set to you. Waiting is close to free. Wait until you know the split.

Two tie-breakers really do hold up. When two asks look about equally likely to land, prefer the one that, if it misses, hands the turn to the opponent with fewer cards, and prefer to fight in half-suits your team is already winning. v0.3 measured that pair at +1.41 sets per deal-pair; with an inference layer good enough to make the primary criterion much sharper, they are still worth +1.08. Better card-reading does not make positional judgement redundant.

The endgame is bookkeeping, so do the bookkeeping. Every one of 200 measured games reached a position where no ask anywhere could succeed: the sets were decided and only the declarations were left. Two consequences for a human. First, the last few turns are not a contest, so spend your attention earlier. Second — and this is the one people get wrong — if your team holds a half-suit outright, *no opponent can legally ask in it*, so it cannot be taken from you. There is never a reason to rush that declaration. In our games, half-suits that reached "ours but we cannot place the split" were nulled 23.4% of the time against 0.92% otherwise, and accounted for 73% of all nulls. Almost every one of those is a self-inflicted wound.

You can talk, and most players do not realise it. You may not ask for a card you hold. So asking for a card *proves to the table* that you do not have it — and when your team already owns the half-suit, the ask cannot lose you anything, because no opponent holds a card of it to give. In that situation an ask is a free, legal message to your partners telling them exactly where a card is not. Use it when you are stuck on a split. (Our engine gains about 20% fewer nulls doing this; it does not gain games, because the turn it spends is worth about as much as the sets it saves. For a human, who is far more likely to misplace a split than an engine is, the trade is probably better.)

Spend your effort on card-tracking, not on cleverness about which ask to make. Three measurements in this study point the same way and none of them is visible alone. The full eleven-term objective ranks candidate asks barely better than $\mathbb{P}$(success) by itself. Nearly every term we added *beside* $\mathbb{P}$(success) is a null. And the largest single effect in the whole study apart from the opponent model itself — +0.340 sets per deal-pair — comes from tripling the sampling budget, which does not touch the objective at all: it only makes the $\mathbb{P}$(success) the objective is already dominated by more accurate.

For a human that is a claim about where to put attention. Knowing more precisely where the cards are beats reasoning more carefully about what to do with what you know. The engine's version of "know more precisely" is more Monte Carlo draws; yours is tracking the failed asks, which are the constraints that do the work. It is not that positional judgement is worthless — the two tie-breakers above are real — but that between an hour spent on card-reading and an hour spent on ask selection, the measurements say the first hour is worth several of the second.

Do not hold back the re-take. A card an opponent has just taken from you is a card you know they hold, so asking for it back is a certain ask, and a certain ask keeps your turn. The case against is that the exchange is public and repeating it tells the table this half-suit is contested while neither side nets a card. Strong players describe doing it. We measured it, and it loses: a penalty on re-taking cost −0.340 sets per deal-pair at a small weight and −1.015 at a large one, the loss growing monotonically with how firmly the policy withholds. Two variants of playing to the score — more variance when behind, more caution when ahead — also failed to beat playing the same way regardless.

The situation is not rare, which is what makes the result usable: a re-take is available at 11% of positions, and at 18% of decision points the acting seat is already two or more exchanges deep with some opponent. So this is advice about a common decision, not a corner case. Take the certain ask.

We are careful about one thing here. What was measured is a penalty applied to *every* re-take, including the first, and the argument for withholding is really about a *repeated* exchange. The engine can now gate the penalty on the exchange recurring, and we can say in advance what that version could be worth at most — +0.159 sets per deal-pair — because the gate touches only the 5% of positions where a re-take is available and no card has yet come back the other way, which is 47% of what the ungated penalty flags. That is small enough that the 200-pair cells this study used for screening could not have resolved it either way, which is why no such cell was run.

It has now been run at 2000 pairs, pre-registered in `jobs/PREREGISTRATION_retake_gate.md` with the design, the five prior failures and the analysis all fixed before a pair was played. The gated penalty scores $-\mathbf{0.004}$ sets per deal-pair $[-0.067, +0.059]$ against the champion — indistinguishable from not having it. Against the ungated -0.340 the difference is $+0.336 \pm 0.169$.

Two readings, and the paper takes the cautious one. The gate does remove the harm: the ungated form's loss does not survive restricting the penalty to a repeated exchange, which is what its own argument always said it should. But the recovery is +0.336 where the gate's model caps it at +0.159, and a gate cannot recover more harm than it removes flags for. One of those numbers is wrong, and the likelier one is the ungated −0.340: it comes from a 200-pair screen whose interval $[-0.665, -0.015]$ barely clears zero, which is precisely the configuration that inflates an estimate — the sixth appearance in this paper of the same selection effect. So the honest statement is that withholding the re-take, in the only form its argument ever justified, neither helps nor hurts; and that the harm the ungated version appeared to do was partly an artefact of the screen that measured it.

The advice does not change, and it is now better supported. Withholding does not pay. Take the certain ask.

And trading hard does not pay either. Every cell above *withholds*. The opposite policy — take the card straight back, trade hard inside the duel, which is what strong players more often describe doing — had never been measured. It has now been, at 2000 pre-registered pairs (`jobs/PREREGISTRATION_retake_bonus.md`), and it returns $+\mathbf{0.013}$ $[-0.052, +0.079]$: the seventh null in this family. Against the gated penalty's −0.004 the difference is $+0.017 \pm 0.046$. The two policies act on *disjoint* slices of the same decision in opposite directions, and both are null, which says something neither says alone — the re-take decision does not repay any policy at all, in either direction.

What that null is about was fixed before the run, because the reachable base rate was measured first. A re-take is a *certain* ask, so the objective already scores it at $P(\text{success}) = 1$ — the term this whole paper says dominates — and it is *already the chosen ask at* 69% of the positions where it is legal. A bonus can therefore act at only 3.5% of positions, and where it can, the median score gap to the leader is **0.000**: at the median reachable position the weight is breaking an exact tie between two asks the objective values equally. Breaking a tie has expected value zero unless the objective is systematically wrong in a way correlated with re-taking. So this is a null about tie-breaking, and it is emphatically *not* a measurement of "trade hard in a duel" as a human would mean it. The pre-registration says all of this before the first pair, which is the only reason the sentence can be written now.

The practical advice is therefore unchanged in content and much better supported in kind: the re-take is a certain ask, so take it, and do not spend any thought on whether to make a policy of taking it or of holding back. Neither policy is worth anything. Spend the attention on the asks that are not certain.

Both of those figures moved, and the reason is worth recording rather than quietly restating. The depth statistic originally counted successful asks between this seat and one opponent in *either direction separately*, so an opponent taking two cards off this seat and giving nothing back scored the same depth as a card going out and coming home — opposite situations, since in the first that opponent has plainly netted two cards. Over 1023 harvested positions, 37.8% of the positions where the gate opened had no two-way exchange at all with the opponent it opened on. Requiring cards to have moved both ways takes the "two or more exchanges deep" rate from 57% to 18% and *raises* what the gate could recover from +0.098 to +0.159, because sparing a first re-take now correctly includes every position where nothing has come back yet. This is the same defect the gate was built to fix, one level down: an implementation that did not match its own stated hypothesis. The unit test that should have caught it was named for exactly this property and its example happened to run both ways, so it passed against code that did not have it.

And a warning about the obvious idea. Scoring asks by "how much does this improve my team's chance of winning this half-suit" — which sounds like the right objective, is measured in the right units, and rests on a well-calibrated model — is a disaster: −7.4 sets per deal-pair against simply asking for the card most likely to be there. In this game, the probability that the ask lands is not one consideration among several. It is the objective, and everything else is a tie-break.

21 Failed and abandoned experiments

Recorded deliberately, because in this project the negative results have repeatedly been the informative ones.

An exact posterior, by enumerating the disjunctions. Since the largest measured effect in this study comes from making the posterior more accurate, an *exact* posterior is the obvious thing to want, and there is a standard route to one. The counting DP is exact over candidate masks and quotas; what forces sampling is the OR constraints the no-bluff rule leaves behind. The complement of an OR is a conjunction of exclusions, which the DP absorbs natively, so inclusion–exclusion over the k OR constraints gives both the exact partition function and exact marginals in 2^k DP passes and no sampling at all.

We measured the two quantities that decide it before writing any of it. On 100 real decisions the median is $k = 6$ OR constraints (max 14; *none* has zero), and one DP pass returning marginals costs 0.86 ms. So the median decision would cost 55 ms against the sampler's 3.8, and only 15% of decisions would cost no more than sampling does.

The reason it fails is structural rather than a constant factor, which is why no amount of optimisation rescues it: the cost of exactness grows as 2^k in the same k that makes sampling hard. Where the sampler struggles — many OR constraints, a tightly coupled support — enumeration is exponentially out of reach, and where enumeration is cheap there is barely a constraint to handle and the sampler is already close to exact. The two methods are easy in the same places and hard in the same places, so neither can rescue the other. Recorded because this is an idea that will occur to the next person too, and it took ten minutes to kill with two measurements rather than a week to kill with an implementation.

Exact inference, on its own. The headline effort of this version — replacing a documented-biased sampler with provably correct inference — bought nothing in play (Section 14.1) and was measurably worse at predicting reality (Section 14.2). We report it as a failure of the *hypothesis*, not of the implementation: the implementation is validated against exhaustive enumeration in four separate ways. What failed was the assumption that the uniform posterior is the right target.

Rejection sampling for the disjunctive clauses. Exactly correct, and unaffordable: measured acceptance mean 0.28, median 0.17, below 0.05 on 16% of positions. Abandoned in favour of importance sampling with closed-form densities.

Folding the clauses into the exact dynamic program. Sound, and tractable in the median case (refined state space 5,760), but with a tail that makes it unusable as a default: p90 230,000 and p99 2.7×10^7. Retained as an option, not as the path.

A naïve batch rewrite of the sampler. Vectorising the draw across the batch gave only 1.4× until the dictionary materialisation of the draws was removed as well; the correct fix was to carry the whole batch as an integer matrix end to end, after which the same change gave 5.1×. Worth recording because the first measurement would have justified abandoning a change that was in fact the right one.

An under-converged fit that looked plausible. The first half-suit value model was fitted by plain gradient descent on unstandardised features and reported a held-out log-loss *worse than the class prior* while producing entirely reasonable-looking coefficients. Feature scales spanning two orders of magnitude were the cause. This is the reason the fitting routine now reports against two baselines (the class prior and a single-feature model) rather than against nothing.

A false positive, caught. Of more than twenty-five screening cells, exactly one produced an interval excluding zero in the improvement direction (+0.490, [+0.184, +0.796], 500 pairs). It failed to replicate twice on fresh seeds (+0.068 and −0.066). We record it as a failed experiment rather than burying it in a methodology note, because it is the shape of the mistake this project is most likely to make next.

Style that responds to the match. Every objective term in v0.4 is a function of one candidate ask and the posterior. None knows the score, and none knows that the last four moves were the same two players passing one card back and forth. Two ideas were built to close that gap, each ablatable to zero, and both are refuted — not as nulls, but with a dose-response, which is the more informative outcome because the gradient confirms the mechanism rather than merely the sign. All cells are 200 duplicate deal-pairs against the champion.

change	set diff	95% CI
break the duel, $w_{\text{retake}} = 0.30$	−0.340	[−0.665, −0.015]
break the duel, $w_{\text{retake}} = 1.00$	**−1.015**	[−1.562, −0.468]
gated on a real duel, 2000 pairs	−0.004	[−0.067, +0.059]
reward the re-take, $w_{\text{retake}} = -0.30$, 2000 pairs	+0.013	[−0.052, +0.079]
tie-breakers up when behind, $w_{\text{behind}} = 0.50$	−0.185	[−0.628, +0.258]
tie-breakers up when behind, $w_{\text{behind}} = 1.00$	**−0.890**	[−1.384, −0.396]
tie-breakers down when behind, $w_{\text{behind}} = -0.50$	−0.075	[−0.552, +0.402]

Breaking the duel penalises taking back a card this seat just lost to that same opponent. Section 3 notes that the state graph is cyclic precisely because two opponents can trade a card back and forth; the question here was whether a player should decline to. They should not, and three times the penalty costs three times the sets. This is the scheduling argument of Proposition 3 arriving from a third direction: a card you watched an opponent take is a card you *know* they hold, so taking it back is a certain ask and a certain ask keeps the turn. Deferring it risks the turn to preserve an option that was not going anywhere, and the information concealed by not asking is worth less than the possession thrown away. It is worth recording that this is a manoeuvre strong players describe using, which the engine could not previously express at all — the measurement is the point, not the outcome.

Score-adaptive tie-breakers scale the weights by how far behind the team is, on the reasoning that a losing side should buy information and a winning side should buy certainty. Scaling them *up* when behind loses monotonically. Scaling them *down* is indistinguishable from the champion. The asymmetry is the finding: the loss surface is steep on the side of trusting the tie-breakers more and flat on the side of trusting them less, which is what one would expect if v0.3's hand-tuned constants already sit at the point where the tie-breakers have given everything they have — and is a third piece of evidence, after the inverted-U sweeps and the substitution result of Section 14.6, that they are tie-breakers rather than an objective.

Neither idea is retried in another form. What both share is that they spend $P(\text{success})$ to buy something else, and this paper's most reproducible finding is that nothing else is worth buying with it.

Belief-space possession-chain search. Not abandoned — this one resolved, and it is listed here because it spent more pairs than anything else in the study and because three of its four rounds belong in a section about what does not work. Three runs of the identical configuration returned +0.570 (200 pairs), −0.044 and +0.386 (500 pairs each); the first and third exclude zero and the second excludes the third's estimate. Two 700-pair cells pre-registered as decisive pooled to $[-0.019, +0.324]$ and settled nothing, because their size had been chosen against an estimate the selected cell inflated. A fourth round, pre-registered and sized against the unbiased figure, finally answered it: +0.104, 95% CI $[+0.020, +0.189]$ over 6000 fresh pairs, with the six blocks agreeing at $\tau = 0$. Section 14.11 has the arithmetic. The lesson kept here is the shape of the estimate over rounds — +0.570, then +0.153, then +0.104 — and that no amount of cleverness in the analysis substituted for pairs.

Inherited from v0.3, and not revisited. Determinized search (PIMC and Information-Set MCTS) lost to its own prior; a value network trained on belief features and applied inside determinized worlds lost 34–6 to a train/inference distribution mismatch; quiescence extension made search worse; and an expected-value claim model predicted an optimal threshold near 0.70 that direct measurement falsified. None of these were re-attempted here, because v0.3's diagnosis — that the binding constraint is the objective, not the depth of search — was the premise of this version and is not contradicted by anything we measured.

22 Limitations

The opponent model is a model. Everything else in the inference stack is deduction, validated against enumeration. Equation (6) is a hypothesis about how people choose, fitted with one parameter. Against opponents whose ask policy is unrelated to their depth in a half-suit it could in principle hurt, and against an opponent who deliberately inverts the heuristic it is exploitable in the ordinary sense that any model is. We measure both cases rather than assert robustness, and γ is a single number that can be set to zero.

Depth is evaluated on the initial deal. The choice model is really about what a player held *when they asked.* Cards move, so the two diverge late in a game. The exact quantity would need a replay of the public history per candidate world, turning an $O(1)$ update into $O(|h|)$.

The uniform-posterior result is about *this* game and *these* opponents. We show that exact inference under an uninformative- choice assumption is worse than a biased sampler that accidentally encodes a choice model. That is a statement about which approximation is closer to the truth, not a general licence to prefer biased inference.

Exact solving is still bounded, and bounded in a way this section previously misstated. The tables solve the *perfect-information* game. Calling them absolute ground truth without that qualifier is an overclaim, and two measurements say how large a one.

First, the coverage. "Reaches positions with one, and now two, unresolved half-suits" describes the solver's reach over positions handed to it, not the game's. A tablebase value is legitimate only where the acting seat's belief pins every live card, which `tablebase4.pinned_state` enforces. Over 2582 decisions from 25 games that gate opens on 2.59% of decisions: never at three or more live half-suits (0 of 188), 12.1% at two, 52.7% at one. It first opens at step 100.2 of 103.5, with 8.1 of 54 cards left, and in 5 of 25 games it never opens at all.

Second, the gap. Applied by determinizing over a belief, the tables overstate the mover's team by 5.294 ± 0.159 sets on the positions play actually reaches, and by 8.18 with all nine half-suits still live, where the closed form $V = \text{sign} \times (2f - m)$ of §9 predicts the team on move takes 8.36 of the nine sets where play delivers +0.27. (That 8.36 is an average over the $m = 9$ positions play reaches, which is not the same as a fresh deal: footholds only shrink, so f has already fallen a little by then. At the deal itself (9) gives +8.794 exactly.) The closed form is not evidence that the Fish endgame is shallow; its own proof turns on "every ask can be made to hit, and a hit retains the turn", which is what perfect information buys and nothing else does. Measured independently, a team given the true deal beats the champion by 17.483 sets per duplicate deal-pair against a maximum of 18.

So the honest statement is not that midgame positions remain out of reach pending a larger solver. It is that exact ground truth covers 2.6% of the decisions a player faces, all in the last 3% of the game, and that extending the perfect-information solver to three half-suits in full — the natural next computational step — would move that coverage by approximately zero, because the positions it would newly cover essentially never arise with every live card pinned. More ground truth requires solving the imperfect-information game, and the smallest Fish position carrying any hidden information already needs three players holding live cards, hence two on one side: a team-coordination problem with no two-player zero-sum reduction beneath it.

No equilibrium claim, and the one exception. We compute no equilibrium. We do now compute an exploitability *lower* bound, in Section 19: a single seat deviating optimally at one endgame decision gains +0.1111 per game with an interval clear of zero, so the all-champion profile is not an equilibrium. That is a bound against a deterministic realisation of the champion rather than against the champion as it plays, and it is a lower bound only — it says the profile is exploitable, not by how much. The appropriate solution concept for a game with pre-play agreement and no in-play communication is TMECor (Celli and Gatti, 2018), and nothing here approximates it. Every strength statement is relative to a named opponent or absolute against an exactly solved subgame.

A rules discrepancy, stated. The printed rules allow a set to be declared at any point, including during an opponent's turn. v0.3's game loop only ever asked the player on move, so the rule was accepted by the engine but unreachable by any policy — which matters most for a seat that has run out of cards and therefore never gets a turn of its own while a teammate still has some. v0.4 adds off-turn declaration, offered identically to both generations so that the comparison stays a comparison of one thing. The primary evaluation is nevertheless run under v0.3's own configuration, so that the numbers are comparable with the previous version, with the variant reported separately.

23 Conclusions

We set out to remove a defect the previous version had documented in its own code: a belief sampler that was not uniform over the deals consistent with the public record. We removed it, exactly. Hidden-state inference in Literature reduces to counting assignments in a degree-constrained bipartite system, and although that is $\#P$-complete in general, the instances the game produces are tiny in the dimension that matters — roughly four distinct candidate sets even when twenty-six cards are unlocated — so a dynamic program returns the exact partition function, exact marginals, exact uniform sampling and exact joint probabilities. The disjunctive constraints that the program cannot absorb are handled by importance sampling with closed-form proposal densities, verified against exhaustive enumeration for support, density and marginals, with a control demonstrating that dropping the weights visibly breaks the estimate.

And it changed nothing. Against the previous champion, at matched seeds over 250 duplicate deal-pairs, provably correct inference moved the set differential by -0.008 per pair.

The instructive part is why, and that we could find out. Because the hidden hands exist in simulation, a posterior can be scored on the question it is actually trying to answer, and that measurement says the exactly-uniform posterior is *worse* at predicting reality than the biased sampler it replaced. Proposition 1 is a correct theorem about a false hypothesis: it assumes players' choices carry no information beyond legality. The old sampler's bias, an artefact of how it seeded the disjunctive constraints, happened to encode a crude model of how players choose. Removing the bias removed the model with it.

Replacing the accident with one line of modelling — a player asks in a half-suit in proportion to how many cards of it they hold — recovers the loss and goes past it, on all three instruments we have: posterior likelihood against the true hands, agreement with an exact solver, and duplicate-deal play. The last of these puts it at $+1.9$ sets per deal-pair against the previous champion, which is larger than the entire objective improvement that separated that champion from its own baseline.

Seven lessons generalise beyond this game.

Correct is not the same as good. An inference procedure is correct relative to a model. When a cheaper, wronger procedure beats an exact one, the exact procedure's *assumptions* are the thing that is failing, and the productive response is to find which assumption and relax it — not to add compute to the correct-but-misspecified thing.

Measure the intermediate quantity, not only the outcome. Every belief metric in the previous version was indirect, and a change that was provably correct and empirically useless would have been very hard to diagnose from win rates alone. Scoring the posterior directly against ground truth turned a puzzling null result into an identified cause in one experiment. Where a simulator gives access to the hidden state, refusing to use it for *evaluation* is a false economy; the discipline that matters is not letting it reach the acting policy.

Know what your experiment could not have found. The measured per-pair standard deviation of the set differential is around 3.8 sets, which means the previous version's 600-pair cells could not resolve effects below about 0.44 sets per pair. Several of its reported nulls were therefore uninformative rather than negative. We report the minimum detectable effect alongside every interval for that reason.

The sharper form of this lesson took most of the study to find, and it is that the noise figure is not a constant of the problem. Under common random numbers a paired trial on which the two arms behave identically contributes exactly zero, so the standard deviation scales with $\sqrt{s}$, where s is the share of trials on which the two arms diverge at all — and across our 46 cells the remaining factor is nearly constant at 3.86 ± 0.29, though Section 13.2 shows it drifts enough at low s to matter where the rule is used. The consequence is that sizing every experiment against a single number taken from two maximally different policies systematically over-sizes the comparisons that matter most, the near-identical ones. A candidate perturbing a tenth of trials needs 9.5× fewer of them than the constant implies. The error is conservative, so no interval we published is too narrow; what it cost is experiments never run and nulls recorded from cells that could not have found anything — and unlike a bad interval, those leave no trace to audit later.

Replicate the winner, not the losers. A screening study over twenty-five cells will hand you an apparent winner under a complete null, and it will arrive with a plausible mechanism attached — ours came with a story about two independently-motivated changes stacking, which is exactly what the previous version had genuinely found for a different pair of terms. The cheap defence is not a multiplicity correction, it is one more run on fresh seeds. Ours cost forty minutes and saved a wrong champion.

And it is not a lesson one learns once. The same error appears five times in this study, each occurrence a level above where the last was caught: a selected screening cell, then a section built on three points, then the power calculation for the experiment correcting the first, then the estimator built to measure the ask objective, and finally — in a shape we did not anticipate — an impossibility claim that closed a research direction. None was caught by being more careful. Each was caught by measuring the quantity that had been reasoned about — which is the next lesson, and the reason it is stated separately.

A diagnostic you can optimise is not the thing you want. The sampler reports an effective sample size, and it is a real quantity honestly computed. Tuning the proposal to raise it worked — 83.7 effective draws became 105.8 — and made the posterior 3.4× worse, because effective sample size measures how *flat* the importance weights are and a proposal covering one corner of the target very well has beautifully flat weights over the corner it covers. Every intermediate metric in a pipeline is a proxy for something further downstream, and the moment it becomes an optimisation target it stops being a measurement of what it proxied. Section 14.3 keeps both halves of that result, and the parameter ships at zero.

An impossibility claim is a claim. We wrote, in this paper and in the code, that the belief tracker *cannot* attach to a determinized mid-game position, and closed the objective-learning line on it. The claim was specific, mechanistic, well-argued and supported by a number, and every one of those properties made it less likely to be checked. It was false: the tracker is anchored on the initial deal but does not have to be handed one, and given the current hand and the public history it back-computes a consistent deal in a few lines. The measurement that had been used to support the diagnosis, repeated with the strong policy actually running, moves from $+0.101$ to $+0.681 \pm 0.142$.

What made it survive is that a negative claim generates no experiment. A positive result invites replication; "this cannot be done" invites moving on, and the sentence that closes a direction is exactly the sentence nobody tests.

Having been caught once, we swept the engine for the rest of them. `fish4/NEGATIVE_CLAIMS.md` registers every load-bearing impossibility claim in the source with a verdict: one refuted (the above), one true in the limit and false where the engine operates (a proposal twist "cannot change what the policy computes" — correct asymptotically, and at the shipped 160 draws it made the marginals 3.4× worse), and six that are either rules theorems or were measured rather than assumed. The rider that sweep produced is worth more than the register: an asymptotic "cannot" and a finite-sample "cannot" are different claims, and only one of them is about the engine that ships, so a measured negative should always carry the range it was measured over. The asymmetry is worth naming because the cost is asymmetric too: an inflated positive wastes the run that eventually corrects it, while an unexamined negative costs everything downstream of the direction it closed, and leaves no anomaly to notice. Our stated defence — measure the quantity you reasoned about — applies verbatim, but we had been applying it only to numbers we wanted to be true.

A coefficient that decays may be a covariate that decays. The choice model's fitted exponent falls from 2.00 at the opening to -0.02 by the endgame, which reads as players changing how they choose. Most of it is regression dilution: the model conditions on depth at the *initial* *deal*, cards move, and the covariate's disagreement with the hand it describes grows monotonically through a game. Noise in a covariate attenuates its own coefficient, so noise that grows with time manufactures a trend out of drift. Refitting on the quantity the model is actually about leaves a decay that is real and half the size. The general form: before explaining a trend in a fitted parameter, check whether the thing it is fitted *to* is degrading at the same rate.

23.1 What we would do next

- **A proper likelihood, not a proxy — two of three steps taken.** This item previously read: the choice model uses depth on the initial deal, and the right object is the acting policy's own probability of the observed action given the hand, evaluated at the time of the ask. Two thirds of that turned out to be cheap.

  At the time of the ask costs a lookup table, not a fixpoint. Cards move only on successful asks and successful asks are public, so at-ask-time depth is initial-deal depth plus a delta the log fixes identically in every world (Section 14.5). It is worth 4,654 nats on 17,005 real decisions.

  The policy's own probability was measured rather than assumed, by fitting the champion's actual half-suit choice as a conditional logit — which is what Table 4 is. The exponent is 2.195 ± 0.033 at the ask, not the 1 the shipped model assumes.

  What remains is the part that is genuinely not free: using that exponent requires the denominator the engine drops. Measured across worlds, the coefficient of variation of that denominator is a V centred on $\alpha = 1$ — 0.0000 there, 0.0352 at 1.207, 0.2255 at 2.195, and 0.0997 at the shipped $\gamma = 0.35$. So the engine currently pays more for the omission than the fitted exponent would, and reaching the exponent the data want means computing the denominator per world: tracking depth in *every* live half-suit per player during the draw rather than only the asked ones. That is a constant-factor cost in the sampler.

  And the free version of it has now been played, which changes what this item is worth. At $\alpha = 1$ the denominator is exactly constant, so the better covariate can be used there at no cost at all. Six pre-registered blocks put that configuration at +0.1015, 95% $[+0.0063, +0.1967]$ — real, below the +0.15 this study fixed in advance as the smallest effect worth adopting, and heterogeneous enough ($\hat{\tau} = 0.150$, the size of the effect itself) that a random-effects reading includes zero. The engine keeps $\gamma = 0.35$.

  That is the useful bound on the expensive version. The cheap form of this idea, at the one exponent where the model is correctly specified, buys about a tenth of a set. Paying a constant factor in the sampler to reach $\alpha = 2.195$ has to beat that by enough to be worth the sampler time it takes away from draws — and draws are the largest measured effect in this study. We would now want a sizing argument for that trade before writing the code, not after.

  A fixpoint is still not required, and that is the substantive update: the opponent being modelled is a policy whose choice rule can be fitted directly from its own play.

- **A coordinated best-responder.** The robustness tests here show the model is not harmed by an opponent built to violate its assumption, which is weaker than showing it cannot be exploited. Training a genuine best response, and reporting the value it extracts, is the only route to an absolute claim about how good the champion is.

- **Cross-play.** Once anything in this engine is *learned* from self-play rather than configured, cross-play between independently seeded runs becomes necessary rather than optional, for the reason Hanabi established: an agent that only wins alongside copies of itself has not solved the game.

- **Belief-space search, past the one point we tested.** Section 12 built the design this section previously proposed — a lookahead over the posterior rather than over sampled determinizations — and Section 14.11 now settles it at +0.104 per deal-pair. Settling it took pairs rather than a cleverer analysis, which is the part worth carrying forward: the harness's intervals are measured to be honest, so the 1200 pairs that started this were never going to separate a tenth of a set from zero, and no amount of pooling was going to fix that. The question answered is narrow: possession-chain value, a single sweep of quota coupling, beam 4, depth 3, entering additively. Proposition 3 says the version without the coupling could never have worked, which removes a whole branch of the space rather than one point in it. What is still open is a leaf evaluator other than banked cards, a converged rather than single-sweep coupling, and a horizon that does not stop at the end of one possession — and, now that the one tested point is known to be worth something, whether any of those is worth more.

A Reproducing this work

The engine, agents, experiments and this paper are in one repository. v0.3's code is untouched, so its results remain reproducible alongside these; v0.4 lives in `fish4/`.

```
py -m pytest tests  -q     # v0.3: rules, fuzz, leakage proofs, belief
                           # soundness, statistics, exact solver
py -m pytest tests4 -q     # v0.4: exact counting vs brute force, sampler
                           # support/density/marginals, batch equivalence,
                           # information-boundary invariance, exact solver v2

py scripts4/duel.py jobs/j6_ladder.json 4      # the strength ladder
py scripts4/duel.py jobs/j11_final.json 6      # headline + ablations
py scripts4/summarise_duels.py                 # every duel, with its MDE

py scripts4/posterior_accuracy.py 14 3   # posteriors scored on the true hands
py scripts4/exact_bench4.py              # agreement with provably optimal play
py scripts4/fit_hsvalue.py 200 3         # fit the half-suit value model
py scripts4/analytics4.py 120 3          # strategy statistics
py scripts4/check_tex.py                 # structural check on this document
```

Every duel appends to `results/v04_duels.jsonl` recording both random seed streams, the full agent specifications, the rule configuration and the result. `summarise_duels.py` prints, alongside every interval, the smallest effect that run could have detected — which is the column that stops a null being read as a negative.

B The champion

```
from fish4.registry4 import make_agent
agent = make_agent(("fishbot4", {"opponent_gamma": 0.35}))
```

Everything else is at its default, and the defaults are deliberately v0.3's own ask objective ($w_{\text{suit}} = 0.06$, $w_{\text{turn}} = 0.60$, $w_{\text{scarce}} = 0.20$). The improvement reported here comes from inference, not from re-tuning the objective on top of it — and Section 14.9 is the record of trying to re-tune it and failing.

C Information-boundary tests

The property asserted is that every quantity v0.4 computes is bit-identical when the hidden cards of the other five seats are permuted into a different layout equally consistent with the public record. Candidate permutations are drawn from the acting seat's own belief sampler — uniform permutation almost never lands on a consistent world, because the public record is very constraining — and each candidate is then validated by an *independent* replay of the whole public history, so the test does not assume the sampler is correct. Checked quantities: posterior marginals under four combinations of the opponent-model parameters, all eleven ask-objective terms, and the end-to-end policy action.

There is a second boundary, and it is worth recording that the first set of tests did not protect it. The permutation tests establish that the *policy* never reads a card it should not; they say nothing about what the *server* puts on the wire. An earlier revision of the multiplayer table included the deal seed in the per-seat view, as a debugging convenience. Every hand is a deterministic function of that integer, so one number defeated the entire property the permutation tests were built to guarantee — and it was readable without joining the table, since a table code alone fetches a view. The lesson generalises past this bug: a leak test that only exercises the component you were thinking about will keep passing while the leak moves to the component you were not. The transport now has its own test, which walks the serialised view recursively and fails on any key or value that could reconstruct the deal, for a seated player and for an anonymous visitor alike.

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